Is There an Optimal Decision Theory?

Within-Problem Conflicts, the Semantics of Menus, and Counterfactual Alternatives

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ABSTRACT. It is often suggested that no decision theory can be optimal on every fair decision problem, because the standards by which theories are judged conflict. This paper makes the claim precise for three successively stronger notions of a decision theory and determines when it holds. If a theory is any assignment of policies to finite problems whose structural representation supplies every counterfactual, ex-ante optimality is always attainable, and every inconsistency among ex-ante optimality, causal dominance, ratifiability, commitment stability, and an evidential criterion has a core of one or two criteria; the conflict between ex-ante optimality and dominance is possible exactly when the state law depends on the prescribed action. If a theory is a rule on causal matrices, whose rows are the belief states induced by the agent's policies and whose diagonal is ex-ante value, common affine invariance again yields only small cores, while invariance under row translations, with neutrality and contraction, forces a rule to choose every option on exactly the domains satisfying a reachability condition, and principal binary independence characterises ex-ante evaluation; both results fail if a smaller menu keeps the belief model of the larger one. If a theory is a computable procedure that reads the source code of its environment and must infer the dependence itself, an optimal function always exists under a designated substitution semantics, and a uniform optimiser exists under a termination promise. An optimal agent that halts on every input, or that selects exactly among computable expected values, must be non-computable, and the oracles that compute one are exactly the PA degrees, some of them strictly below the halting problem; prescribing answers off the admissible class raises the requirement to $\mathbf{0}'$ or $\mathbf{0}''$. Problems built against a theory exist for every theory the environments can compute, but not only for those. Further negative results come from the ways of recognising the agent in code. An axiomatisation of counterfactual operators shows that calibration together with an extensionality that identifies computations by their actual behaviour forces causal evaluation and is inconsistent with ratifiability, while an extensionality that compares computations under every substitution admits ex-ante evaluation satisfying every desideratum. Each impossibility concerns how the agent's alternatives are specified, not how to choose among them. A closing discussion argues that the specification is fixed, where it is fixed at all, by the mechanism coupling the agent to its predictors, which the standard thought experiments leave undetermined.

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1. Introduction

Is there a decision theory that gets every decision problem right? For half a century the question has been pursued case by case. A theory is proposed; a problem is constructed on which its verdict seems wrong; the theory is amended or a rival proposed; and the rival meets its own counterexample. The impression this history leaves is that no theory can be optimal everywhere, because the standards by which theories are judged pull in different directions. This paper tries to turn that impression into theorems. It asks, for precisely defined criteria of optimality and precisely defined classes of problems, when a decision theory satisfying them exists and when it does not, and it tries to close the question in the sense of determining every inconsistency among the criteria considered and exhibiting a witness for every consistent combination.

1.1. From dominance to Newcomb's problem

The expected-utility theories of Ramsey (1931), von Neumann and Morgenstern (1944), and Savage (1954) model an agent's options as functions from states of the world to consequences, with the states not depending on what the agent does. Under that assumption, the principle of dominance—prefer an option that is better in every state—is a consequence of maximising expected utility, and the two principles never conflict. Jeffrey (1965) replaced the act-independent states with a single algebra of propositions, valuing an act by the news it would bring; Bolker (1966, 1967) supplied the representation theorem. In Jeffrey's theory the probability of a state may depend on the act, and dominance and expected utility can come apart.

Nozick (1969) made the possibility vivid. A reliable predictor has placed a large sum in an opaque box if and only if it predicted that the agent would take only that box; a transparent box contains a small sum. Taking both boxes dominates, since the contents are fixed; taking only the opaque box maximises expected utility conditional on the act. Nozick credited the problem to the physicist William Newcomb and proposed restricting dominance to cases where the states are probabilistically independent of the acts. Gibbard and Harper (1978) gave the conflict its canonical form by distinguishing two expected utilities: one computed with the probabilities of counterfactual conditionals, which measures what an act would causally bring about, and Jeffrey's, which measures what it indicates. Causal decision theory, as the first came to be called, takes both boxes; evidential decision theory takes one. Gibbard and Harper also introduced Death in Damascus, in which Death, having predicted the agent's destination, waits there: whichever city the agent chooses, the other looks better. Lewis (1979b) observed that the prisoner's dilemma played against a sufficiently similar partner is a Newcomb problem, and Lewis (1981b) unified the causal theories of Gibbard and Harper, Skyrms (1980, 1982), and Sobel (see Sobel 1994) by means of dependency hypotheses. Against the charge that causal decision

theorists end up poorer, Lewis replied that Newcomb's problem rewards irrationality (Lewis 1981a). On the evidential side, the tickle defence of Eells (1982) argued that a self-aware agent's own beliefs and desires screen off the correlations on which one-boxing relies, and Spohn (1977) argued that deliberation crowds out prediction, so that an agent cannot coherently assign probabilities to its own current options. Joyce (1999) gave causal decision theory a representation theorem, and by the end of the century it had become the orthodox view (Weirich 2024).

1.2. Instability, ratifiability, and the counterexamples

Death in Damascus exposed a second fault line. Jeffrey (1981, 1983) proposed that a rational choice should be *ratifiable*: optimal relative to the beliefs the agent would have upon making it. In Death in Damascus no pure option is ratifiable. Harper (1986) proposed maximising causal expected utility among ratifiable options and admitting mixed acts so that such options exist; Richter (1984), Weirich (1985), and Rabinowicz (1988) studied the instability that makes every option look worse once chosen; and Skyrms (1990b) modelled deliberation as a dynamical process whose rest points are deliberational equilibria, connecting ratifiability with equilibrium in games (Skyrms 1990a). Sobel (1994) collected much of this work.

Egan (2007) reopened the debate with cases in which an act is evidence about the very causal structure that determines its outcome—shooting as evidence of a lesion that would make one miss, pressing a button that kills all psychopaths as evidence that one is a psychopath—and in which causal decision theory seems to give the wrong verdict. The responses define the current landscape. Deliberational theories recommend equilibria of a process of self-updating (Arntzenius 2008; Joyce 2012, 2018); benchmark theory assesses each option against a state-dependent benchmark (Wedgwood 2013); Spohn (2012) argued that a properly formulated causal theory one-boxes; Ahmed (2012, 2014a,b) revived the evidential theory with cases in which causal reasoning fails; Hare and Hedden (2016) classified decisions as self-reinforcing or self-frustrating and criticised the stability-based remedies; Gallow (2020) proposed a causal theory that manages the news an act carries; and Gallow (2026) argues for comparativism, on which instrumental value is fundamentally a matter of pairwise comparison. The argument from wealth was reconsidered as well: Bales (2018) and Spencer and Wells (2019) argued that it compares agents who face different problems, and Wells (2019) constructed a problem with equal opportunities in which causal reasoners predictably end up richer. The collection edited by Ahmed (2018) surveys the state of play.

1.3. Policies, plans, and fairness

A third line of work evaluates not acts but policies. Kuhn's analysis of extensive games introduced perfect recall and the equivalence of mixed and behavioural strategies under it (Kuhn 1953). Hammond (1988) derived expected utility from consequentialism and dynamic consistency in decision trees; McClennen (1990) defended resolute choice, the execution of an ex-ante plan even when it is not optimal ex post; and

Gauthier (1986, 1994) argued that dispositions which are rational to adopt can make it rational to carry out assurances and threats. Piccione and Rubinstein (1997) separated planning-stage from action-stage optimality in the absent-minded driver, and Aumann et al. (1997) showed that the planning-optimal strategy is action-optimal although the converse fails; Arntzenius et al. (2004) described sequences of individually rational choices whose sum is disastrous and from which only binding escapes.

In artificial intelligence the policy view became a family of decision theories. Updateless decision theory (Dai 2009) selects the input–output map that is optimal under the agent's prior; timeless decision theory (Yudkowsky 2010) treats a decision as the output of an abstract computation; and Soares and Fallenstein (2015) and Yudkowsky and Soares (2017) developed these ideas into functional decision theory, with a peer-reviewed presentation by Levinstein and Soares (2020). Functional decision theory is claimed to do well on *fair* problems, those in which the outcome depends only on the agent's behaviour in the problem at hand; its authors leave a formal criterion of fairness as future work and suggest that a general-purpose theory may match or outperform all rivals on fair problems if one can be found. MacAskill (2019) objected with the Bomb, a problem in which the policy-optimal action is to take a box known to contain a bomb, violating the principle that under certainty one should choose the best outcome. Oesterheld and Conitzer (2021) showed how to extract money from causal decision theorists by predictive offers; Everitt et al. (2015) formulated sequential, act-based and policy-based versions of the causal and evidential theories; Bell et al. (2021) showed that value-based reinforcement learners in Newcomb-like environments converge, if at all, to ratifiable policies, and that continuous environments always admit one; Halpern (2020) gave conditions for the existence of ratifiable mixed acts; MacDermott et al. (2023) characterised the theories with mechanised causal graphs; and Oesterheld and Conitzer (2024) and Cooper et al. (2024) studied when self-locating beliefs make act-level causal or evidential reasoning agree with the ex-ante optimal policy. Wolpert and Benford (2013) argued that the two Newcomb arguments presuppose incompatible formalisations of how choice and prediction are related.

1.4. Programs, proofs, and logical counterfactuals

The functional theories treat a decision as the output of a computation, and so inherit a question that the earlier theories could leave implicit: what would have happened had that computation produced a different output? An idea of this kind is already in Hofstadter's *superrationality*, on which agents who know that they reason identically should cooperate in a symmetric prisoner's dilemma (Hofstadter 1983). Game theory made programs the objects of choice. Tennenholtz (2004) introduced program equilibrium, in which players submit programs that can read one another's code, and Bárász et al. (2014) showed that agents defined in provability logic cooperate robustly with one another, while Critch (2019) proved a resource-bounded version of Löb's theorem underlying such results.

Within the functional programme, Soares and Fallenstein (2015) identified reasoning about counterpossibles—conditionals whose antecedent is a non-actual output of

a deterministic computation—as the central open problem, and gave a proof-based formalisation that queries a provability oracle. Such agents can act on spurious counterfactuals, provable only because the agent's own action is provably different. The difficulty was raised in discussion of timeless decision theory (Drescher 2009), given an explicit Löbian form for proof-searching agents (Slepnev 2010, 2012), and analysed by Benson-Tilsen (2014); it is now known as the "five-and-ten" problem (Demski and Garrabrant 2019; Fallenstein 2013). The "chicken rule", on which an agent takes any action it can prove it will not take, was proposed to block such proofs (Slepnev 2011). Fallenstein developed modal versions of updateless decision theory, proved that for every provably extensional problem some sound extension of arithmetic makes it optimal (Fallenstein 2014b,c), constructed problems on which every fixed modal theory fails (Fallenstein 2014a), and proposed a partial semantics of "third-person counterfactuals" (Fallenstein 2015a,b). Logical induction (Garrabrant et al. 2016) gave a computable criterion for assigning probabilities to logical sentences; its use for decisions requires exploration and meets obstacles of its own (Garrabrant 2017). Demski and Garrabrant (2019) placed these problems within a wider account of agents embedded in the world they reason about, and Taylor (2020) proposed alternatives to logical counterfactuals altogether. In general reinforcement learning, Leike and Hutter (2015a) showed that natural notions of optimality for universal agents are either dependent on the prior or trivial, and Leike and Hutter (2015b) located AIXI and exact policy selection from computable values in the arithmetical hierarchy; the degree results of Section 16 answer the analogous question for decision functions on disjoint problems. Two classical results of computability theory frame all of this: Rice's theorem (Rice 1953), that every non-trivial property of the functions that programs compute is undecidable, and Kleene's second recursion theorem (Kleene 1938; Moschovakis 2010), which allows a program to refer to its own code.

In philosophy, the same conditionals are *counterpossibles*. On the orthodox semantics of Lewis (1973) they are all vacuously true; non-vacuists evaluate them at impossible worlds (Brogaard and Salerno 2013; Nolan 1997); and the exchange between Williamson (2015, 2018) and Berto et al. (2018) concerns whether a semantics can treat them non-vacuously while respecting the identifications that ordinary counterfactuals respect. Lewis (1976) proved the classical triviality theorem for a related identification, of conditional probabilities with probabilities of conditionals.

1.5. Impossibility results

Four results are of the form this paper pursues. Skalse (2021) proposed a general counterexample to any decision theory: an opaque box that is empty exactly when the target theory permits taking it. Fallenstein (2014a) proved the corresponding result in provability logic: every fixed modal decision theory fails on some provably extensional problem. Briggs (2010) proved that no completely general decision theory satisfies both a Pareto condition across the evaluations of an agent's possible post-decision selves and a self-sovereignty condition on which the comparison of two options depends only on how those two options evaluate all options, modelling decision-theoretic paradoxes on voting paradoxes and Sen's liberal paradox.

Podgorski (2022) proved that, for theories assigning each option an overall value, independence of irrelevant alternatives forces the comparison of two options to depend only on their values by their own lights, and so singles out the evidential theory; his tournament decision theory escapes the result by producing no overall value.

These results draw on the theory of social choice, whose methods the present paper also uses. Arrow (1951) established that no aggregation rule satisfies a handful of individually innocuous conditions on an unrestricted domain. The contraction condition on choice functions goes back to Chernoff (1954); Arrow (1959) showed that choice from all finite subsets is rationalised by an ordering exactly when it satisfies the weak axiom of revealed preference; and Sen (1970, 1971, 1977, 1993) named and analysed the conditions α, β, and γ and the limits of internal consistency of choice. Contraction consistency conflicts with majority choice through the binary cycle, a fact central to the theory of tournament solutions (Brandt et al. 2016; Brandt and Harrenstein 2011; Laslier 1997; Moulin 1986), and it continues to generate impossibilities (Basu Mallik and Bhattacharya 2026; Blair et al. 1976).

1.6. The question and the answer

The thesis this paper set out to prove was that no decision theory is optimal on all fair decision problems, the obstruction coming from mutually incompatible criteria of optimality rather than from any demand for a general rule. Whether it is true depends on what a decision theory is required to be, and the paper answers the question three times, for three successively stronger requirements. Each strengthening moves the obstruction, and at each stage the obstruction turns out to lie in the specification of the agent's alternatives rather than in the choice among them.

A theory as an assignment (Part I). Section 2 defines a class of finite problems with prediction in which every counterfactual the criteria need is supplied by a structural representation: for every policy, the law of the world's response is part of the problem. A decision theory is any assignment of policies to problems, lookup tables included. The thesis is then false: choosing an ex-ante maximiser in each problem is a decision theory that is ex-ante optimal everywhere (Theorem 3.6). The answer is cheap because the model hands the agent exactly what causal, evidential, and functional theories disagree about. What remains true is shallow. Among ex-ante optimality, structural causal dominance, ratifiability, commitment stability, and an evidential criterion, every inconsistency has a core of one or two criteria (Theorems 5.7 and 5.10), the central pair being Nozick's. A conflict between ex-ante optimality and dominance is possible exactly when the state law depends on the prescribed action (Theorem 4.2); it survives imperfect prediction and randomisation, with an exact trade-off between the two kinds of regret (Proposition 6.1). Every within-problem conflict reduces to the choice of the law under which an alternative is evaluated: the law induced by the alternative, or the law induced by the actual choice.

A theory as a rule (Part II). Requiring one rule across problems needs a common description of them. Each problem is reduced to its causal matrix, whose rows are the belief states induced by the agent's possible policies and whose diagonal is ex-ante

value; every real square matrix arises (Theorem 7.2). With common affine invariance, the cross-problem axioms again reduce to singleton and pairwise cores, one of which is Briggs's theorem (Theorem 8.1). With invariance under row translations—the matrix form of evaluating options against the baseline of each belief state—a genuine four-axiom impossibility appears: neutrality, row-translation invariance, and contraction force a rule to choose every option (Theorem 9.3), on exactly those domains satisfying a reachability condition (Theorem 9.7). Ex-ante evaluation, in turn, is characterised by principal binary independence together with neutrality, affine invariance, ordering, non-triviality, and monotonicity (Theorem 11.1 and Corollary 11.2), whose core is Podgorski's theorem. Both results depend on how a smaller menu is modelled. If removing options also removes the beliefs that those options would induce, the results hold; if the belief model is held fixed, row-baseline-invariant evaluation becomes menu coherent, and contraction no longer forces ex-ante evaluation (Section 10 and Theorem 11.4). The impossibilities concern the preservation of comparisons when the belief perspectives associated with removed options disappear, not the preservation of choices when options disappear.

A theory as a procedure (Part III). Finally, the supplied dependence is removed. An agent is a computable function that receives the source code of its environment and must itself determine what would happen under each of its outputs. With a designated substitution interface, an optimal decision function exists on both admissible classes considered, fixed copies of arbitrary programs included (Theorem 15.3), and under a termination promise one algorithm is optimal everywhere (Theorem 15.5), the uniform counterpart of Theorem 3.6. The negative results that have nothing to do with counterfactual inference are three: queries about other problems (Proposition 15.4); fixed copies of the agent, which defeat every agent that halts on all inputs (Theorem 15.6, Skalse's construction made well defined); and the uncomputability of exact selection among computable expected values, which holds with no predictor at all (Theorem 15.10). They can be measured exactly. The oracles that compute an optimal agent halting on every input, or one selecting exactly among computable expected values, are the PA degrees (Theorem 16.3), so an optimal theory need not compute the halting problem, and some optimal theories lie strictly between the computable degree and $\mathbf{0}'$ (Corollary 16.4); prescribing a default off the admissible class raises the requirement to $\mathbf{0}'$ or $\mathbf{0}''$ (Proposition 16.6). Problems built against a theory exist for every theory that the environments can compute, but non-computable theories can also be predictably wrong (Proposition 16.8). The genuinely counterfactual question is how an agent should identify the parts of a closed program that depend on its output. Literal, local, and global matching of the agent's code each fail in a characteristic way (Theorems 17.2–17.5), and proof-based counterfactuals are spurious or undefined for ordinary programs (Theorem 17.7 and Proposition 17.9). Axiomatising counterfactual operators gives a complete consistency map (Theorem 18.12). Calibration on independent problems, together with an extensionality that identifies computations by their actual behaviour, forces causal evaluation (Lemma 18.9), and with ratifiability it forms a minimal inconsistent triple whose witness is Death in Damascus (Theorem 18.11). Under an extensionality that compares computations under every substitution, a

single operator, ex-ante evaluation, satisfies every desideratum (Corollary 18.13). The two evaluations of Part I thus reappear as two notions of when computations are the same.

The conclusion of the paper unifies the three answers. *No impossibility found here is an impossibility of choosing well among specified alternatives. Each concerns how the alternatives are specified: under which law an alternative is evaluated, which belief perspectives survive when options are removed, and how a change in the agent's output is transported through the computations that depend on it.*

Several results are known and are presented as such: the Newcomb conflict (Nozick), the absence of ratifiable pure acts (Gibbard and Harper; Jeffrey), mixed existence under continuity (Halpern; Bell and co-authors), the Pareto–self-sovereignty conflict (Briggs), the core of the binary characterisation (Podgorski), the fixed-copy diagonal (Skalse; Fallenstein in modal form), the spurious-proof mechanism behind five-and-ten, modal optimality (Fallenstein), the recursion-theoretic ingredients of the exact-selection and degree theorems (PA degrees and the basis theorems of Scott and of Jockusch and Soare), and the complexity of almost-sure termination (Kaminski and Katoen). The consistency maps, the characterisation of when dominance and ex-ante optimality can conflict, the trade-off identity, the analysis of noisy commitment, the circulant theorem and its exact domain criterion, the comparison of menu semantics, the cardinality results on families of rules, the matching theorems, and the consistency map for counterfactual operators, and the degree classification of optimal agents were not found in the literature searched; their ingredients are classical, and no claim of priority is made.

Part I (Sections 2–6) treats single problems: the model, the criteria and the possibility theorem, the witness problems, commitment and the within-problem maps, and robustness. Part II (Sections 7–13) treats rules: the causal matrix, the pairwise reduction, row translations and contraction, menu semantics, the characterisation of ex-ante evaluation, problem types, and mixed policies. Part III (Sections 14–19) treats procedures: problems as programs, optimal agents and computable optimisation, the degree of non-computability that optimality requires, the inference of dependence from source, the axioms for counterfactual operators, and the witness problems revisited together with the literature on counterpossibles. Part IV interprets the results. Section 20 argues that the disagreement among decision theories concerns what the agent's alternatives are, that it is settled, where it is settled at all, by the mechanism coupling the agent to whatever depends on it, and that the usual thought experiments leave that mechanism unspecified; it also records what the results leave untouched, the assumption that agents have real-valued utilities. Section 21 states the final thesis and the questions left open, and Appendix A lists every stipulation made.

Part I.
Single problems

2. Decision problems with prediction

This section fixes the class of problems on which decision theories are compared. The aim is a model rich enough to contain the problems that drive the debate (Newcomb's problem, the twin prisoner's dilemma, Death in Damascus, the Bomb, counterfactual mugging) and precise enough that every criterion to be considered has a determinate verdict on every problem. Each place where an informal notion—a counterfactual, "the same computation", "holding the state fixed"—has to be settled by stipulation is marked as such; the stipulations are collected in Appendix A.

2.1. Problems, policies, and structural plans

Definition 2.1 (Decision problem). A *decision problem* D is a finite extensive-form tree with three kinds of non-terminal nodes and real payoffs at its leaves:

(i) *nature nodes*, with fixed probability distributions over their successors;

(ii) *prediction nodes*, nature nodes whose distribution may depend on the agent's policy, but only through the actions that the policy prescribes at specified information sets;

(iii) *agent nodes*, partitioned into a finite set H_D of information sets, each $h \in H_D$ carrying a finite nonempty set A_h of actions.

It is required that no information set occurs twice on any path of the tree.

Perfect recall implies the last requirement, which is all that will be used. A *deterministic policy* for D is a choice of action at every information set, and the policy space is

$$\Pi_D = \prod_{h \in H_D} A_h,$$

a finite nonempty set. Mixed policies are deferred to Sections 6 and 13.

The policy-dependent kernels of the prediction nodes determine the distribution of actual play, but they do not by themselves settle the causal counterfactuals that several criteria require: what would have happened had the agent acted otherwise, *with the environment's responses held fixed*. To make these counterfactuals explicit, the tree is supplemented with a structural description of the environment.

Definition 2.2 (Structural representation). A *structural representation* of D consists of

(i) a finite nonempty set S_D of *environment plans*, each of which specifies the response of every nature and prediction node, including nodes on branches not taken and responses that depend on earlier physical actions;

(ii) for every policy $\pi \in \Pi_D$, a probability distribution $\mu_\pi \in \Delta(S_D)$;

(iii) the payoff $U_D(s, \pi)$ obtained by traversing the tree with plan s and policy π.

The *ex-ante value* of π is

$$V_D(\pi) = \sum_{s \in S_D} \mu_\pi(s)\, U_D(s, \pi).$$

The representation separates two kinds of dependence. *Ordinary causal dependence* of the environment on earlier actions is carried by the responses written into each plan s. *Predictive dependence* on the policy is carried by the map $\pi \mapsto \mu_\pi$. A tree with policy-dependent kernels can always be given a structural representation by supplying potential outcomes for its nature and prediction nodes; how the outcomes on alternative branches are coupled is an additional modelling choice, which matters for the causal criteria and is fixed by the representation. Conversely, every structural representation is implemented by a tree whose root is a prediction node selecting s according to μ_π, followed by deterministic traversal.

2.2. *Well-definedness and fairness*

Two conditions on problems are distinguished.

Definition 2.3 (Well-definedness). D is *well defined* if μ_π is a probability distribution on S_D for *every* $\pi \in \Pi_D$.

Definition 2.4 (Fairness). D is *fair* if S_D, U_D, and the entire map $\pi \mapsto \mu_\pi$ are fixed features of D: the evaluation of an agent may depend on the actions its policy prescribes, but not on the identity, name, or source code of the decision theory that produced the policy. Fixed algorithms may occur as constants of the environment ("the predictor simulates the algorithm g_0"), provided g_0 does not vary with the agent being evaluated.

Fairness excludes problems whose evaluation depends on the identity of the decision theory that produced the policy. It does not exclude a diagonal against a fixed theory T, which pays an agent exactly when its action differs from that of T: there T enters as a constant. Remark 3.7 shows that such a problem fails well-definedness instead, and Theorem 15.6 that it becomes well defined once problems are programs. Well-definedness is a separate requirement. A verbally described "perfect predictor" may satisfy an action-based fairness condition while its promised behaviour is logically inconsistent for some policy; Section 4.4 gives an example. The presence of a fixed algorithm in the environment does not by itself make a self-referential specification well defined (Remark 3.7). Fixed algorithms are harmless here because a problem is a finite object whose data include their outputs; when problems are given as source code and the agent must compute those outputs, they are not (Section 15).

Write $\mathcal{F}^*$ for the class of well-defined, fair finite problems with a fixed structural representation, closed under the commitment construction of Section 5.

Definition 2.5 (Decision theory). A *decision theory* on a class $\mathcal{F}$ of problems assigns to every $D \in \mathcal{F}$ a policy $T_D \in \Pi_D$. No uniformity, computability, or generality is required: every lookup table is a decision theory.

Totality is part of the definition. A proposed criterion that recommends nothing on some problem therefore fails to define a decision theory on any class containing that problem; it does not define a theory that is silent there. The permissive definition is the weakest sense of "decision theory", and it is the first of three. Part II requires a theory to be a single rule on a common reduced form of problems, and Part III requires it to be a computable procedure that receives a description of the problem and must itself work out what the structural representation supplies here.

2.3. Reach and local causal deviation

For a policy π and an information set h, let

$$R_h(\pi) = \{s \in S_D \,:\, h \text{ is reached in the traversal with } (s, \pi)\}, \qquad r_\pi(h) = \mu_\pi\big(R_h(\pi)\big).$$

The number $r_\pi(h)$ is the actual probability that h is reached under π; h is *positively reached* under π if $r_\pi(h) > 0$. For $a \in A_h$, let $\pi[h{\leftarrow}a]$ denote the policy that agrees with π everywhere except at h, where it prescribes a.

Lemma 2.6. *For every π, h, and $a \in A_h$:*

(i) $R_h(\pi[h{\leftarrow}a]) = R_h(\pi)$;

(ii) if $s \notin R_h(\pi)$, *then* $U_D(s, \pi[h{\leftarrow}a]) = U_D(s, \pi)$.

Proof. With the plan s fixed, the path through the tree is determined by the prescriptions at the information sets it visits. Up to its first visit to h, the path under $(s, \pi[h{\leftarrow}a])$ uses only prescriptions at information sets other than h, which are unchanged; so it coincides with the path under (s, π), and h is reached under one exactly when it is reached under the other. If h is not reached, the two paths coincide entirely. Since no information set recurs on a path, a visit to h is the first and only one. □

The *causal value* of a at a positively reached h, with beliefs induced by π, is

$$Q_\pi(h, a) = \frac{1}{r_\pi(h)} \sum_{s \in R_h(\pi)} \mu_\pi(s)\, U_D\big(s, \pi[h{\leftarrow}a]\big).$$

This fixes the causal counterfactual completely: the distribution over environmental responses is the one the actual policy induces, the action at h is changed, and the policy's other prescriptions are kept. The contrast with ex-ante evaluation lies in a single subscript: $Q_\pi(h, a)$ averages with μ_π, whereas $V_D(\pi[h{\leftarrow}a])$ averages with $\mu_{\pi[h\leftarrow a]}$. Every within-problem conflict studied below is a consequence of that difference.

3. Optimality criteria and the possibility of ex-ante optimality

3.1. Within-problem criteria

The criteria below are the precise forms of the principles that the main decision theories are usually taken to satisfy. Each is a condition on the policy T_D that a theory T selects in a problem D; the commitment criterion C4 and the evidential criterion C6 are introduced in Section 5, once the constructions they require are available.

Definition 3.1 (Ex-ante optimality, C1). T satisfies C1 on D if

$$T_D \in \arg\max_{\pi \in \Pi_D} V_D(\pi).$$

Definition 3.2 (Structural causal dominance, C2). T satisfies C2 on D if, at every information set h positively reached under $\pi = T_D$, there is no $a \in A_h$ such that

$$U_D\big(s, \pi[h{\leftarrow}a]\big) > U_D\big(s, \pi[h{\leftarrow}\pi(h)]\big) \quad \text{for every } s \in R_h(\pi).$$

The quantifier ranges over *structural* states: all plans that reach h, including those to which μ_π assigns probability zero. This is a deliberate resolution of an ambiguity in "every state compatible with h", and it matters. If compatibility meant only the states in the support of μ_π, then in Death in Damascus with an infallible predictor (Section 4.3) the single supported state places Death in the chosen city, so switching would strictly improve the payoff in every supported state and C2 itself would be unsatisfiable. Likewise, on trees where prediction nodes follow h, "hold fixed the outcomes of the nodes preceding h" leaves open how the later prediction nodes are to be evaluated; freezing their policy dependence while averaging over their outcomes can make dominance collapse into an unsatisfiable best-response requirement. The structural plan fixes both matters at once.

Definition 3.3 (Ratifiability, C3). T satisfies C3 on D if, at every information set h positively reached under $\pi = T_D$,

$$\pi(h) \in \arg\max_{a \in A_h} Q_\pi(h, a).$$

Definition 3.4 (Existence, C5). T satisfies C5 on D if its own criterion is satisfiable there, that is, if T_D is defined.

Since decision theories are total by Definition 2.5, C5 holds for every theory and imposes no restriction; it records the requirement that a criterion be satisfiable on every problem of the class. Were partial theories admitted, totality would become an independent axiom and the consistency maps of Section 5 would change.

Proposition 3.5. C3 *implies* C2.

Proof. Suppose h is positively reached under $\pi = T_D$ and some a satisfies $U_D(s, \pi[h{\leftarrow}a]) > U_D(s, \pi)$ for every $s \in R_h(\pi)$. Since $r_\pi(h) > 0$, the conditional

distribution $\mu_\pi(\cdot \mid R_h(\pi))$ is a probability measure carried by $R_h(\pi)$, and a strict pointwise inequality on $R_h(\pi)$ gives $Q_\pi(h,a) > Q_\pi(h,\pi(h))$. Thus $\pi(h)$ is not a maximiser of $Q_\pi(h,\cdot)$, and C3 fails. □

3.2. Ex-ante optimality is always attainable

The opening question admits a direct answer once "decision theory" is taken in the permissive sense of Definition 2.5.

THEOREM 3.6 (Universal ex-ante selection). *On every class of well-defined finite problems there is a decision theory satisfying* C1 *on every problem of the class.*

Proof. Fix a tie-breaking order on each Π_D and set $T_D^{\mathrm{ex}} = \text{first arg max}_{\pi\in\Pi_D} V_D(\pi)$. Each Π_D is finite and nonempty, and V_D is real valued on it because D is well defined, so the maximum exists. □

REMARK 3.7 (Self-referential problems). Skalse (2021) proposes, for a fixed target theory X, a choice between a transparent box containing 1000 and an opaque box whose content is 0 if X permits choosing it in that problem and 2000 otherwise, and argues that another theory can outperform X. The target theory is an environmental constant, so this is not discrimination by the identity of the evaluated agent. Translated into the present model, however, the construction requires a fixed point: for an ex-ante optimiser, an empty opaque box makes it choose the transparent box, which requires the opaque box to be full, and conversely. Neither content yields a well-defined instance. In the same way, the diagonal family D_b with $u(a) = \mathbf{1}\{a \neq b\}$, $b \in \{0,1\}$, would need $b = T(D_b)$, but an ex-ante optimiser has $T(D_0) = 1$ and $T(D_1) = 0$. The schema does not refute Theorem 3.6 on well-defined problems; its additional self-referential realisability assumption is where it leaves that class. Lewis (1981a) analysed a related failure: an announced reward for supposedly irrational choice can render the proposed scenario inconsistent. The verdict depends on the problem being a finite object whose data include the constant's output. When problems are given as programs, so that the agent must compute that output itself, the same construction is well defined and defeats every agent that halts on all inputs (Theorem 15.6).

The unqualified thesis that no decision theory is optimal on all fair problems is therefore false. What can be true is that particular *combinations* of criteria are jointly unsatisfiable, and the remaining sections determine exactly which. A first observation shows how little such a result can depend on generality or uniformity. Call a criterion C *local* if whether a theory T satisfies C on D depends only on T_D; C1, C2, C3, and C5 are local, while C4 is not. For a local criterion C_i let $A_i(D) \subseteq \Pi_D$ be the set of policies that satisfy it on D.

LEMMA 3.8 (Pointwise reduction). *For local criteria $C_1, \dots, C_k$ and a class $\mathcal{F}$,*

$$\exists T\ \forall D \in \mathcal{F}\ \bigwedge_i C_i(T,D) \quad\Longleftrightarrow\quad \forall D \in \mathcal{F}\ \bigcap_i A_i(D) \neq \emptyset.$$

Proof. From left to right, T_D lies in the intersection. From right to left, let T_D be the first element of the intersection under a fixed order. □

Because lookup tables are admitted, an impossibility for local criteria is never more than the exhibition of a single problem on which their acceptable sets are disjoint. No difficulty about general rules, computability, or self-reference is involved; those enter only when a theory must be a single rule on a common description of problems (Part II) or a computable procedure that reads descriptions (Part III). This is the precise sense in which the within-problem impossibilities of the next two sections are shallow; non-local criteria, and the cross-problem axioms of Sections 7–11, are where deeper structure can appear.

4. The witness problems

This section works through the problems that separate the criteria. Payoffs are scaled for arithmetic convenience; nothing depends on the particular numbers beyond the inequalities stated.

4.1. Newcomb's problem

Let O denote taking only the opaque box and B taking both boxes. A predictor of symmetric error $\varepsilon \in [0, 1]$ forecasts the agent's prescription at its single information set, which is reached with probability one; the opaque box is full if and only if the forecast is O. Payoffs are

	opaque box full	opaque box empty
O	2	0
B	3	1

with $\Pr(\text{full} \mid O) = 1 - \varepsilon$ and $\Pr(\text{full} \mid B) = \varepsilon$. The structural states are the two contents of the opaque box.

Proposition 4.1 (Nozick 1969). *In Newcomb's problem $V(O) = 2(1 - \varepsilon)$ and $V(B) = 1 + 2\varepsilon$, so $V(O) - V(B) = 1 - 4\varepsilon$ and* C1 *uniquely requires O whenever $0 \le \varepsilon < \frac{1}{4}$. Since $U(s, B) - U(s, O) = 1$ in both structural states,* C2 *and* C3 *uniquely require B for every ε. Hence no theory satisfies* C1 *and* C2 *on any class containing Newcomb's problem with $\varepsilon < \frac{1}{4}$.*

With opaque prize m and transparent bonus c, the same conflict holds exactly when $m(1 - 2\varepsilon) > c$. Any symmetric predictor better than chance therefore produces it for suitable stakes. The conflict is Nozick's original one between maximising expected utility and dominance; the content of Proposition 4.1 is only that the formal criteria reproduce it.

The proposition shows that a conflict *can* arise. The next result characterises exactly when it can, and shows that neither accurate prediction nor large stakes is needed.

THEOREM 4.2 (Policy-dependent state laws characterise the conflict). *Let A be a finite set of actions with $|A| \geq 2$, S a finite set of states, and $\mu_a \in \Delta(S)$ the state law induced by prescribing a at a single always-reached choice. The following are equivalent:*

(i) for every payoff function $u : A \times S \to [0,1]$, some ex-ante maximising action is not strictly dominated state by state;

(ii) the laws μ_a, $a \in A$, are all equal.

Proof. (ii)⇒(i). Under a common law μ, if a strictly dominates b in every state then $\mathbb{E}_\mu u(a,\cdot) > \mathbb{E}_\mu u(b,\cdot)$, so no ex-ante maximiser is strictly dominated; a maximiser exists because A is finite.

(i)⇒(ii). Suppose $\mu_a \neq \mu_b$. There is an event $E \subseteq S$ on which they differ; exchanging the names of a and b if necessary, $\delta = \mu_b(E) - \mu_a(E) > 0$. Choose $0 < c < \delta/(1+\delta)$ and set

$$u(b,s) = (1-c)\,\mathbf{1}_E(s), \qquad u(a,s) = (1-c)\,\mathbf{1}_E(s) + c,$$

with payoff 0 for every other action. All payoffs lie in $[0,1]$, and a strictly dominates b by c in every state. Yet

$$V(b) - V(a) = (1-c)\mu_b(E) - (1-c)\mu_a(E) - c = (1-c)\delta - c > 0,$$

and $V(a) \geq c > 0$ exceeds the value 0 of every other action. So b is the unique ex-ante maximiser, and it is strictly dominated. □

Any non-constant dependence of the preceding state law on the prescription thus suffices for some bounded payoff assignment to separate C1 from C2; for fixed payoffs, the size of the dependence determines whether the conflict occurs.

4.2. The twin prisoner's dilemma

Two copies of the same policy play a prisoner's dilemma with temptation 3, reward 2, punishment 1, and sucker's payoff 0. The twin is represented as a prediction node that copies the agent's prescription at its single information set; it is *not* a second, independently assignable coordinate of Π_D, which would remove the stipulated identity of the two decisions. Then $V(\text{cooperate}) = 2$ and $V(\text{defect}) = 1$, so C1 requires cooperation, while with the twin's action held fixed, defection gains one unit whether the twin cooperates or defects, so C2 and C3 require defection. The connection between this problem and Newcomb's is due to Lewis (1979b). What the tree does not supply is the identity of the two decisions: "the same computation" enters only as a stipulated dependence of the prediction node on the agent's prescription. Sections 17 and 18 ask whether an agent given the twin's source code could infer that dependence, and Section 19 records how each candidate answer decides this problem.

4.3. Death in Damascus

Death predicts which of two cities the agent will choose and waits there (Gibbard and Harper 1978). Let actions and states both be $\{1,2\}$, with $U(s,a) = \mathbf{1}\{a \neq s\}$ and $\mu_a = \delta_a$ (infallible prediction).

PROPOSITION 4.3. *In Death in Damascus no deterministic policy satisfies* C3*, whereas both deterministic policies satisfy* C2*. The failure of* C3 *persists for every symmetric error* $\varepsilon < \frac{1}{2}$.

Proof. Under the policy a, the induced state is a; staying has causal value 0 and switching has causal value 1. With error ε, staying has causal value ε and switching $1 - \varepsilon$, and $\varepsilon < 1 - \varepsilon$. For C2, neither city is better than the other in *both* structural states, so no action is dominated. □

Consequently C3, read as a requirement on deterministic policies, is individually unsatisfiable on any class containing this problem: it fails to define a decision theory at all. The absence of ratifiable pure acts in this problem is well known from the ratifiability literature (Skyrms 1990a); Section 6.3 records how far randomisation repairs it.

4.4. The Bomb

MacAskill (2019) describes a predictor of error 10^{-24} that places a bomb in the left box if and only if it predicts that the agent will take the right box, and leaves a note saying so. The agent sees the note and the bomb. Taking the left box is free; taking the right box costs 1; burning to death costs $M = 10^9$ in the scaling used here.

Once the agent observes whether a bomb is present, there are *two* information sets, h_{bomb} and h_{clear}, and four policies $(a_{\text{bomb}}, a_{\text{clear}}) \in \{L, R\}^2$. This has a consequence for well-definedness.

PROPOSITION 4.4 (The Bomb as stated is not well defined). *No predictor whose forecast determines the observation ("predicts R: bomb; predicts L: no bomb") and whose target is the agent's* eventual *action can have an error bound below one uniformly over the four policies.*

Proof. Under the policy (L, R), a forecast of R produces a bomb and hence the action L; a forecast of L produces no bomb and hence the action R. The forecast is wrong in every case. □

A coherent version is obtained by letting the predictor target the fixed coordinate $\pi(h_{\text{bomb}})$, whether or not h_{bomb} is eventually reached:

$$\Pr\big(\text{bomb} \mid a_{\text{bomb}} = L\big) = \varepsilon, \qquad \Pr\big(\text{bomb} \mid a_{\text{bomb}} = R\big) = 1 - \varepsilon.$$

PROPOSITION 4.5. *In the repaired Bomb with death cost* $M > 1$*, the ex-ante values are*

policy $(a_{\text{bomb}}, a_{\text{clear}})$	V
(L, L)	$-M\varepsilon$
(L, R)	$-M\varepsilon - (1 - \varepsilon)$
(R, L)	$-(1 - \varepsilon)$
(R, R)	-1

For $0 < \varepsilon < 1/(M+1)$, C1 *uniquely selects* (L, L), *while* C2 *and* C3 *uniquely select* (R, L).

Proof. The values follow by conditioning on the presence of the bomb. (L, L) beats (R, L) exactly when $M\varepsilon < 1 - \varepsilon$, that is $\varepsilon < 1/(M+1)$; it beats (L, R) and (R, R) a fortiori. For $\varepsilon > 0$ both information sets are positively reached under every policy. At h_{bomb}, every structural state reaching it contains the bomb, and R yields $-1 > -M$; at h_{clear}, L yields $0 > -1$. Hence (R, L) is the only policy satisfying C2, and, since it is causally optimal at both information sets, it satisfies C3. □

MacAskill's numbers satisfy the inequality. Two qualifications are essential. At $\varepsilon = 0$, h_{bomb} is unreached under (L, L), and the reached-only criteria C2 and C3 do not oppose that policy. And if the problem is instead taken to be the already-conditioned decision "a bomb is visibly present", its two options have values $-M$ and -1, and C1 itself selects R. The visible-bomb challenge is MacAskill's; the complete contingent-policy kernel above is the additional assumption under which it becomes a theorem.

4.5. Counterfactual mugging

A fair coin is tossed. On heads the agent receives r if and only if an infallible predictor forecasts that the agent's prescription on tails is to pay; on tails the agent is asked to pay c, with $r > c > 0$ (Nesov 2009). There is one information set, reached on tails. Then

$$V(\text{Pay}) = \tfrac{1}{2}(r - c) > 0 = V(\text{Refuse}),$$

so C1 requires paying, while at the tails information set paying costs c in every structural state reaching it, so C2 and C3 require refusing. The problem separates ex-ante from ex-post evaluation by a prediction about an information set that is unreached on half the paths; it is used in Section 5.3 to locate evidential criteria and in Section 6.4 to test restrictions on what predictors may target.

5. Commitment, evidence, and the within-problem map

5.1. The commitment extension

Reflective stability asks what a theory would do if, before any prediction is made, it could choose which policy it will run. To state this as a criterion, the choice of a policy must itself be a decision problem.

Definition 5.1 (Commitment extension). Let D be a problem and λ a *coupling* of the laws $(\mu_q)_{q \in \Pi_D}$, that is, a probability distribution on $\prod_{q \in \Pi_D} \operatorname{supp} \mu_q$ whose marginal on the q-th coordinate is μ_q. The *commitment extension* $K_\lambda(D)$ is the problem in which

(i) the agent, at a single information set, chooses a policy $q \in \Pi_D$;

(ii) this choice irrevocably installs q;

(iii) the environment draws $\omega = (s_q)_{q\in\Pi_D}$ from λ, independently of the choice, and the agent receives $U_K(\omega, q) = U_D(s_q, q)$.

Unless stated otherwise the structural states of $K_\lambda(D)$ are $S_K = \prod_q \operatorname{supp}\mu_q$, and $K(D)$ denotes the product coupling $\lambda = \bigotimes_q \mu_q$. The constructor is tagged and well founded, so that $K(D)$, $K(K(D))$, … have unambiguous decompositions, and $\mathcal{F}^*$ is closed under it.

The extension models a *causal* choice of policy made before prediction: changing the commitment changes the environment's subsequent response, because the coordinate s_q that is paid out depends on q, while the law λ of the whole vector does not.

Proposition 5.2. *For every coupling λ and every $q \in \Pi_D$, $V_{K_\lambda(D)}(q) = V_D(q)$.*

Proof. $V_{K_\lambda(D)}(q) = \mathbb{E}_\lambda\big[U_D(s_q, q)\big] = \mathbb{E}_{\mu_q}\big[U_D(s, q)\big] = V_D(q)$, since the q-th marginal of λ is μ_q. □

Definition 5.3 (Commitment stability, C4). T satisfies C4 on D if the commitment $q = T_{K(D)}$ that T chooses in $K(D)$ satisfies $q(h) = T_D(h)$ at every information set h positively reached under T_D.

Unlike the other criteria, C4 relates a theory's behaviour on two different problems. It is the first cross-problem condition in this paper, and the only one stated without passing to the reduced form of Section 7.

5.2. *The map of* C1–C5

The consistent combinations need explicit total witnesses, not merely examples on a few problems.

Proposition 5.4 (A coherent ex-ante theory). *There is a decision theory on $\mathcal{F}^*$ satisfying* C1 *and* C4.

Proof. On problems that are not commitment extensions, let $T_D = T_D^{\mathrm{ex}}$. Recursively, in $K(D)$ let T commit to the policy T_D already selected for D. By Proposition 5.2, T_D maximises $V_{K(D)}$, so C1 holds on $K(D)$; C4 holds with exact agreement. □

C1 alone does not imply C4: on a problem in which both actions yield zero, an ex-ante maximising lookup table may choose different tied actions in D and in $K(D)$.

Proposition 5.5 (A dominance-respecting theory). *For each D fix a full-support distribution ν_D on S_D, for instance the uniform one, and let*

$$T_D^\nu = \operatorname{first}\operatorname*{arg\,max}_{\pi\in\Pi_D} \sum_{s\in S_D} \nu_D(s)\, U_D(s, \pi).$$

Then T^ν satisfies C2 *on every problem.*

Proof. Let $\pi = T_D^{\nu}$ and suppose that at an information set h positively reached under π some a satisfies $U_D(s, \pi[h{\leftarrow}a]) > U_D(s, \pi)$ for all $s \in R_h(\pi)$. By Lemma 2.6(ii) the payoff is unchanged on $S_D \setminus R_h(\pi)$. The set $R_h(\pi)$ is nonempty because $r_\pi(h) > 0$, and ν_D gives it positive weight, so $\pi[h{\leftarrow}a]$ strictly improves the objective that π maximises, a contradiction. □

T^{ν} is an existence witness, not a recommended theory: its fixed weights ignore the predictive dependence altogether.

Proposition 5.6. *No theory satisfies* C2 *and* C4 *on a class containing Newcomb's problem with* $\varepsilon = 0$ *and its commitment extension.*

Proof. In D, C2 requires B (Proposition 4.1). At $\varepsilon = 0$ every support is a singleton, so $K(D)$ has the single structural state (opaque box full under O, empty under B), and committing to O yields 2 while committing to B yields 1. Committing to O therefore strictly dominates, and C2 in $K(D)$ requires it; C4 requires the commitment to agree with B at the always-reached information set. □

Theorem 5.7 (Within-problem map). *On* $\mathcal{F}^*$ *the inclusion-minimal inconsistent subsets of* $\{C1, \dots, C5\}$ *are exactly*

$$\{C3\}, \qquad \{C1, C2\}, \qquad \{C2, C4\},$$

and the maximal consistent subsets are exactly $\{C1, C4, C5\}$*, witnessed by Proposition 5.4, and* $\{C2, C5\}$*, witnessed by Proposition 5.5.*

Proof. The three sets are inconsistent by Propositions 4.3, 4.1, and 5.6; their proper subsets are consistent by the two witnesses (C5 holds for every theory). A consistent set cannot contain C3. If it contains C2, it contains neither C1 nor C4, so it lies in $\{C2, C5\}$. Otherwise its members lie in $\{C1, C4, C5\}$. □

Among C1–C5, then, there is no minimally inconsistent family of three or more individually satisfiable criteria. The requested minimality is attained only by the pairs $\{C1, C2\}$ and $\{C2, C4\}$, whose singletons have explicit total witnesses, and the first is a direct pointwise disagreement in the sense of Lemma 3.8.

5.3. An evidential criterion

An evidential criterion requires a joint law of acts and states. Under a deterministic policy, conditioning on an alternative action is ordinarily conditioning on a null event, so the expression $\mathbb{E}[\text{payoff} \mid h, \text{agent takes } a]$ cannot be read off the actual distribution of play. The following completion makes it determinate.

Definition 5.8 (Reference-policy evidential criterion, C6$^{\kappa}$). For each D fix a full-support distribution κ_D on Π_D and define the joint law $J_D(\Pi = \pi, S = s) = \kappa_D(\pi)\mu_\pi(s)$, with $Y = U_D(S, \Pi)$. For $h \in H_D$ and $a \in A_h$ let

$$e_h(a) = \mathbb{E}_{J_D}\big[Y \mid h \text{ reached},\ \Pi(h) = a\big],$$

with $e_h(a) = -\infty$ when the conditioning event is null. T satisfies C6$^\kappa$ on D if $T_D(h) \in \arg\max_a e_h(a)$ at every h positively reached under T_D.

This is a specified evidential criterion, not an identification with every version of evidential decision theory; in sequential problems it evaluates continuation behaviour with the reference distribution over policies. It is individually satisfiable: choose an e_h-maximiser at every h with $J_D(h \text{ reached}) > 0$ and anything elsewhere. If h is positively reached under T_D, then $J_D(h \text{ reached}) \geq \kappa_D(T_D) r_{T_D}(h) > 0$, so the constraint is met, and some action has a finite evidential value there.

Proposition 5.9. *(i) In Newcomb's problem $e(O) = V(O)$ and $e(B) = V(B)$, so* C6$^\kappa$ *requires O for $\varepsilon < \frac{1}{4}$ and conflicts with* C2.

(ii) In counterfactual mugging C6$^\kappa$ *requires refusing, since $e(\mathit{Pay}) = -c < 0 = e(\mathit{Refuse})$ at the tails information set, and conflicts with* C1.

(iii) In the commitment extension of counterfactual mugging, $e(q) = V_D(q)$, so C6$^\kappa$ *commits to paying while refusing in the original problem, and conflicts with* C4.

Proof. (i) With one always-reached information set, conditioning on the action identifies the policy, so $e(a) = \mathbb{E}_{\mu_a} U(\cdot, a) = V(a)$. (ii) Conditional on tails and on the tails prescription being "pay", the payoff is $-c$; on "refuse" it is 0. (iii) In $K(D)$ the commitment is the policy variable and λ does not depend on it, so $e(q) = \mathbb{E}_\lambda U_D(s_q, q) = V_D(q)$, and $V_D(\text{Pay}) > V_D(\text{Refuse})$; the tails information set is reached with probability $\frac{1}{2}$ under the refusing policy. □

Theorem 5.10 (Map with the evidential criterion). *On $\mathcal{F}^*$ the inclusion-minimal inconsistent subsets of* $\{\text{C1}, \dots, \text{C5}, \text{C6}^\kappa\}$ *are*

$$\{\text{C3}\}, \quad \{\text{C1}, \text{C2}\}, \quad \{\text{C2}, \text{C4}\}, \quad \{\text{C1}, \text{C6}^\kappa\}, \quad \{\text{C2}, \text{C6}^\kappa\}, \quad \{\text{C4}, \text{C6}^\kappa\},$$

and the maximal consistent subsets are $\{\text{C1}, \text{C4}, \text{C5}\}$, $\{\text{C2}, \text{C5}\}$, *and* $\{\text{C6}^\kappa, \text{C5}\}$.

Proof. Among C1, C2, C4, C6$^\kappa$, every pair except $\{\text{C1}, \text{C4}\}$ is inconsistent by Propositions 4.1, 5.6, and 5.9, and each listed maximal set has a total witness. □

A different evidential semantics can change this table; Section 13.3 compares two. The theorem is a statement about the displayed definitions.

5.4. Noisy commitment and couplings

Proposition 5.6 uses exact prediction and, implicitly, a choice of coupling. Both matter, and they matter differently for structural and for expectational comparisons. Let the opaque prize be m and the transparent bonus c, with $m > c > 0$, and let the symmetric error be $\varepsilon \in [0, 1]$. Write $g = m(1 - 2\varepsilon) - c$ for the ex-ante advantage of O.

Let X indicate a full opaque box under commitment to O, and Y a full opaque box under commitment to B, so that $\Pr(X = 1) = 1 - \varepsilon$ and $\Pr(Y = 1) = \varepsilon$. Every coupling is determined by $t = \Pr(X = 1, Y = 1)$, which ranges over $0 \leq t \leq \min(\varepsilon, 1 - \varepsilon)$:

(X, Y)	probability	payoff of O	payoff of B	difference $O - B$
$(1, 1)$	t	m	$m + c$	$-c$
$(1, 0)$	$1 - \varepsilon - t$	m	c	$m - c$
$(0, 1)$	$\varepsilon - t$	0	$m + c$	$-m - c$
$(0, 0)$	t	0	c	$-c$

Proposition 5.11 (Noisy commitment in Newcomb's problem). *(i) For every coupling, the expected causal value of committing to q is $V_D(q)$ whatever the agent's beliefs about its own commitment; every criterion that ranks commitments by expected causal value requires commitment to O exactly when $g > 0$, while the same criterion requires B in D.*

(ii) With structural states $S_K = \operatorname{supp} \lambda$, commitment to O strictly dominates commitment to B if and only if $\varepsilon = 0$; commitment to B strictly dominates if and only if $\varepsilon \geq \frac{1}{2}$ and $t = 1 - \varepsilon$.

(iii) With structural states $S_K = \prod_q \operatorname{supp} \mu_q$, neither commitment dominates for $0 < \varepsilon < 1$, whatever the coupling.

Consequently the C2–C4 *conflict of Proposition 5.6 occurs, on this pair of problems, only at $\varepsilon = 0$ and under every coupling, while the reversal of expectational comparisons between D and $K_\lambda(D)$ occurs whenever $g > 0$, independently of λ.*

Proof. (i) Since λ does not depend on the commitment, the causal and the ex-ante value of committing to q coincide and equal $V_D(q)$ by Proposition 5.2. In D the bonus c is gained in every state, so every expectational causal comparison favours B. (ii) O dominates only if the three rows with negative difference are null: $t = 0$ and $\varepsilon - t = 0$, so $\varepsilon = 0$; conversely, at $\varepsilon = 0$ only the row $(1, 0)$ remains. Directly: at positive error O pays 0 with positive probability whereas B always pays at least $c > 0$, so no coupling makes O even weakly dominant. B dominates exactly when the row $(1, 0)$ is null, $t = 1 - \varepsilon$, which the constraint $t \leq \min(\varepsilon, 1 - \varepsilon)$ allows only for $\varepsilon \geq \frac{1}{2}$. (iii) For $0 < \varepsilon < 1$ all four rows are structural states, and the differences have both signs. □

Changing a probability law cannot change C2 while its structural state set is held fixed; the coupling matters for C2 only through the convention $S_K = \operatorname{supp} \lambda$. The three couplings one might call natural give

coupling	t
independent (product)	$\varepsilon(1 - \varepsilon)$
shared predictor-correctness bit	0
comonotone realised payoffs	$\min(\varepsilon, 1 - \varepsilon)$

and none of them restores dominance of O at positive error. Couplings do matter in other problems: for a shared Bernoulli variable Z, the payoffs $U = 1 + 100Z$ and $W = 100Z$ satisfy $U > W$ everywhere on the support of the shared coupling, whereas the product coupling admits the reversing pair $U = 1$, $W = 100$. Which couplings produce dominance is settled in general by the following criterion, a finite form of Strassen's theorem on stochastic order (Strassen 1965).

Proposition 5.12 (Dominance under some coupling). *Let ν_U, ν_W be finitely supported laws on $\mathbb{R}$ with distribution functions F_U, F_W.*

(i) There is a coupling with $U \geq W$ almost surely if and only if $F_U(x) \leq F_W(x)$ for all x.

(ii) There is a coupling with $U > W$ almost surely if and only if $F_U(x) \leq F_W(x^-)$ for every $x \in \operatorname{supp} \nu_U$.

Proof. Necessity follows from the inclusions $\{U \leq x\} \subseteq \{W \leq x\}$ and $\{U \leq x\} \subseteq \{W < x\}$, which hold almost surely under the respective couplings. For sufficiency, let V be uniform on $(0, 1]$ and set $U = F_U^{-1}(V)$, $W = F_W^{-1}(V)$, with $F^{-1}(v) = \min\{y : F(y) \geq v\}$. In case (i), $F_U \leq F_W$ gives $F_W^{-1} \leq F_U^{-1}$ pointwise. In case (ii), let $x = F_U^{-1}(v) \in \operatorname{supp} \nu_U$; then $F_W(x^-) \geq F_U(x) \geq v$, and since ν_W is finitely supported, $F_W(y) \geq v$ for some $y < x$, so $F_W^{-1}(v) < x$. □

A payoff coupling lifts to a coupling of the underlying states by distributing mass within payoff fibres according to the conditional state laws. For a general problem, the residual content of C4 is then an agreement condition: some C2-permitted original policy must agree, on the information sets it positively reaches, with a commitment that no other commitment structurally dominates. Proposition 5.12 decides which commitments are dominated; it does not by itself decide whether C2 and C4 are consistent on a whole class of problems with imperfect prediction, which remains open (Section 21).

6. Robustness of the within-problem conflict

The central conflict between C1 and C2 was obtained with deterministic policies, a single always-reached choice, and a predictor targeting that choice. This section asks which of these features it needs.

6.1. Imperfect prediction

By Proposition 4.1, the conflict holds for every $0 \leq \varepsilon < \frac{1}{4}$ at the scaled payoffs, and for every $\varepsilon < \frac{1}{2}$ at suitable stakes, since $m(1 - 2\varepsilon) > c$ can be arranged whenever $\varepsilon < \frac{1}{2}$. At $\varepsilon = \frac{1}{2}$ the symmetric predictor carries no policy dependence and this mechanism disappears. Theorem 4.2 gives the general statement: what is needed is not accuracy but any difference between the state laws that two prescriptions induce.

6.2. Randomisation does not reconcile ex-ante and causal evaluation

Let the agent one-box with probability p. Suppose first that the predictor observes the distribution, filling the box with probability $\varepsilon + (1 - 2\varepsilon)p$, and that the action is drawn independently given p. Expanding,

$$V(p) = p \cdot 2\big(\varepsilon + (1 - 2\varepsilon)p\big) + (1 - p)\big(2\varepsilon + 2(1 - 2\varepsilon)p + 1\big) = 1 + 2\varepsilon + (1 - 4\varepsilon)p.$$

The same formula holds if the predictor instead observes the sampled action and predicts it with symmetric error ε. For $\varepsilon < \frac{1}{4}$, C1 requires $p = 1$ and C2 requires $p = 0$. The conflict also admits an exact quantitative form. Put $g = 1 - 4\varepsilon > 0$, let $R_1(p) = \max_q V(q) - V(p)$ be ex-ante regret, and let $R_{\text{causal}}(p)$ be the expected regret relative to the best action in the realised state.

PROPOSITION 6.1 (Exact trade-off). *In randomised Newcomb, $R_1(p) = g(1-p)$ and $R_{\text{causal}}(p) = p$, so that*

$$\frac{R_1(p)}{g} + R_{\text{causal}}(p) = 1.$$

Simultaneous tolerances $R_1 \le \delta_1$ and $R_{\text{causal}} \le \delta_2$ are achievable if and only if $\delta_1/g + \delta_2 \ge 1$. Under infallible prediction, every policy has regret at least $\frac{1}{2}$ by one of the two measures.

Proof. V is affine with slope g and maximal at $p = 1$, which gives R_1. In every realised state two-boxing is better by exactly one unit, so the causal regret is 1 when the agent one-boxes and 0 otherwise. The tolerances require $1 - \delta_1/g \le p \le \delta_2$, a nonempty interval exactly under the stated inequality. At $\varepsilon = 0$, $g = 1$ and $\max(R_1, R_{\text{causal}}) \ge \frac{1}{2}$. □

6.3. Existence of optimal and ratifiable mixed policies

Admitting mixed policies does not by itself make either C1 or C3 satisfiable; regularity of the predictor's dependence on the distribution is what matters.

PROPOSITION 6.2. *(i) Let a binary reward, independent of the realised action, succeed with probability $q(p) = p$ for $p < 1$ and $q(1) = 0$. Then $V(p) = q(p)$ has supremum 1 and no maximiser.*

(ii) Let Death choose city 1 if and only if $p \ge \frac{1}{2}$, where p is the probability of city 1, with payoff 1 for avoiding Death. No mixture is ratifiable, although Death observes only the distribution.

Proof. (i) is immediate. (ii) If $p < \frac{1}{2}$, Death is in city 0, and city 1 is the unique causal best response, so a ratifiable p would put all its mass on city 1; if $p \ge \frac{1}{2}$, city 0 is the unique best response. Either way p is not supported on best responses to itself. □

THEOREM 6.3 (Existence under continuity). *Let actions and states be finite, let the action be sampled independently given the mixed policy $p \in \Delta(A)$, and let $p \mapsto \mu_p \in \Delta(S)$ be continuous. Then a mixed policy satisfying C1 exists, and so does a mixed policy all of whose supported actions are causal best responses to μ_p.*

Proof. $V(p) = \sum_a p_a \sum_s \mu_p(s) u(s,a)$ is continuous on the compact simplex, so it attains its maximum. For ratifiability, let $Q(a;p) = \sum_s \mu_p(s) u(s,a)$ and $B(p) = \Delta\big(\arg\max_a Q(a;p)\big)$. The correspondence B has nonempty, compact, convex values, and its graph is closed because the finitely many functions $Q(a;\cdot)$ are continuous. By Kakutani's theorem (Kakutani 1941) there is $p \in B(p)$. □

Theorem 6.3 is the one-shot core of two published existence results with stronger structure: Halpern's theorem on ratifiable acts, which assumes costless statewise mixing and conditional state laws that are affine in the mixture (Halpern 2020), and the theorem of Bell et al. (2021) that every continuous Newcomblike decision process has a strongly ratifiable policy. Neither identifies ratifiable policies with ex-ante maximisers, and Proposition 6.1 shows that they cannot be identified.

Remark 6.4 (Predicting a realised sample). If Death independently samples the agent's distribution, the usual symmetric model admits $p = \frac{1}{2}$. If Death instead predicts the agent's realised private sample, the verdict depends on further semantics: whether the agent knows its sample, and whether a deviation preserves or replaces the randomness the predictor tracks. Conditioning on a revealed, perfectly predicted sample restores the deterministic instability. A marginal test of support-optimality can still accept $p = \frac{1}{2}$ while ignoring the correlation between the prediction and the sample, and should not be confused with a criterion that preserves that correlation when the unchanged policy is evaluated. Skyrms already cautioned against an unconditional randomisation rescue (Skyrms 1990a).

6.4. Restrictions on what predictors may target

Some responses to Newcomb-type problems restrict prediction to information sets that are in some sense actual. Newcomb's problem has one information set, reached with probability one under every policy, so its conflict survives every such restriction except the last:

admissible prediction targets	Newcomb conflict survives
information sets positively reached under the evaluated policy	yes
information sets reached under every admissible policy	yes
information sets on the realised path	yes
information sets already reached before the prediction occurs	no

The last restriction removes anticipatory dependence altogether: a "prediction" of an action already taken is an ordinary causal response, which belongs in the structural plan, so μ_π no longer depends on π. In that case ex-ante and causal evaluation agree.

Proposition 6.5. *If $\mu_\pi = \mu$ does not depend on π, then every* C1*-optimal policy satisfies* C3*, and therefore* C2*.*

Proof. By Lemma 2.6,

$$\begin{aligned} V_D(\pi[h{\leftarrow}a]) - V_D(\pi) &= \sum_{s \in R_h(\pi)} \mu(s)\big(U_D(s, \pi[h{\leftarrow}a]) - U_D(s,\pi)\big) \\ &= r_\pi(h)\big(Q_\pi(h,a) - Q_\pi(h,\pi(h))\big). \end{aligned}$$

If $r_\pi(h) > 0$ and $\pi(h)$ is not causally optimal at h, the deviation raises the ex-ante value. □

Removing predictions about unreached information sets eliminates particular witnesses, such as counterfactual mugging, and so changes the evidential part of the map in Theorem 5.10. It leaves the central Newcomb incompatibility intact.

Part II.
Rules across problems

7. The causal matrix and cross-problem axioms

Theorem 5.7 shows that within-problem criteria yield only singleton and pairwise obstructions. If a non-trivial impossibility in the style of Arrow exists, it must therefore come from requirements that relate a theory's choices on *different* problems: invariance, independence, contraction. Such requirements need a common description of problems, which the following reduced form provides.

7.1. Reduced form

Definition 7.1 (Causal matrix). The *causal matrix* of a problem D is the square matrix indexed by Π_D,

$$M_D[\pi, \sigma] = \sum_{s \in S_D} \mu_\pi(s)\, U_D(s, \sigma),$$

the value of running σ while the environment responds as it would to π.

Row π is the causal evaluation of every policy under the beliefs that π induces; column σ records how σ fares under every such belief state; and the diagonal is the ex-ante value, $M_D[\pi, \pi] = V_D(\pi)$. The matrix compresses the problem: it retains expectations under each induced law, and it forgets the structural states. Two consequences should be kept in view. The column-dominance axiom below compares expected values, whereas C2 compares structural outcomes, including states of probability zero; and "π maximises its own row" is a global form of ratifiability, which differs from the local, information-set form C3 in multi-stage problems. For problems with a single choice, the matrix and the structural criteria agree, and all the witnesses below are of that kind.

Theorem 7.2 (Realisability). *Every finite real square matrix is the causal matrix of a fair, well-defined problem with a single information set. Deleting a set of indices together with their rows and columns is realised by deleting the corresponding actions while keeping the remaining laws.*

Proof. Given $M \in \mathbb{R}^{I\times I}$, let the agent choose $i \in I$ at a single information set; let the states be $S = I$; let an infallible predictor forecast the chosen index, $\mu_i = \delta_i$; and let $U(s, j) = M_{sj}$. Then $M_D[i, j] = U(i, j) = M_{ij}$. The problem is fair and well defined, and deleting actions leaves the laws of the others unchanged. □

In this realisation C1 selects the maximal diagonal entries, and C3 accepts exactly the indices i whose diagonal entry is maximal in row i. Rules on matrices therefore have an unrestricted domain, as social welfare functions do in Arrow's theorem (Arrow 1951), and the framework contains every finite recommendation structure.

7.2. Rules and axioms

Fix an infinite universe $\mathbb{L}$ of labels. A *matrix* is a pair (I, M) with $I \subseteq \mathbb{L}$ finite and nonempty and $M \in \mathbb{R}^{I\times I}$; for $\emptyset \neq K \subseteq I$, $M|_K$ is the *principal restriction* to $K \times K$, which keeps labels. A bijection $\sigma : I \to I'$ acts by $(\sigma \cdot M)_{\sigma(i)\sigma(j)} = M_{ij}$.

Definition 7.3 (Rule). A *rule* assigns to every matrix a reflexive relation $\succeq_M$ on I, with strict part $\succ_M$, and recommends its maximal elements,

$$G(M) = \{i \in I \,:\, \text{there is no } j \in I \text{ with } j \succ_M i\},$$

which are required to be nonempty for every M.

The convention makes the comparison axioms below constrain the same relation that determines choice, and it prevents a rule that lacks an ordering from recommending nothing. A decision theory satisfies *reduction* (A0) if its choice on D depends only on the labelled matrix M_D, so that it selects from $G(M_D)$ for some rule. A *single-valued* deterministic selector cannot be neutral: on the zero 2×2 matrix, swapping the indices fixes the input but moves either possible answer. Choice correspondences avoid this separate obstruction.

Definition 7.4 (Cross-problem axioms). A rule may satisfy the following conditions, for all matrices M and all indices i, j.

N *Neutrality.* For every bijection $\sigma : I \to I'$: $i \succeq_M j \iff \sigma(i) \succeq_{\sigma\cdot M} \sigma(j)$.

F *Common affine invariance.* $\succeq_{aM+b\mathbf{1}\mathbf{1}^\top} = \succeq_M$ for $a > 0$, $b \in \mathbb{R}$.

R *Row translations.* $\succeq_{M+b\mathbf{1}^\top} = \succeq_M$ for every $b \in \mathbb{R}^I$ (adding b_k to every entry of row k).

P *Column Pareto.* If $M_{k\sigma} \geq M_{k\tau}$ for every $k \in I$, strictly for some k, then $\sigma \succ_M \tau$.

L *Restricted Pareto.* The same, with k ranging only over $\{\sigma, \tau\}$.

D *Diagonal optimality.* $G(M) \subseteq \arg\max_i M_{ii}$.

J *Global ratifiability.* Every $i \in G(M)$ satisfies $M_{ii} \geq M_{ij}$ for all j.

B *Binary independence.* $i \succeq_M j \iff i \succeq_{M|_{\{i,j\}}} j$.

S *Self-sovereignty.* If $M, M' \in \mathbb{R}^{I\times I}$ have the same rows i and j, then $i \succeq_M j \iff i \succeq_{M'} j$.

C *Contraction.* $G(M) \cap K \subseteq G(M|_K)$ for every nonempty $K \subseteq I$.

O *Ordering.* $\succeq_M$ is complete and transitive.

Q *Non-triviality.* For some I there are $M, M' \in \mathbb{R}^{I \times I}$ with $G(M) \neq G(M')$.

P is the expectational counterpart of dominance; D is C1; J is ratifiability of pure choices; S is the condition of Briggs (2010, §6), under which the comparison of two options depends only on how each of them evaluates all options; C is Chernoff's condition, Sen's property α (Chernoff 1954; Sen 1971); B is an independence of irrelevant alternatives for binary comparisons, of the kind that Podgorski (2022) imposes on overall-value theories, required here across menu sizes. Neutrality is stated for bijections between different label sets, because the proofs below transport choices from one menu to another; invariances are imposed on the relation, not only on its maximal elements. Ordering concerns the actual output relation at each input. Requiring only that each $G(M)$ be rationalisable by some weak order would be vacuous, since any nonempty set can be placed in a top indifference class; requiring one order to rationalise the choices from all restrictions of a given matrix is stronger and already entails contraction (Arrow 1959).

LEMMA 7.5. B *implies* S *and* C.

Proof. If rows i and j coincide in M and M', so does the principal block on $\{i, j\}$, and B transfers the comparison. For C, let $i \in G(M) \cap K$. Any $j \in K$ with $j \succ_{M|_K} i$ would satisfy $j \succ_{M|_{\{i,j\}}} i$ and hence $j \succ_M i$, contradicting the maximality of i in M. □

The following monotonicity condition is used in Section 11. Write E_{ii} for the matrix with a single entry 1 in position (i, i).

DEFINITION 7.6 (Diagonal monotonicity, Mon). If $M' = M + \delta E_{ii}$ with $\delta \geq 0$, then $i \in G(M)$ implies $i \in G(M')$, and every true comparison $i \succeq_M j$ or $i \succ_M j$ remains true in M'.

7.3. What row-translation invariance means

The axiom R will turn out to be decisive, and it is tempting to read it as the matrix form of causal evaluation. It deserves a precise interpretation.

THEOREM 7.7 (Factorisation). *A rule satisfies* R *if and only if* $\succeq_M$ *depends on* M *only through the row-normalised matrix* $Z(M)_{ij} = M_{ij} - M_{ii}$*, equivalently through the within-row differences of* M*.*

Proof. Row translations leave Z unchanged. Conversely, if $Z(M) = Z(M')$, then $M'_{ij} - M_{ij} = M'_{ii} - M_{ii}$ is constant along each row, so M' is a row translation of M. □

On a restricted domain, R is factorisation through the image of Z; the zero-diagonal representative need not itself be admissible. With this, familiar criteria can be classified once a matrix semantics is fixed for each.

criterion	matrix semantics	satisfies R?
fixed-prior causal decision theory	rank by $w^\top M$ for a fixed weight vector w	yes
pure global ratifiability	admit i iff $M_{ii} \geq M_{ij}$ for all j	yes, as an admissibility condition
mixed ratifiability	$p \in \mathrm{BR}(p^\top M)$	yes, as an equilibrium set
deliberational causal decision theory	requires initial beliefs, update dynamics, and a final selection	depends on these
evidential theory with reference-policy law	maximise M_{ii}, with $P(i,s) = \rho_i \mu_i(s)$	no
ex-ante policy maximisation	maximise $M_{ii} = V(i)$	no

Ratifiability's invariance does not establish that ratifiable options exist or supply a ranking. For deliberational theories, fixed-belief causal comparisons with difference-based update dynamics preserve row baselines, but arbitrary update dynamics need not; Joyce (2012) distinguishes equilibrium credences from the act subsequently chosen, and Arntzenius (2008) identifies the update rule as additional structure needed to define equilibrium. An "FDT-style" theory is ex-ante maximisation under stipulated functional counterfactuals and so fails R under that stipulation (Yudkowsky and Soares 2017).

PROPOSITION 7.8. R *does not characterise causal rationality: ranking by* $r_i = M_{ii} - \max_j M_{ij}$ *satisfies* N, F, R, O, *yet on*

$$M = \begin{pmatrix} 0 & 10 & -1 \\ 10 & 0 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

it chooses option 3, *which both other columns strictly dominate.*

Proof. The scores are $(-10, -10, -1)$. Column 3 is $(-1, -1, -1)^\top$, below both $(0, 10, 0)^\top$ and $(10, 0, 0)^\top$ in every row. □

So R means exactly invariance to row-specific baselines. It captures a feature shared by several causal procedures, but it supplies neither causal expected-utility maximisation nor dominance.

8. A PAIRWISE REDUCTION UNDER COMMON AFFINE INVARIANCE

The first cross-problem result shows that, with common affine invariance as the only invariance, the axioms of Definition 7.4 behave like the within-problem criteria: every inconsistency has a core of size one or two.

THEOREM 8.1 (Pairwise reduction). *On the domain of all finite real square matrices, under principal restriction, the inclusion-minimal inconsistent subsets of* $\{\mathsf{N}, \mathsf{F}, \mathsf{P}, \mathsf{L}, \mathsf{D}, \mathsf{J}, \mathsf{B}, \mathsf{S}, \mathsf{C}, \mathsf{O}, \mathsf{Q}\}$ *are exactly*

$$\{\mathsf{L}\}, \quad \{\mathsf{J}\}, \quad \{\mathsf{P}, \mathsf{D}\}, \quad \{\mathsf{P}, \mathsf{B}\}, \quad \{\mathsf{P}, \mathsf{S}\}, \quad \{\mathsf{P}, \mathsf{C}\},$$

and the maximal consistent subsets are exactly

maximal consistent subset	witness rule
$\{\mathsf{N}, \mathsf{F}, \mathsf{P}, \mathsf{O}, \mathsf{Q}\}$	rank by column sums $\sum_k M_{ki}$
$\{\mathsf{N}, \mathsf{F}, \mathsf{D}, \mathsf{B}, \mathsf{S}, \mathsf{C}, \mathsf{O}, \mathsf{Q}\}$	rank by diagonal entries M_{ii}

The proof consists of five obstructions and a containment argument.

PROPOSITION 8.2 (Restricted Pareto alone). *No rule satisfies* L.

Proof. Consider

$$K = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

In rows 1 and 2, column 1 reads $(0, 1)$ and column 2 reads $(-1, 0)$, so L forces $1 \succ 2$. Likewise rows $2, 3$ force $2 \succ 3$ and rows $1, 3$ force $3 \succ 1$. Every option has a strict defeater, so $G(K) = \varnothing$. □

If bare cyclic relations without a nonempty recommendation were allowed when O is absent, the obstruction would instead be the pair $\{\mathsf{L}, \mathsf{O}\}$; the convention of Definition 7.3 makes it a singleton.

PROPOSITION 8.3 (Global ratifiability alone). *No rule satisfies* J.

Proof. In $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ neither option maximises its own row. This is Death in Damascus in reduced form. □

PROPOSITION 8.4 (Pareto against the diagonal). *No rule satisfies* P *and* D.

Proof. In $N_0 = \left(\begin{smallmatrix} 2 & 3 \\ 0 & 1 \end{smallmatrix}\right)$, column 2 strictly dominates column 1, so P gives $2 \succ 1$ and $G(N_0) = \{2\}$, while option 1 uniquely maximises the diagonal. This is Newcomb's problem in reduced form. □

PROPOSITION 8.5 (Pareto against self-sovereignty; Briggs 2010). *No rule satisfies* P *and* S*, and no rule satisfies* P *and* B.

Proof. Let

$$M^{+} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 1 & 0 & -1 \end{pmatrix}, \qquad M^{-} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}.$$

In M^+, column 1 weakly dominates column 2, strictly in row 3, so P gives $1 \succ 2$; in M^- it gives $2 \succ 1$. Rows 1 and 2 are the same in both matrices, contradicting S; so is the principal block on $\{1, 2\}$, contradicting B. □

This is an explicit instance of Briggs's theorem, which states the incompatibility of a Pareto condition and self-sovereignty for decision-theoretic rankings and does not require neutrality or transitivity (Briggs 2010, pp. 21–26). Causal decision theory satisfies her Pareto condition and violates self-sovereignty; evidential decision theory and Wedgwood's benchmark theory (Wedgwood 2013) satisfy self-sovereignty and violate Pareto. The conflict needs at least three options: on two-option problems the benchmark theory satisfies both. The pair {P, S} is therefore not new.

Proposition 8.6 (Pareto against contraction). *No rule satisfies* P *and* C.

Proof. Return to the matrix K of Proposition 8.2. Its binary restrictions are

$$K|_{\{1,2\}} = K|_{\{2,3\}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad K|_{\{1,3\}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

and P forces $G(K|_{\{1,2\}}) = \{1\}$, $G(K|_{\{2,3\}}) = \{2\}$, $G(K|_{\{1,3\}}) = \{3\}$. Each option of K is rejected in one binary restriction containing it, so by C none can be chosen from K. □

The mechanism is the classical binary cycle that makes contraction consistency incompatible with majority-type choice (Brandt and Harrenstein 2011; Sen 1970); the modelling choice that gives it force here is that restriction deletes the rows of the removed options together with their columns. Section 10 shows that it disappears when the rows are retained.

Proof of Theorem 8.1. The six sets are inconsistent by Propositions 8.2–8.6, and each proper subset of each pair is consistent by the witnesses. Column sums satisfy N, F (scores are multiplied by a and shifted by $|I|b$), P (a weakly dominant column with one strict entry has a strictly larger sum), O, and Q. Diagonal ranking satisfies N, F, D, B (and hence S and C by Lemma 7.5), O, and Q. A consistent set contains neither L nor J. If it contains P, it excludes D, B, S, and C, and lies in the column-sum set; otherwise it lies in the diagonal set. □

Two of the reference rules one might expect to appear do not. Maximin over rows violates the strong Pareto axiom: in $\left(\begin{smallmatrix} 0 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, column 1 weakly dominates column 2, but both column minima are 0. Positive fixed-weight column averages satisfy P, and uniform weights add N, but fixed unequal weights require persistent labels or additional input. The selector T^{ν} of Proposition 5.5 is in general not a rule in the sense of A0: it uses structural information that M_D does not retain.

With common affine invariance alone, then, there is no Arrow-style family among these axioms.

9. Row-baseline invariance and contraction

Adding row-translation invariance changes the picture. It produces a genuine impossibility among four independent axioms, none of which concerns Pareto, ordering, or ex-ante value.

9.1. The circulant theorem

The proof embeds an arbitrary matrix in a matrix with a transitive symmetry group. It rests on a distinct-differences lemma of the kind used by Fink and Ruiz (1986) to embed graphs in circulants.

Lemma 9.1 (Distinct differences). *Let $k \geq 2$, $v_i = 2^i$ for $i = 0, \dots, k-1$, and $m > 2(2^{k-1} - 1)$. The $k(k-1)$ ordered differences $v_j - v_i$, $i \neq j$, are distinct and nonzero modulo m.*

Proof. For $j > i$, $v_j - v_i = 2^i(2^{j-i} - 1)$; the power of two and the odd part determine i and $j - i$, so the positive differences are distinct integers, and so are their negatives. Every difference has absolute value at most $D = 2^{k-1} - 1$, so two distinct differences differ by a nonzero integer of absolute value at most $2D < m$, and no difference is divisible by m. □

Lemma 9.2 (Circulant embedding). *Every matrix $Z \in \mathbb{R}^{k \times k}$ with zero diagonal is a principal submatrix of a circulant matrix $A_{uv} = f(v - u \bmod m)$ on $\mathbb{Z}/m\mathbb{Z}$ with $f(0) = 0$.*

Proof. With v_i and m as in Lemma 9.1, set $f(0) = 0$ and $f(v_j - v_i) = Z_{ij}$ for $i \neq j$, which is well defined by the lemma, and give f arbitrary values on the unused residues. Then $A_{v_i v_j} = f(v_j - v_i) = Z_{ij}$. □

Theorem 9.3 (Neutrality, row translations, and contraction force universal choice). *On the domain of all finite real square matrices, every rule satisfying* N, R, *and* C *(principal restriction) has $G(M) = I$ for every matrix (I, M). Consequently* {N, R, C, Q} *is inconsistent.*

Proof. Let (I, M) be given with $|I| = k \geq 2$ (the case $k = 1$ is trivial), and let $Z = Z(M)$, which has zero diagonal; by R, $G(Z) = G(M)$. By Lemma 9.2 and N (relabelling the vertices of the circulant so that the embedded ones carry the labels of I), Z is the principal restriction to I of a matrix A whose symmetry group acts transitively on its indices: each cyclic shift $u \mapsto u + 1$ of $\mathbb{Z}/m\mathbb{Z}$, transported by the relabelling, maps A to itself. By N, $G(A)$ is invariant under these symmetries, and since it is nonempty it contains every index. By C, $G(Z) = G(A|_I) \supseteq G(A) \cap I = I$. Hence $G(M) = I$, and no two matrices on the same index set have different choice sets. □

Without O the conclusion is universal *choice*, not universal indifference: incomparabilities may remain. The symmetry step is standard in the theory of tournament

solutions, where neutral, nonempty solutions must select every vertex of a vertex-transitive tournament (Brandt 2011; Brandt et al. 2016); the embedding has the precedent noted above. What the theorem adds is the use of contraction to carry universal choice from symmetric matrices to all matrices, and of row-baseline invariance to reach every matrix from a zero-diagonal one. No published statement of this conclusion was found in the literature searched; its ingredients are established, and no priority is claimed.

Proposition 9.4 (Independence). *Each three-element subset of* {N, R, C, Q} *is satisfiable, so the family is minimally inconsistent:*

omitted axiom	rule satisfying the other three
N	responsive fixed-label rule
R	diagonal ranking
C	row-range ranking, $r_i = \max_j M_{ij} - \min_j M_{ij}$
Q	universal indifference

All four rules also satisfy F *and* O.

Proof. Row ranges are unchanged by row translations, multiplied by a under F, and permuted under relabelling; the rule is nonconstant. Diagonal ranking satisfies C by Lemma 7.5. The *responsive fixed-label rule* fixes two labels u, v and a well-order of $\mathbb{L}$ in which u, v form an adjacent block above all other labels. When both occur, u and v are compared by the sign of $M_{uu} - M_{uv}$ (a tie if it is zero); all other comparisons follow the fixed order. The comparison of u and v uses a within-row difference of their principal block, so the rule satisfies R, F, and B, hence C; it is responsive, so Q holds, but it is not neutral. □

If neutrality is regarded as part of the reduction A0 rather than as an axiom, the same result is a minimal three-axiom family {R, C, Q} relative to that background.

Proposition 9.5. {N, R, B, Q} *is minimally inconsistent, and so is* {R, D}.

Proof. Suppose $i \succ_M j$ in some matrix. By B the same holds in $M|_{\{i,j\}}$, and by R in its row normalisation $H = \left(\begin{smallmatrix} 0 & x \\ y & 0 \end{smallmatrix}\right)$, with i first. Let

$$C = \begin{pmatrix} 0 & x & y \\ y & 0 & x \\ x & y & 0 \end{pmatrix}.$$

Its principal blocks on $(1,2)$, $(2,3)$, and $(3,1)$ all equal H, so N and B give $1 \succ 2$, $2 \succ 3$, and $3 \succ 1$, and $G(C) = \emptyset$. Hence no strict comparison ever occurs and $G(M) = I$ always, contradicting Q. Minimality follows from the witnesses of Proposition 9.4, with row-range ranking now serving as the rule that omits B. For {R, D}: adding 2 to the second row of $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ changes the unique diagonal maximiser from option 1 to option 2, while R keeps the choice fixed. □

Unlike Theorem 9.3, this argument uses the link between comparisons and nonempty maximal choice fixed in Definition 7.3.

THEOREM 9.6 (Map with row translations). *Adding* R *to the family of Theorem 8.1 introduces exactly three new minimal inconsistent subsets,*

$$\{\mathsf{R},\mathsf{D}\}, \qquad \{\mathsf{N},\mathsf{R},\mathsf{B},\mathsf{Q}\}, \qquad \{\mathsf{N},\mathsf{R},\mathsf{C},\mathsf{Q}\},$$

and the maximal consistent subsets become

witness rule	maximal consistent subset
column sums	{N, F, P, R, O, Q}
diagonal ranking	{N, F, D, B, S, C, O, Q}
row ranges	{N, F, R, S, O, Q}
responsive fixed-label rule	{F, R, B, S, C, O, Q}
universal indifference	{N, F, R, B, S, C, O}

Proof. Column sums satisfy R because a row translation shifts every column sum by $\sum_k b_k$; row ranges satisfy S because each score depends only on its own row. Let a set avoid all minimal inconsistent subsets. If it contains P, its pairwise conflicts confine it to the first row. If it omits P and R, it fits the second. If it contains R, it omits D; if it contains both N and Q, the two four-element obstructions exclude B and C, leaving the third row; without N it fits the fourth, and without Q the fifth. □

9.2. An exact domain condition

Theorem 9.3 needs matrices of unbounded size: with a fixed maximum size n, the rule that chooses everything below size n and maximises row ranges at size n satisfies N, R, C, and Q. The exact requirement on the domain can be stated without quantifying over rules. Let $\mathcal{D}$ be a set of matrices closed under relabelling, and interpret each invariance as applying whenever both the matrix and its transform belong to $\mathcal{D}$. Form a directed graph on the *pointed matrices* $\mathcal{P}_{\mathcal{D}} = \{(M,i) : M \in \mathcal{D},\ i \in I_M\}$, with

(i) edges in both directions between (M,i) and $(\sigma \cdot M, \sigma(i))$ for admissible relabellings;

(ii) edges in both directions between (M,i) and $(M + b\mathbf{1}^\top, i)$ for admissible row translations;

(iii) extension edges $(M|_K, i) \to (M,i)$ for $i \in K$, whenever $M|_K$ and M are in $\mathcal{D}$.

Let $\mathrm{Reach}(x)$ be the set of pointed matrices reachable from x (including x), and let $\Phi_M = \{(M,i) : i \in I_M\}$ be the fibre of M.

THEOREM 9.7 (Exact domain criterion). *On $\mathcal{D}$, the axioms* N, R, C, Q *are jointly inconsistent if and only if*

$$\textit{for every } x \in \mathcal{P}_{\mathcal{D}} \textit{ there is } M \in \mathcal{D} \textit{ with } \Phi_M \subseteq \mathrm{Reach}(x). \tag{DR}$$

Proof. Say (M,i) is *rejected* if $i \notin G(M)$. Rejection propagates along relabelling and translation edges by N and R, and along extension edges by the contrapositive

of C. Under (DR), a single rejection at x would reject the entire fibre of some M, contradicting $G(M) \neq \emptyset$; so nothing is ever rejected, and Q fails. Conversely, if (DR) fails at $x = (M_0, i_0)$, set

$$G_x(M) = \{i : (M, i) \notin \mathrm{Reach}(x)\}.$$

Each $G_x(M)$ is nonempty because no fibre lies in $\mathrm{Reach}(x)$. Since relabelling and translation edges are bidirectional, $\mathrm{Reach}(x)$ is closed under them, which gives N and R. If $i \in G_x(M) \cap K$ but $i \notin G_x(M|_K)$, the extension edge would put (M, i) in $\mathrm{Reach}(x)$; so C holds. Finally $i_0 \notin G_x(M_0)$, so $G_x(M_0)$ is a nonempty proper subset of I_{M_0}, and some relabelling of M_0 on the same index set moves it, which gives Q. Tying chosen options above rejected ones yields O as well. □

The circulant condition is sufficient for (DR) but not necessary. If F is also imposed, admissible affine transformations are added as bidirectional edges.

9.3. Physical restrictions on problems

Does the richness that (DR) demands survive natural restrictions on the problems themselves?

Proposition 9.8 (Bounded payoffs). *On the domain of $[0,1]$-valued matrices, $\{\mathsf{N}, \mathsf{R}, \mathsf{C}, \mathsf{Q}\}$ is consistent; adding F makes it inconsistent again.*

Proof. Fix $r \in [\frac{1}{2}, 1)$ and put $i \to j$ if $M_{ii} - M_{ij} > r$ and $M_{ji} - M_{jj} > r$. An edge forces $M_{ii} > r \geq \frac{1}{2} \geq 1 - r > M_{jj}$, so no vertex has both an incoming and an outgoing edge and the graph is acyclic. Choose the options with no incoming edge, tied above the rejected ones, which are tied among themselves. The conditions use only within-row differences, giving R; edges depend only on principal 2×2 blocks and persist under restriction, giving C; the rule is neutral, ordered, and responsive (it chooses $\{1\}$ from $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ and both options from the zero matrix); and raising a diagonal entry only removes incoming and adds outgoing edges, so it satisfies Mon. It fails F, since scaling by a small $a > 0$ removes every edge.

With F, given $M \in [0,1]^{n \times n}$ and small $\alpha > 0$, pass from M to αM by F and then to $A_{ij} = \frac{1}{2} + \alpha(M_{ij} - M_{ii})$ by R; every intermediate matrix stays in $[0,1]$. Embed A in a circulant as in Lemma 9.2, filling unused residues with $\frac{1}{2}$, and run the proof of Theorem 9.3. □

Proposition 9.9 (Prediction bounded away from accuracy). *Let $P_n = (1-\eta)I + \frac{\eta}{n}\mathbf{1}\mathbf{1}^\top$ with $0 < \eta < 1$, a prediction channel whose probability of a correct forecast is $1-\eta+\eta/n \leq 1-\eta/2$ for $n \geq 2$. With unrestricted real payoffs, every $M \in \mathbb{R}^{n \times n}$ is the causal matrix of a problem whose predictor uses P_n. With payoffs in $[0,1]$ and F, the circulant argument survives, uniformly in n.*

Proof. P_n is invertible, with $P_n^{-1} = \frac{1}{1-\eta} I - \frac{\eta}{(1-\eta)n}\mathbf{1}\mathbf{1}^\top$. Let the state be the forecast label, drawn from row i of P_n when i is chosen, and let $U = P_n^{-1} M$ be the state-by-action payoff table; then the causal matrix is $P_n U = M$. Since $P_n^{-1}\mathbf{1}\mathbf{1}^\top = \mathbf{1}\mathbf{1}^\top$, the

constant matrix $\frac{1}{2}\mathbf{1}\mathbf{1}^\top$ is mapped to itself, and the absolute row sums of P_n^{-1} are at most $(1+\eta)/(1-\eta)$; so a neighbourhood of the constant matrix whose radius does not depend on n is mapped into $[0,1]$, and the compressed matrices of Proposition 9.8 remain realisable. □

An accuracy bound alone, then, does not destroy richness. Completely uninformative prediction is different: if all induced laws coincide, the channel loses rank, every row of the causal matrix is the same, and the problem is an ordinary causal one.

Proposition 9.10 (Bounded action sets). *For every $k \geq 2$, the symmetric extensions required by Theorem 9.3 are realisable by problems with perfect recall and at most k actions per information set, provided contraction applies to arbitrary sets of policies.*

Proof. Take a complete binary decision tree of depth m, with the agent observing its own past actions and the prediction hidden until the end. It has $2^m - 1$ information sets and $N = 2^m$ action paths, and each path is produced by exactly $L = 2^{2^m-1-m}$ policies, since prescriptions off the path are free. Let a predictor forecast the eventual path and let the payoff depend on the forecast and the actual path; any path matrix C is then realisable, and the policy matrix is $C \otimes J_L$, with J_L the all-ones $L \times L$ matrix. If C is circulant, rotating path blocks and permuting duplicate policies within blocks act transitively on the policies. Any target matrix embeds in a circulant whose size is a power of two (Lemma 9.2), and is recovered by restricting to one representative policy from each designated block. □

The proviso matters: arbitrary sets of surviving policies are generally not obtained by pruning local action sets, which produces product-shaped sets. Restricting contraction to literal pruning is a different axiom (Section 10).

Proposition 9.11 (A single bounded choice). *With one information set and an unbounded action set, every square matrix is realisable. With one information set and at most $k \geq 2$ actions, the rule that is universally indifferent below size k and ranks by $M_{ii} - \frac{1}{k}\sum_j M_{ij}$ at size k satisfies* N, F, R, C, O, Q, *and* Mon.

Proof. The first claim is Theorem 7.2. For the second, every proper restriction of a size-k matrix lands where every option is chosen, so C holds; the score is a within-row difference, invariant under F up to scale, neutral, responsive, and increasing in M_{ii}. □

No finite size ceiling, therefore, forces triviality under the existential form of Q. If non-triviality is required specifically on binary menus, the three-option circulant $\left(\begin{smallmatrix} 0 & x & y \\ y & 0 & x \\ x & y & 0 \end{smallmatrix}\right)$ already blocks it as soon as three options are available.

10. The semantics of a smaller menu

Theorems 8.1 and 9.3 suggest a dichotomy: menu-coherent evaluation forces ex-ante evaluation, while causal evaluation cannot be menu-coherent without becoming trivial. Both theorems, however, rest on one way of removing options from a problem: principal restriction, which deletes the removed options' rows together with their columns. This section examines that step. It turns out to be where the dichotomy fails.

10.1. Three operations

A smaller menu can be modelled in at least three ways.

Principal restriction (C_{pr}). Delete the removed options' rows and columns. A smaller menu then also removes the beliefs that the agent would have if it chose the removed options.

Column restriction (C_{col}). Keep every row and delete only the columns. The input becomes a rectangular matrix of belief states by available options.

Fixed-reference restriction (C_{ref}). Evaluate options by averaging rows with a reference distribution over belief states, as in fixed-prior causal decision theory, and keep that distribution when options are removed.

Proposition 10.1 (Law-preserving removal is principal). *Let D' be obtained from D by making only the policies in $K \subseteq \Pi_D$ available, and suppose that their response laws and consequences are unchanged: $\mu'_i = \mu_i$ and $U'(s,i) = U(s,i)$ for $i \in K$. Then $M_{D'} = M_D|_{K\times K}$.*

Proof. Immediate from Definition 7.1: each entry of $M_{D'}$ involves only laws and consequences of policies in K. □

Principal restriction is thus forced by the definition of the causal matrix once law-preserving removal is stipulated. Two qualifications apply. First, arbitrary subsets K do not arise by pruning local action sets, which produces product-shaped sets of policies; an arbitrary K is realised by treating the surviving policies as macro-actions of a new one-choice predictive problem, which preserves the response laws but not the extensive form. Second, an intervention that actually announces a smaller menu may change how a predictor responds; the model does not require $\mu_i^{D'} = \mu_i^D$, and law preservation is an additional cross-problem assumption. Removal by macro-actions must also be distinguished from commitment before prediction (Definition 5.1), which changes the counterfactual structure and need not yield a principal submatrix. Spencer and Wells (2019) make the related point that converting a simultaneous problem into a sequential one can create a different decision problem, and Gallow (2026) explicitly defines a binary sub-decision by removing the other options *and conditioning credences on the surviving disjunction*, which is yet another operation.

10.2. Rectangular and fixed-reference domains

Definition 10.2 (Rectangular inputs). A *rectangular input* is a triple (J, A, M) with $J \subseteq \mathbb{L}$ finite, $\varnothing \neq A \subseteq J$, and $M \in \mathbb{R}^{J\times A}$. Here J indexes the retained belief perspectives and A the available options, and option $a \in A$ has its *own row a.* Column restriction is $(J, A, M)|_K = (J, K, M_{J\times K})$ for $\varnothing \neq K \subseteq A$.

The own-row association is what keeps diagonal optimality, self-sovereignty, and monotonicity meaningful. The off-menu rows are retained descriptions of the old conditional or counterfactual laws; they are additional data, not rows of the canonical causal matrix of the smaller problem, which remains square. Column restriction can therefore describe a fair augmented problem, but it does not follow from reduction to $M_{D'}$ alone.

The axioms extend to rectangular inputs as follows: relabellings act on J (and hence on A); R adds b_j to row j for every $j \in J$; P quantifies over all rows in J; D maximises M_{aa} over $a \in A$; S compares inputs with the same J and A; Q asks for two inputs with the same J and A and different choice sets; and contraction becomes

$$\mathsf{C}_{\mathrm{col}} : \qquad G(J, A, M) \cap K \subseteq G(J, K, M_{J\times K}).$$

Binary independence splits into two inequivalent axioms:

B_{pr} the comparison of a, b agrees with their comparison in the principal square input $(\{a, b\}, \{a, b\}, M_{\{a,b\}\times\{a,b\}})$;

$\mathsf{B}_{\mathrm{col}}$ the comparison of a, b agrees with their comparison in $(J, \{a, b\}, M_{J\times\{a,b\}})$, which keeps every row.

Both imply $\mathsf{C}_{\mathrm{col}}$, by the argument of Lemma 7.5; B_{pr} implies S, while $\mathsf{B}_{\mathrm{col}}$ does not, because it never compares inputs with different belief universes.

For fixed-reference restriction, supply a probability vector $w \in \Delta(J)$, evaluate $v_a(M, w) = \sum_j w_j M_{ja}$, and restrict by $(J, A, M, w)|_K = (J, K, M_{J\times K}, w)$; under relabelling, w is relabelled too, and uniform weights give a rule that depends on the rectangular matrix alone. A reference distribution over belief perspectives need not be the agent's probability over its *available* choices: if it is meant to be, eliminating choices ordinarily requires conditioning, and keeping positive weight on eliminated choices requires a different interpretation. The next result shows that the reference cannot be updated while preserving contraction.

Theorem 10.3 (Weight preservation). *Let w, w' be probability vectors on the same set J, let at least two columns survive a restriction, and suppose the update $w \mapsto w'$ does not depend on the payoff entries. Weighted averaging ranks every payoff matrix identically under w and w' if and only if $w' = w$, and contraction holds for every payoff matrix under the update only if $w' = w$.*

Proof. The difference of two columns can be any $x \in \mathbb{R}^J$, so identical rankings require $\operatorname{sign}(w^\top x) = \operatorname{sign}(w'^\top x)$ for all x; the two functionals are then positive multiples of each other, and normalisation gives $w' = w$. If $w' \neq w$, choose x with $w^\top x = 0 < w'^\top x$ and replace it by $x - \epsilon\mathbf{1}$ for small $\epsilon > 0$, so that $w^\top x < 0 < w'^\top x$.

Realise x as the difference of two surviving columns a, b, so that b is preferred under w and a under w', and set every removed column low enough that b is a full-menu maximiser under w. After restriction b is no longer chosen, and contraction fails. □

Deleting rows and renormalising their weights therefore preserves all rankings only when the deleted rows had total weight zero; with full-support weights, any proper row deletion changes the reference.

operation	available columns	belief rows and reference	canonical law-preserving matrix?
principal	delete unavailable options	delete the corresponding rows	yes, given law preservation
column	delete unavailable options	retain every row	needs augmented reference data
fixed reference	delete unavailable options	retain J and w	needs a supplied reference
delete and renormalise	delete unavailable options	change the reference	a different belief update

10.3. Consistency maps under column restriction

THEOREM 10.4 (Column map with principal binary independence). *On all finite rectangular inputs, the inclusion-minimal inconsistent subsets of*

$$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{P}, \mathsf{D}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$$

are exactly

$$\{\mathsf{P}, \mathsf{D}\}, \quad \{\mathsf{P}, \mathsf{B}_{\mathrm{pr}}\}, \quad \{\mathsf{P}, \mathsf{S}\}, \quad \{\mathsf{R}, \mathsf{D}\}, \quad \{\mathsf{N}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{Q}\},$$

and the maximal consistent subsets are

maximal consistent subset	witness
$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{P}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	column sums $\sum_{j \in J} M_{ja}$
$\{\mathsf{N}, \mathsf{F}, \mathsf{D}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	diagonal ranking
$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	full-menu relative scores
$\{\mathsf{F}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}, \mathsf{Q}\}$	responsive fixed-label rule
$\{\mathsf{N}, \mathsf{F}, \mathsf{R}, \mathsf{B}_{\mathrm{pr}}, \mathsf{S}, \mathsf{C}_{\mathrm{col}}, \mathsf{O}\}$	universal indifference

All the witnesses also satisfy Mon.

Proof. The *full-menu relative-score rule* ranks by $s_a = M_{aa} - \frac{1}{|A|} \sum_{c \in A} M_{ac}$ when $A = J$ and ties every option when $A \subsetneq J$. It is neutral and invariant under F and R; on full menus each score depends only on its own row, and inputs compared by S have the same J and A, so S holds; every proper column restriction produces a non-full menu, on which everything is chosen, so $\mathsf{C}_{\mathrm{col}}$ holds; and s_a increases with M_{aa}. Column sums are unchanged on surviving columns by column restriction, which gives $\mathsf{C}_{\mathrm{col}}$, and shift uniformly under R. The other witnesses are as in Theorem 9.6.

The obstructions {P, D}, {P, S}, and {R, D} are proved as in Propositions 8.4, 8.5, and 9.5, with square inputs; {P, B_{pr}} follows from $B_{pr} \Rightarrow S$; and {N, R, B_{pr}, Q} is proved by the three-cycle of Proposition 9.5, applied to square inputs. For completeness: a set containing P excludes D, B_{pr}, and S, and the column-sum witness covers it. Without P and without R, diagonal ranking covers it. With R, D is excluded and at least one of N, B_{pr}, Q is omitted, and the last three witnesses cover these cases. □

The obstruction {P, C} of Theorem 8.1 has disappeared, and so has {N, R, C, Q}. The matrix

$$W = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

makes the difference visible. Its column sums tie. Under principal restriction, P successively forces 1 over 2, 2 over 3, and 3 over 1 in the binary sub-menus. Under column restriction, the third row remains in every binary sub-menu and blocks those Pareto judgements, and column averaging continues to tie.

THEOREM 10.5 (Column map with column binary independence). *Replacing* B_{pr} *by* B_{col}*, the minimal inconsistent subsets are exactly*

$$\{P, D\}, \qquad \{P, S\}, \qquad \{R, D\},$$

and the maximal consistent subsets are {N, F, R, P, B_{col}, C_{col}, O, Q} *(column sums),* {N, F, D, B_{col}, S, C_{col}, O, Q} *(diagonal ranking), and* {N, F, R, B_{col}, S, C_{col}, O, Q} *(the binary-universe rule below).*

Proof. The *binary-universe rule* ranks by $s_a = M_{aa} - M_{ab}$ when $J = A = \{a, b\}$ and ties everything when $|J| \geq 3$; singleton menus are trivial. It satisfies Q on a two-row universe, and B_{col} never connects different belief universes, so it does not force the two-row behaviour to extend to three rows. Column sums satisfy B_{col} because deleting other columns leaves the two sums unchanged. The containment argument is as before: P excludes D and S; without P and R, diagonal ranking applies; with R and without P, the binary-universe rule applies. □

The weakness here lies in the existential form of Q. Fix a belief universe J with $|J| \geq 3$, allow all its menus, and let Q_J require non-triviality on that universe.

THEOREM 10.6 (A five-axiom family). {N, R, B_{col}, S, Q_J} *is a minimal inconsistent family.*

Proof. If every binary comparison on J were a tie, B_{col} would make every comparison on J a tie, and every menu would be chosen entire, contradicting Q_J; so some strict binary comparison occurs on J. On a binary menu $\{a, b\}$, S makes the comparison depend only on the principal block of a and b, and R lets its diagonal be normalised to zero. Neutrality, applied to the relabellings of J that permute three of its elements cyclically, replicates the strict comparison around a three-cycle of binary menus, and B_{col} transfers the three comparisons to the menu of those three options, all of which

are then strictly beaten. The witnesses omitting N, R, B_{col}, S, Q_J are, respectively, the responsive fixed-label rule, diagonal ranking, full-menu relative scores, column sums, and universal indifference; they can additionally satisfy F, C_{col}, O, and Mon. □

The strongest replacement thus depends on whether non-triviality is required on a universe large enough to support the obstruction.

10.4. What survives of the circulant theorem

PROPOSITION 10.7. *Uniform column averaging satisfies* N, F, R, P, B_{col}, C_{col}, O, Q, *and* Mon. *The same holds for weighted averaging with any fixed full-support reference* w *under* C_{ref} *(with* N *understood to relabel* w*).*

Proof. Each option's score is unaffected by the other columns and by their removal; row translations shift all scores equally; positive weights give P; raising M_{aa} raises only the score of a. □

Under column or fixed-reference restriction, then, no impossibility can be obtained from any subset of these axioms beyond those listed in Theorems 10.4 and 10.5. In particular, row-baseline-invariant evaluation *can* be menu coherent when the belief model is held fixed, and the first half of the proposed dichotomy fails. The impossibility of Theorem 9.3 concerns the preservation of comparisons when the belief perspectives associated with removed options disappear, not the preservation of choices when options disappear.

11. WHAT CHARACTERISES EX-ANTE EVALUATION

The second half of the dichotomy claimed that menu coherence forces ex-ante evaluation. This section shows that principal binary independence does force it, given symmetry, ordering, and monotonicity, and that contraction does not.

11.1. Binary independence: the Self-Esteem theorem

THEOREM 11.1 (Relational Self-Esteem theorem). *On the domain of all finite real square matrices, a rule satisfies* N, B_{pr}, *and* O *if and only if there is a complete transitive relation* $\succeq^*$ *on* $\mathbb{R}$ *such that, for every* M *and all* i, j,

$$i \succeq_M j \iff M_{ii} \succeq^* M_{jj}.$$

Proof. Consider

$$H = \begin{pmatrix} p & r & p \\ q & s & u \\ p & t & p \end{pmatrix}.$$

The principal block on $\{1, 3\}$ is constant, so the transposition of 1 and 3 fixes it; by N and B_{pr} the comparison of 1 and 3 is symmetric, and by completeness $1 \sim 3$. By transitivity, 1 and 3 compare in the same way with 2. By B_{pr} the first comparison is

that of the first index with the second in $\left(\begin{smallmatrix} p & r \\ q & s \end{smallmatrix}\right)$, and the second is that of the first index with the second in the principal block on $(3, 2)$, which is $\left(\begin{smallmatrix} p & t \\ u & s \end{smallmatrix}\right)$. Since r, q, t, u are arbitrary, the comparison of two options does not depend on the off-diagonal entries of their block, and by B_{pr} and N it is a function of the ordered pair of diagonal values. Define $x \succeq^* y$ if the first index is weakly preferred in $\mathrm{diag}(x, y)$. Completeness is inherited, and transitivity follows by applying B_{pr} to $\mathrm{diag}(x, y, z)$. Conversely, ranking by a weak order of the diagonal values satisfies N, B_{pr}, and O, and has nonempty maximal sets on finite menus. □

This is the relational form of the "Self-Esteem Theorem" of Podgorski (2022): if the comparison of two options is a fixed function of the four values $V_X(X)$, $V_Y(Y)$, $V_X(Y)$, $V_Y(X)$ and overall verdicts respect independence of irrelevant alternatives, the comparison depends only on each option's value by its own lights, $V_X(X)$ and $V_Y(Y)$. Podgorski concludes that among overall-value views only the evidential one respects that independence, and his tournament decision theory escapes the theorem by producing no single overall value. Here the common comparison function is supplied by N and B_{pr}, and the scalar overall value by O; the relational version assumes no real-valued representation of $\succeq^*$.

Corollary 11.2. *Adding* F *leaves exactly three rules: increasing diagonal ranking, decreasing diagonal ranking, and universal indifference. Adding* Q *and* Mon *leaves only increasing diagonal ranking.*

Proof. Every pair $x > y$ is mapped to every other such pair by a positive affine transformation, so under F all unequal pairs are oriented alike; equal values tie by reflexivity and N. Q excludes indifference. Under decreasing ranking, raising M_{11} in the zero 2×2 matrix removes option 1 from the choice set, violating the choice clause of Mon. □

Since $M_{ii} = V_D(i)$,

$$\mathsf{N} + \mathsf{F} + \mathsf{B}_{\mathrm{pr}} + \mathsf{O} + \mathsf{Q} + \mathsf{Mon}$$

characterises ex-ante policy ranking under the stated matrix semantics. Under the structural axioms N, F, B_{pr}, O, a single binary benchmark with unequal diagonal entries identifies which of the three rules applies; no finite benchmark, however, certifies that the structural axioms themselves hold everywhere.

Proposition 11.3 (Exact domain threshold). *For domains consisting of all matrices up to a maximum size, the characterisation holds if and only if matrices of size three are included.*

Proof. With every matrix of sizes one to three, the proof of Theorem 11.1 goes through, and B_{pr} propagates the conclusion to any larger sizes present. With sizes at most two, B_{pr} is vacuous, and ranking by $s_i = M_{ii} - M_{ij}$ satisfies N, F, O, Q, and Mon; on $\left(\begin{smallmatrix} 1 & 100 \\ 0 & 0 \end{smallmatrix}\right)$ it chooses option 2. □

Merely having some three-option problems is not enough; a sufficient richness condition is that the domain contain all binary matrices, all the clone extensions H above, and their relabellings (and, for the affine conclusion, the relevant affine transforms). The argument also works on $[0,1]$-valued matrices of sizes up to three, with F interpreted internally.

11.2. Contraction does not force the diagonal

Can contraction replace binary independence? Under every menu semantics, the answer is no.

THEOREM 11.4 (A contraction-consistent family without diagonal optimality). *Write $d_i = M_{ii}$. For $\lambda \geq 0$ let $i \to_\lambda j$ if $d_i - d_j > \lambda |M_{ij} - M_{ji}|$; choose the options with no incoming edge, tied at the top, and rank the rejected options below them by their diagonal entries. Every such rule satisfies* N, F, C_{pr}, C_{col}, O, Q, *and* Mon. *At $\lambda = 0$ it is increasing diagonal ranking; for every $\lambda > 0$ it sometimes recommends an option whose diagonal entry is not maximal.*

Proof. Every edge strictly lowers the diagonal, so the edge relation is acyclic and the choice set is nonempty. Whether $i \to_\lambda j$ depends only on $M_{ii}, M_{jj}, M_{ij}, M_{ji}$, which survive both principal and column restriction; so a chosen option acquires no incoming edge under restriction, which gives both forms of contraction. Raising d_i can only remove edges into i and add edges out of i; hence i's chosen status can only improve, other options' chosen status can only worsen, and if i stays rejected its rank among the rejected improves, which gives Mon. N and F are clear, since $|M_{ij} - M_{ji}|$ scales with a and is translation invariant. For $\lambda > 0$, in $\left(\begin{smallmatrix} 1 & 0 \\ 100/\lambda & 0 \end{smallmatrix}\right)$ there is no edge, and both options are chosen. □

Even an ordered, neutral, affine-invariant, monotone, contraction-consistent rule, therefore, need not maximise ex-ante value. The family also defeats finite certification: given finitely many benchmark matrices, choose $\lambda > 0$ so small that every option with non-maximal diagonal in every benchmark receives an edge from a diagonal maximiser. The whole ranking then coincides with diagonal ranking on every benchmark, while a new binary matrix with a large off-diagonal asymmetry relative to its diagonal gap violates D.

PROPOSITION 11.5 (Maximal cardinality). *Let $\mathfrak{c} = |\mathbb{R}|$. There are $2^{\mathfrak{c}}$ rules satisfying* N, F, C_{pr}, C_{col}, O, Q, *and* Mon.

Proof. For $E \subseteq \mathbb{R}$, put $i \to j$ if $d_i > d_j$, $s = M_{ji} - M_{ij} > 0$, and either $d_i - d_j > s$, or $d_i - d_j = s$ and $(d_j - M_{ij})/s \in E$; complete as in Theorem 11.4. Edges descend the diagonal and depend only on principal blocks, so the choice set is nonempty and both contractions hold. Raising d_i only decreases the gaps $d_k - d_i$, so it never creates an edge into i (at the boundary it may delete one), and only increases the gaps $d_i - d_j$, so it never deletes an edge out of i; this gives Mon. The boundary parameter $(d_j - M_{ij})/s$

is invariant under positive affine maps. On

$$M_r = \begin{pmatrix} r+1 & 0 \\ 1 & r \end{pmatrix}$$

we have $d_1 - d_2 = s = 1$ and boundary parameter r, so the choice is $\{1\}$ if $r \in E$ and $\{1, 2\}$ otherwise. Distinct sets E give distinct rules, and $2^{\mathfrak{c}}$ is the number of all rules on finite real matrices over a countable label universe. □

The class cannot, therefore, be parameterised by finitely many real numbers, or even countably many.

11.3. Fixed-reference scores

Suppose that only the score vector $v_a = \sum_j w_j M_{ja}$ is available, and that restriction deletes coordinates. Even then rigidity does not return. The rule choosing the top $k \geq 2$ scores, including boundary ties, $G_k(v) = \{a : \#\{b : v_b > v_a\} < k\}$, with the chosen options tied above the others and the others ranked by score, satisfies N, F, contraction, O, Q, and Mon.

Proposition 11.6. *There are $2^{\mathfrak{c}}$ rules on finite score vectors satisfying* N, F, *contraction,* O, Q, *and* Mon, *already distinguished on vectors of length four.*

Proof. Fix a set E of normalised shapes of triples. For a candidate j and a triple T of other options, let $g = (v_i - v_j)_{i \in T}$. Reject j if some triple has $\sum g > 0$, or has $\sum g = 0$, $g \neq 0$, and sorted normalised shape in E; choose the rest, and complete the ranking as above. A score maximiser is never rejected, since all its gaps are non-positive. Deletion removes witnesses, so contraction holds. Raising v_j lowers every gap sum for j, so a zero-sum rejection can only turn into a negative sum, and raises the relevant sums for other candidates; this gives Mon. Positive affine maps preserve signs and normalised shapes. On $v = (0, -1, t, 1 - t)$ with $0 < t < \frac{1}{2}$, the option of score 0 has the single triple $(-1, t, 1 - t)$, of sum zero, and is rejected exactly when its shape lies in E; these shapes are distinct for distinct t. □

With a fixed full-support w, raising M_{jj} raises only v_j, so the matrix form of Mon follows. The upshot is that the dichotomy fails in both directions: contraction under any of the three semantics is compatible with a rich family of non-ex-ante rules, and row-baseline-invariant evaluation is compatible with contraction once the belief model is fixed. What does force ex-ante evaluation is principal binary independence, which is a strong requirement: it imposes on every comparison the beliefs that the two compared options would induce, and no others.

12. Types, parameters, and certification

A recurring hope in the debate is that the relevant problem types are few: that a decision theory could be settled by its behaviour on a small catalogue of representative

problems. Whether this is so depends on what counts as the same type, and on which structural axioms a theory is already known to satisfy.

12.1. Problems

Boolean classifications are automatically finite. For k local criteria, the profile $\mathcal{S}(D) = \{I \subseteq \{1, \dots, k\} : \text{some policy satisfies every criterion in } I\}$ takes at most 2^{2^k} values. A finite classification by joint-satisfiability profiles thus exists trivially; it need not help to solve any problem, and C4, being a relation between problems, is not one of these local predicates.

Exact recommendation structures are not finite. Call two problems equivalent if a bijection of their policy spaces preserves the set of policies acceptable under each criterion.

Proposition 12.1. *Even with two actions and one always-reached information set, the continuous mixed model contains infinitely many pairwise inequivalent problems.*

Proof. For distinct $r_1, \dots, r_n \in (0, 1)$, let a reward $Y \in \{0, 1\}$, independent of the realised action, satisfy $\Pr(Y = 1 \mid p) = 1 - \prod_i (p - r_i)^2$, a continuous probability since each factor lies in $[0, 1]$. The value is maximised exactly at $p \in \{r_1, \dots, r_n\}$, so exactly n policies satisfy C1, while every policy satisfies C2 and C3. A bijection preserving the C1-sets cannot identify problems with different n. □

For deterministic policies, Theorem 7.2 already shows that every finite recommendation structure occurs. The permissive definition of a theory has a further, direct consequence.

Proposition 12.2 (No finite benchmark suite). *No finite set of problems can establish that an arbitrary decision theory satisfies* C1 *everywhere.*

Proof. Modify T^{ex} only on one two-action problem outside the suite, with a strictly better action, so as to select the worse action. The two theories agree on every benchmark, and only one satisfies C1 everywhere. □

This holds even if the benchmarks and the omitted problem are of the same intuitive type. Representative testing requires an invariance requirement on theories. Note also that infinitely many equivalence classes do not by themselves preclude a finite-dimensional parameterisation: a single real parameter indexes infinitely many cases.

12.2. Rules

The question is sharper for rules on matrices, where invariance axioms are available. Finitely many parameters must then be read cardinally: a finite-dimensional input space does not make an arbitrary rule on it a finite-dimensional family.

PROPOSITION 12.3. *There are $2^{\mathfrak{c}}$ rules satisfying* A0, N, F, *and* O. *Finite benchmark suites cannot certify* P *or* D *for them.*

Proof. For $0 < t < 1$ let $N_t = \left(\begin{smallmatrix} 0 & -1 \\ t+1 & t \end{smallmatrix}\right)$. These lie in distinct orbits of relabelling combined with positive affine maps, and each has trivial stabiliser: without a swap, preserving the entries 0 and -1 forces the identity map; with a swap, the diagonal $(t, 0)$ must be mapped to $(0, s)$, which requires a negative scale. For each $E \subseteq (0, 1)$, choose option 1 on N_t when $t \in E$ and option 2 otherwise, transport this ranking equivariantly over each orbit, and use diagonal ranking elsewhere. This gives $2^{\mathfrak{c}}$ distinct rules, whereas finite tuples of reals form a set of cardinality $\mathfrak{c}$. In N_t, column 1 strictly dominates column 2 and the second diagonal entry is larger; so modifying column sums, or diagonal ranking, on one untested orbit produces a violation of P, or of D, that no finite suite (even an adaptive one) detects. □

The construction uses the permission to have arbitrary lookup rules; it does not carry over unchanged to classes with continuity or measurability requirements.

PROPOSITION 12.4. *There are $2^{\mathfrak{c}}$ rules satisfying* N, F, *and* B *(without* O*), and finite suites cannot certify* D *for them.*

Proof. For $E \subseteq \mathbb{R}$ and $M_{ii} > M_{jj}$, declare $i \succ_M j$ if $(M_{ij} - M_{jj})/(M_{ii} - M_{jj}) \in E$; never prefer a lower diagonal strictly to a higher one, and tie all other pairs. The ratio is invariant under positive affine maps and depends only on the principal block, and strict comparisons descend the diagonal, so choice sets are nonempty. On $M_t = \left(\begin{smallmatrix} 1 & t \\ 0 & 0 \end{smallmatrix}\right)$ the choice is $\{1\}$ if $t \in E$ and $\{1, 2\}$ otherwise. Starting from $E = \mathbb{R}$ (diagonal ranking) and removing one untested ratio defeats any finite suite. □

PROPOSITION 12.5. *There are $2^{\mathfrak{c}}$ rules satisfying* N, F, S, *and* O *that agree on every binary matrix.*

Proof. Let $r_i = (M_{ii} - \min_j M_{ij})/(\max_j M_{ij} - \min_j M_{ij})$, with $r_i = 0$ for a constant row, and for $E \subseteq (0, 1)$ rank by $s_i = \mathbf{1}\{r_i \in E\}$. These rules satisfy N, F, S, and O, even under separate positive affine transformations of rows. On binary matrices every r_i is 0 or 1, so every score is zero; on $\left(\begin{smallmatrix} t & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$ the scores are $(\mathbf{1}\{t \in E\}, 0, 0)$. □

Together with Corollary 11.2, these results give the following picture.

assumptions beyond A0	size of the family; certification
N, F, O	$2^{\mathfrak{c}}$; finite suites certify neither P nor D
N, F, B	$2^{\mathfrak{c}}$; finite suites do not certify D
N, F, S, O	$2^{\mathfrak{c}}$; all binary behaviour can leave the rule undetermined
N, F, C, O, Q, Mon	$2^{\mathfrak{c}}$; finite suites do not certify D (Section 11)
N, F, B, O	exactly three rules; one unequal-diagonal test distinguishes them

"The relevant types are few" is thus true in exactly one of these regimes, and there only conditionally: when principal binary independence and ordering are known to hold globally, a single binary test decides among the three possible rules. Finite observation never verifies those structural axioms themselves.

13. Mixed policies and evidential semantics

13.1. A canonical mixed extension

A pure causal matrix does not determine how prediction responds to a mixed policy, and continuity alone does not specify the extension. One precise extension takes the predictor's response to be linear in the policy and the action to be drawn independently of the response given the announced distribution:

$$\mu_p = \sum_i p_i \mu_i, \qquad g(p) = p^\top M, \qquad V(p) = p^\top M p,$$

where $g(p)_j$ is the causal value of option j under the beliefs induced by p. This differs from a predictor of the realised sample, which can instead yield the linear value $\sum_i p_i M_{ii}$. Define

D_m p maximises $p^\top M p$;

J_m every action in the support of p maximises $p^\top M$;

P_m p assigns probability zero to every weakly dominated column.

Proposition 13.1. *In the canonical extension,* P_m *and* J_m *are jointly satisfiable on every matrix.*

Proof. For $0 < \eta < 1/|I|$ let $\Delta_\eta = \{p : p_i \geq \eta \text{ for all } i\}$. The correspondence $p \mapsto \arg\max_{q \in \Delta_\eta} p^\top M q$ on the compact convex set Δ_η has nonempty convex values and a closed graph, so Kakutani's theorem gives $p^\eta \in \arg\max_{q \in \Delta_\eta} (p^\eta)^\top M q$. If column a weakly dominates column b, full support makes $(p^\eta)^\top M e_a > (p^\eta)^\top M e_b$, so $p^\eta_b = \eta$; otherwise moving mass from b to a would improve the response. Let p be a limit point as $\eta \downarrow 0$. It assigns zero mass to every dominated column. For any q, put $q^\eta = (1 - |I|\eta)q + \eta \mathbf{1} \in \Delta_\eta$; then $(p^\eta)^\top M p^\eta \geq (p^\eta)^\top M q^\eta$, and in the limit $p^\top M p \geq p^\top M q$. Hence p is supported on best responses to itself. □

Proposition 13.2. $\{\mathsf{P}_m, \mathsf{D}_m\}$ *and* $\{\mathsf{J}_m, \mathsf{D}_m\}$ *are inconsistent.*

Proof. For $M = \left(\begin{smallmatrix} 2 & 3 \\ 0 & 1 \end{smallmatrix}\right)$ and $p = (x, 1-x)$, $V(p) = 1 + x$, so D_m selects $x = 1$, while column 2 dominates and is the unique best response to every p, so P_m and J_m select $x = 0$. □

Proposition 13.3. *For rules that select a single lottery,* N *and* D_m *are inconsistent.*

Proof. For $M = I_2$, the swap fixes M, so a neutral single lottery is $(\frac{1}{2}, \frac{1}{2})$; but $V(p) = p_1^2 + p_2^2$ is maximised only at the two pure endpoints. □

A neutral *correspondence* of maximising lotteries exists; a single neutral maximising lottery need not. The pure consistency maps therefore cannot simply be relabelled as mixed maps.

13.2. Contraction for lotteries

For a rule L that selects one lottery, the natural strengthening of contraction is conditional-renormalisation consistency:

$$\mathsf{C}_m: \qquad L(M|_K) = \frac{L(M)|_K}{L(M)(K)} \quad \text{whenever } L(M)(K) > 0.$$

Proposition 13.4. $\{\mathsf{N}, \mathsf{R}, \mathsf{C}_m, \mathsf{Q}\}$ *is a minimal inconsistent family for single-lottery rules.*

Proof. In the circulant extension of Theorem 9.3, neutrality forces the uniform lottery; conditioning forces the uniform lottery on every embedded normalised matrix; row translations then force the uniform lottery everywhere. For minimality, take the uniform distribution over the choice set of, respectively, the responsive fixed-label rule, diagonal ranking, row-range ranking, and universal indifference. The first two maximise a binary-independent weak order, so that $G(M|_K) = G(M) \cap K$ whenever the right-hand side is nonempty, which is exactly what C_m requires of uniform lotteries; row-range ranking is the witness that omits C_m. □

Support contraction alone is much weaker: $p_i \propto 1 + r_i/r_{\max}$, with r_i the row ranges and the uniform lottery when $r_{\max} = 0$, has full support everywhere, satisfies N, F, R, and is nonconstant. A complete mixed consistency map requires choosing among genuinely different output and contraction semantics; the results above are the extensions established under the stated ones.

13.3. Two evidential semantics

The matrix does not supply an evidential criterion until the conditioning experiment is identified. Two natural choices give different answers.

Reference-policy law. With a full-support reference ρ and $P(I = i, s) = \rho_i \mu_i(s)$, conditioning on the policy index i yields μ_i, and the evidential value of i is $\mathbb{E}[U(s, i) \mid I = i] = M_{ii}$. This semantics gives diagonal optimality, and it is the matrix form of $\mathrm{C6}^\kappa$ on single-choice problems.

Conditioning on the realised action under a known mixed policy. In the canonical extension the environment has law μ_p, and observing a positive-probability realised action does not change it, so its conditional value is $g_i(p)$. With an explicit completion for zero-probability alternatives, the resulting support-optimality condition is ratifiability.

The two condition on different variables under different joint laws, and neither can be silently substituted for the other. This is one precise sense in which "evidential decision theory" names a family rather than a criterion, as the literature on the tickle defence and on ratifiability has long made clear (Eells 1982; Jeffrey 1983).

Part III.
Procedures that read descriptions

14. Problems as programs

Every result so far has been proved on problems that *supply* their counterfactuals. A problem of Section 2 comes with the law μ_π for every policy π, and with structural plans that fix what each response would be on every branch. The decision theories that the debate compares disagree precisely about where such laws come from: causal, evidential, and functional theories differ over which parts of the world would have been different had the agent chosen otherwise. A model that hands this dependence to the agent has already settled what the theories dispute. This is why ex-ante optimality came so cheaply in Theorem 3.6, and why the deeper results of Part II concerned the shape of an evaluation rule rather than the identification of alternatives.

This part removes the supply. An agent is now a computable procedure that receives a description of its situation—by default the source code of the environment—and must itself determine what would happen under each of its possible outputs. The question is whether some such procedure is optimal on every fair problem, and, where none is, whether an optimal function exists at all and how far from computable it must be. Answering it requires saying what "optimal" means once the comparison is no longer given by a supplied law, and that is where most of the work lies.

14.1. Agents and environments

Fix an effective language of *descriptions*, finite strings from which a finite nonempty set $A(x)$ of legal actions can be computed for each description x. For the negative results a fixed binary action set suffices, so nothing depends on how defaults are chosen when action sets vary.

Definition 14.1 (Agents). An *agent* is a partial computable function T from descriptions to actions with $T(x) \in A(x)$ whenever $T(x)$ is defined. It is *globally total* if it halts on every well-formed description, and *promise-total* on a class $\mathcal{F}$ of environments if it halts on every description belonging to $\mathcal{F}$.

The two notions differ. "A total computable function" normally means the first; "an agent that halts on every problem in the class" permits the second. Both receive answers below, and the distinction turns out to separate a positive theorem from a negative one. Optimality itself is a property of functions, computable or not (Definition 15.2), and Section 16 measures how non-computable an optimal one must be.

Definition 14.2 (Program environment). A *program environment* E is a program with a finite set H_E of information sets, each $h \in H_E$ carrying a finite nonempty action set $A_{E,h}$ and an *owned query* $q(E,h)$, a description from which E and h can be recovered. Its policy space is $\Pi_E = \prod_{h \in H_E} A_{E,h}$, and its root description $d(E)$ is

one of its owned queries. Given access to an agent, E produces a payoff. For a table $\pi \in \Pi_E$, E^π denotes the execution in which every owned query is answered by the corresponding entry of π.

Four stipulations make owned queries behave like the information sets of Definition 2.1: a query identifies its parent environment and its information set; distinct environments have disjoint owned-query namespaces; repeated queries receive the same answer; and, when the fairness condition $\mathcal{F}_{\text{self}}$ below is imposed, the environment consults the agent only through its owned queries. The disjointness matters: if two environments may both call one query their own, the conflict of Proposition 15.4 below reappears inside a class that is ostensibly self-only.

Three representations of payoffs must be kept apart. An execution may return an *exact rational*, or an explicit finite distribution with rational probabilities and payoffs; it may return a *computable real*, given by a program producing rational approximations of any required accuracy; or it may be a random execution returning a *sampled* payoff whose expectation is the value. The positive results use the first; the strongest negative result (Theorem 15.10) uses only sampled payoffs in $\{0, 1\}$, so it does not exploit exotic real-valued outputs.

14.2. Replacement and anchored evaluation

To ask whether T is optimal on E one must say what E would pay another agent A. Some ways of saying it define a single payoff function $U(E, A)$ common to all agents. Others first run or inspect a reference agent T, freeze part of the resulting situation, and only then consider alternatives; these define an *anchored kernel* $U_K(E, T; A)$, where K names the convention. Every kernel is required to satisfy the baseline identity

$$U_K(E, T; T) = U(E, T), \tag{1}$$

and the corresponding notion of optimality is

$$U(E, T) \geq U_K(E, T; A) \qquad \text{for every admissible agent } A. \tag{2}$$

In general there is no function W with $U_K(E, T; A) = W(E, A)$ for every reference T. Some versions of the question therefore ask for *the best agent in a common environment*, others for *an agent that is a fixed point of its own counterfactual evaluations*. The difference is the program-level form of the difference between C1 and C3.

Definition 14.3 (Replacement conventions). Let E be an environment with a designated *action port*, the place at which the agent's actual action on $d(E)$ is received.

O_{const} Replace the action at the port by A's action, and freeze every other response of the environment at its value under the reference T; fixed embedded algorithms are left unchanged.

O_{oracle} Replace every designated oracle call, the port included, by a call to A.

O_{syn} Replace specified, typed occurrences of program components whose syntax equals the reference program.

O_{ext} Replace specified program components that pass a stated test of extensional equivalence with the reference agent (Section 17.2).

Under O_{syn} and O_{ext} the description $d(E)$ supplied to the agent is held fixed during replacement.

The last clause is a choice. Supplying the rewritten source $d(E[A])$ instead would change the agent's observation as well as the program occupying the agent's role; that is a legitimate model, but a different one (Proposition 18.1 shows one of its consequences). Raw substring replacement is not an adequate reading of O_{syn}, since it can alter quoted data, comments, or unrelated identifiers; typed occurrences or explicit component interfaces are needed. Logical counterfactuals, O_{log}, form a family of semantics rather than a fifth replacement operation, and are treated in Section 17.3.

14.3. Fairness

The fairness condition of Definition 2.4 splits into several conditions, which overlap rather than partition the environments.

$\mathcal{F}_{\text{self}}$ Dependence on the evaluated agent is confined to its answers on the owned queries $q(E, h)$.

$\mathcal{F}_{\text{any}}$ The environment may consult the agent on arbitrary descriptions.

$\mathcal{F}_{\text{ext}}$ Replacing agents by extensionally equal agents preserves the payoff kernel: $T \equiv T'$ and $A \equiv A'$ imply $U_K(E, T; A) = U_K(E, T'; A')$.

$\mathcal{F}_{\text{const}}$ Fixed algorithms, including copies of a particular agent, may occur as constants of the environment.

Two locality requirements should also be separated. *Executed-query locality* says that actual executions call agents only on owned queries. *Kernel locality* says that the whole counterfactual kernel, including the rule that decides what to replace, depends on the reference agent only through its answers on owned queries. Global program-equivalence matching (Section 17.2) satisfies the first while violating the second.

In Section 2 a fixed algorithm inside a problem was harmless: its output on the relevant input is a number that is part of the problem, and the problem is a finite object specified by such numbers. In a program environment the algorithm's code is part of the description and its output is not; the agent must compute it, and may be the very thing being computed. This is what makes $\mathcal{F}_{\text{const}}$ consequential here (Theorem 15.6).

14.4. Programs and causal matrices

The conventions O_{oracle} and O_{const} have exact counterparts in the causal matrix of Section 7. Let E be a finite oracle environment in which every execution with table answers halts with an exact rational payoff, and let the owned queries be answered at two kinds of position: the action ports, and the remaining oracle calls. For $\sigma, \tau \in \Pi_E$

let $M_E[\sigma, \tau]$ be the payoff of the execution in which the oracle calls are answered by σ and the ports by τ.

Proposition 14.4 (Programs induce causal matrices). *For agents T and A write π_T and π_A for the tables of their answers on the owned queries. Then*

$$U_{\text{oracle}}(E, A) = M_E[\pi_A, \pi_A], \qquad U_{\text{const}}(E, T; A) = M_E[\pi_T, \pi_A].$$

M_E is the causal matrix of the finite problem whose structural states are the oracle tables, with $\mu_\pi = \delta_\pi$ and $U(\sigma, \tau) = M_E[\sigma, \tau]$, and every finite real square matrix indexed by a product of finite action sets arises in this way.

Proof. Under O_{oracle} every owned query, port or not, is answered by A, which is the execution indexed by (π_A, π_A). Under O_{const} the ports receive A's answers while every other response is frozen at its value under T; since the only responses depending on the agent are the oracle calls, this is the execution indexed by (π_T, π_A). The problem described has causal matrix $\sum_\rho \mu_\sigma(\rho) U(\rho, \tau) = M_E[\sigma, \tau]$. For the last claim, an environment that calls the oracle on all owned queries, then reads the ports, and pays a prescribed entry $M[\sigma, \tau]$ of a given matrix realises M, exactly as in Theorem 7.2. □

Oracle substitution thus recovers the diagonal of the matrix, ex-ante value, and port-only intervention recovers a row, causal evaluation at the baseline π_T. Both require a designation of which positions are ports and which are calls; a program in which the agent's code occurs inside other programs, with no designated interface, does not carry that designation, and the agent must construct it. Section 18 shows that the two kinds of evaluation reappear, in this form, as the only two coherent ways of doing so.

14.5. Comparing the conventions

Replacing more components does not make optimality logically stronger or weaker.

Proposition 14.5 (No implications between conventions). *For any two of O_{const}, O_{oracle}, O_{syn}, O_{ext} there are terminating environments on which a given agent is optimal under the first and not under the second, and environments on which the reverse holds.*

Proof. Let a typed environment contain an action port, an additional oracle call, a literal copy of the reference agent's code, and an independently encoded copy computing the same function, each queried at $d(E)$; let the reference agent be the constant 0 and the alternative the constant 1. The four conventions then produce the following patterns of the four values.

evaluation	port	oracle call	literal copy	encoded copy
baseline	0	0	0	0
O_{const} alternative	1	0	0	0
O_{oracle} alternative	1	1	0	0
O_{syn} alternative	1	1	1	0
O_{ext} alternative	1	1	1	1

For two conventions producing distinct patterns v and w from the baseline pattern z, set the payoff to 1 at z, 0 at v, 2 at w, and 0 elsewhere. The reference agent is optimal under the first convention and not under the second; exchanging the payoffs at v and w gives the reverse. □

The conventions coincide only under conditions that make their replacement sets coincide. Oracle and syntactic substitution agree under a slot-preserving compilation of oracle calls into designated code occurrences (Section 17.1); oracle and extensional substitution agree when the extensional test selects exactly the designated dependencies; syntactic and extensional substitution agree when their replacement masks coincide. For logical semantics no comparison is possible until a particular semantics is fixed: a guarded provability semantics is partial (Definition 17.8), a proof of the value of each policy-substituted execution simply evaluates an independently supplied substitution (Proposition 17.6), and the scores of a logical inductor are finite-time estimates. Treating these as one fifth payoff function would conceal the main issue.

15. Optimal agents and computable optimisation

This section asks what changes when the decision theory of Theorem 3.6 must be a single function of descriptions, and then a single algorithm. For functions, nothing changes: an optimal function exists on both admissible classes considered, fixed copies of arbitrary programs included. For algorithms, with a fixed substitution semantics and a termination promise, nothing changes either: one algorithm is optimal everywhere. Obstructions appear once the algorithm must halt on every input or the class admits exact comparisons among computable reals, and none of them concerns counterfactual inference. Section 16 measures them exactly.

15.1. Optimal decision functions

Throughout this section and the next, the semantics is O_{oracle} with embedded programs held fixed: every designated call is answered by the evaluated agent, and every other program in the environment runs as an ordinary computation, its running time counting as part of the environment's execution.

Definition 15.1 (Admissible classes). $\mathcal{P}_{\mathbb{Q}}$ is the class of program environments E such that E^{π} halts and returns an exact rational value $V_E(\pi)$ for every $\pi \in \Pi_E$. $\mathcal{P}_{\text{as}}$ is the class of probabilistic program environments, branching with exact rational probabilities and returning terminal rewards that are rationals in $[0, 1]$, such that E^{π} terminates with probability 1 for every $\pi \in \Pi_E$; then $V_E(\pi)$ is the expected reward.

Both definitions use table oracles, so the admissibility of an environment is not defined in terms of whichever optimiser is eventually constructed. In $\mathcal{P}_{\text{as}}$ the reward bound is enforced by the language, so that it holds also on programs that fail the termination promise, and almost-sure termination is required of the whole execution,

embedded computations included; nothing is required of embedded programs on inputs the environment never supplies, nor of executions of probability zero.

Definition 15.2 (Decision functions). A *decision function* is a total function T, not necessarily computable, from descriptions to actions with $T(x) \in A(x)$. On a class $\mathcal{F}$ of admissible environments, T is *optimal* if $U(E, T) \geq U(E, A)$ for every $E \in \mathcal{F}$ and every decision function A.

Under O_{oracle} the evaluated agent enters an execution only through its answers on owned queries, so $U(E, A) = V_E(\pi_E^A)$ with $\pi_E^A(h) = A(q(E, h))$. Every table $\pi \in \Pi_E$ extends to a globally total computable agent, hence

$$\max_A U(E, A) = \max_{\pi \in \Pi_E} V_E(\pi), \tag{3}$$

whether A ranges over decision functions or over computable agents. Optimality is therefore a property of a function, and the questions of this section and the next concern how hard that function is to compute.

Theorem 15.3 (Optimal decision functions exist). *On $\mathcal{P}_{\mathrm{Q}}$ and on $\mathcal{P}_{\text{as}}$ there is an optimal decision function, and it can be chosen to return a fixed default action on every description that is not an owned query of an admissible environment.*

Proof. Fix an effective order on each finite policy space, and for admissible E let π_E^* be the first maximiser of V_E, which exists because Π_E is finite and every value is defined (in $\mathcal{P}_{\text{as}}$, finite because rewards are bounded). Put $T^*(q(E, h)) = \pi_E^*(h)$ for admissible E, and the default elsewhere. Since owned-query namespaces are disjoint, no description receives prescriptions from two environments, so T^* is well defined, and $U(E, T^*) = V_E(\pi_E^*) = \max_A U(E, A)$ by (3). □

The definition is explicit and uses no choice principle. It covers environments containing fixed copies of arbitrary programs, including programs that compute candidate theories: such a copy is a fixed feature of V_E, whatever it computes. Requiring each problem to fix how the world responds to every policy therefore does not exclude problems that refer to a particular theory; it excludes dependence on the *evaluated* agent other than through its owned queries. The hard-coded counterpart of Remark 3.7, the family D_b, needed a fixed point that does not exist; the program counterpart below needs none.

Disjoint ownership is sufficient for existence, not necessary. The exact condition is compatibility: a class $\mathcal{F}$ has an optimal decision function iff

$$\bigcap_{E \in \mathcal{F}} \{T \,:\, \pi_E^T \in \arg\max V_E\} \neq \emptyset.$$

Without ownership the condition fails at once.

Proposition 15.4 (Foreign queries). *There are two terminating environments such that no decision function is optimal on both under O_{oracle}.*

Proof. Fix a binary query q distinct from both environments' own descriptions, and let E_0 pay $\mathbf{1}[A(q) = 0]$ and E_1 pay $\mathbf{1}[A(q) = 1]$. Each has optimal value 1, attained by a constant agent, and no decision function attains 1 in both. □

This is a cross-problem consistency contradiction requiring neither self-reference nor undecidability, and it eliminates optimal functions of every Turing degree. It is excluded by $\mathcal{F}_{\text{self}}$ with disjoint namespaces, and it disappears under O_{const}, which freezes the foreign response.

15.2. A uniform optimiser

THEOREM 15.5 (Uniform optimisation on the promise class). *There is a computable agent that is promise-total on $\mathcal{P}_{\mathbb{Q}}$ and optimal under O_{oracle} on every $E \in \mathcal{P}_{\mathbb{Q}}$.*

Proof. On input $q(E, h)$, let P recover E and its finite policy space, run E^{π} for every $\pi \in \Pi_E$, select the first maximiser π_E^* of V_E in a fixed effective order, and return $\pi_E^*(h)$. Every simulation halts by the promise, there are finitely many, and exact rational comparison terminates. All owned queries of E recover the same environment and use the same tie-breaking, so the answers form the single policy π_E^*, and P agrees with T^* of Theorem 15.3 on every admissible environment. □

P never asks its live self what it would do during a simulated execution; it supplies table answers instead. This is Theorem 3.6 made uniform: the lookup table is replaced by an algorithm, and no knowledge of which parts of E are predictors is needed, because oracle substitution designates them. By Proposition 14.4 the theorem maximises the diagonal of M_E.

An explicit finite policy header is sufficient for the argument but not necessary. A finitely branching oracle machine that halts for *every* total oracle has a finite tree of consistent oracle answers, since otherwise Kőnig's lemma (Kőnig 1927) gives an infinite path, whose answers extend to a total oracle on which the machine runs forever; exhaustive exploration of the tree then yields a finite optimisation problem. Halting against every *computable* oracle does not justify this compactness argument, since an infinite computable tree can lack a computable infinite path.

Dependence of termination on the oracle is harmless as long as every policy terminates. The theorem fails when some alternatives are undefined: if action 0 returns 0 and action 1 simulates machine e and returns 1 if it halts, the environment lies outside $\mathcal{P}_{\mathbb{Q}}$ unless e halts. Adding the convention that divergence pays -1 would make an exact optimiser choose 1 exactly when e halts, and so decide the halting problem. The obstruction then comes from the added divergence semantics, not from the finite-policy argument.

15.3. The fixed-copy diagonal

The optimiser P halts only on the promise. Requiring an agent to halt on every input brings in a diagonal that again involves no counterfactual subtlety.

THEOREM 15.6 (The fixed-copy diagonal). *No globally total computable agent is optimal on $\mathcal{P}_{\mathbb{Q}}$, even on its deterministic, binary environments with payoffs in $\{0, 1\}$ that consult the evaluated agent only on their own description.*

Proof. Fix a globally total computable T, and construct an environment E_T containing the fixed algorithm $g_0 = T$. With $d = d(E_T)$, it computes $c = g_0(d)$, receives the evaluated agent's action a at the port, and pays $\mathbf{1}[a \neq c]$. The algorithm g_0 does not change when the evaluated agent changes, so $U(E_T, A) = \mathbf{1}[A(d) \neq T(d)]$. Since T is globally total, $g_0(d)$ halts, so E_T terminates under either action and lies in $\mathcal{P}_{\mathbb{Q}}$. T receives 0, and the agent taking $1 - c$ at d receives 1. □

In a programming formalism with no built-in parameter for the environment's own description, Kleene's second recursion theorem (Kleene 1938) supplies the self-reference; see Moschovakis (2010) for a modern statement and proof. It produces a program with a specified behaviour involving its own index; it does not prove that the behaviour terminates, and global totality is what supplies termination. The construction is uniform in an index of T. The proof holds under O_{const}, and under O_{oracle} whenever g_0 is an ordinary fixed program rather than a designated oracle call. The transparent-and-opaque-box construction of Skalse (2021) is the same mechanism with different payoffs, and a footnote of Yudkowsky and Soares (2017, §9, n. 15) already discusses an environment that punishes agreement with a fixed copy of a particular program. The proof that AIXI is not computable (Leike and Hutter 2015b, Thm. 14) builds an environment that simulates a given computable policy and acts against it, the same diagonal in reinforcement learning; informal fixed-simulation problems appear in drnickbone (2012).

Remark 3.7 found that construction ill defined in the model of Section 2, and the reason for the difference is the following. There a problem was a finite object whose data included the constant's output, so the construction required a problem D_b with $b = T(D_b)$, and an ex-ante optimiser has no such fixed point. Here the description of E_T encodes a computation involving T without supplying its result, and no fixed point is needed: whatever T outputs on $d(E_T)$, the copy outputs the same. Once that output c is known, the environment is the ordinary one-shot problem with payoffs $\mathbf{1}[a \neq c]$, and its maximiser is $1 - c$. The agent cannot uniformly select it while remaining globally total, because finding c means running T on the very description it is reading.

The diagonal cannot be built against an optimal agent, and for the promise-total optimiser this can be proved rather than suggested.

PROPOSITION 15.7 (Escape of the promise optimiser). *Let E_P be the diagonal environment of Theorem 15.6 built from the optimiser P of Theorem 15.5. Then $P(d(E_P))$ diverges, so $E_P \notin \mathcal{P}_{\mathbb{Q}}$.*

Proof. Suppose $P(d(E_P))$ halted with output b. Then E_P would halt under both actions with payoffs in $\{0, 1\}$, so it would be admissible, and P, optimal on admissible environments, would output the maximiser $1 - b$, not b. □

The escape depends on the evaluation convention. Assigning a payoff to divergence and admitting E_P changes both the class and the optimisation problem, as the halting example above shows.

A non-computable optimal function escapes in a different way. No embedded program computes it globally, but a computable program can agree with it on any finite set of inputs, and non-computability does not forbid such agreement. What optimality forbids is agreement at one particular input. For a fixed program p, let the *anti-prediction environment* E_p compute $c = p(d(E_p))$, read the action at its port, and pay $\mathbf{1}[a \neq c]$.

Proposition 15.8 (Anti-prediction). *If T is optimal on $\mathcal{P}_{\mathbb{Q}}$ and $p(d(E_p))$ halts with an action, then $T(d(E_p)) = 1 - p(d(E_p))$. In particular, no fixed program is accurate for an optimal decision function on the input of its own anti-prediction environment.*

Proof. If $p(d(E_p))$ halts, E_p is admissible with optimal value 1, attained only by the action $1 - p(d(E_p))$. □

Hard-coding does not evade this. An environment that hard-codes a bit k as its prediction is the anti-prediction environment of the constant program k, and the proposition forces $T(d) = 1 - k$ on its description d; for the two environments E_0, E_1 with constant predictions 0 and 1, an optimal function satisfies $T(d(E_0)) = 1$ and $T(d(E_1)) = 0$, so neither satisfies $k = T(d(E_k))$. The observation that inserting the bit changes the description is not by itself the argument; the argument is that the required self-consistency fails for every optimal function. The immunity belongs to optimal functions, not to non-computable ones (Proposition 16.8).

A last baseline is the program form of Death in Damascus.

Proposition 15.9 (Self-prediction held fixed). *Let an oracle environment call the agent twice on d, treating one answer as a prediction p and the other as the action a, and pay $\mathbf{1}[a \neq p]$. Under O_{oracle} every agent receives 0 and every agent is optimal. Under O_{const} every agent has an improving deviation.*

Proof. Coherent substitution answers both calls alike. An intervention changing only the action, with the prediction frozen at the reference agent's answer p, earns 1 with $a = 1 - p$. □

By Proposition 14.4, this is the matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, in which no diagonal entry is maximal in its row; it is Proposition 4.3 again, a ratifiability obstruction and not a computability result. Whether the prediction is linked to the agent, as under O_{oracle}, or fixed, as under O_{const}, is exactly what separates it from the fixed-copy diagonal. If the embedded copy of Theorem 15.6 is declared linked, replacing T by A changes both occurrences, the payoff becomes $\mathbf{1}[A(d) \neq A(d)] = 0$, and every agent is optimal there. Actual behaviour cannot settle the declaration: a fixed program computing 0 and a linked call to an agent that outputs 0 agree on the actual execution and differ under replacement.

15.4. Exact selection

The following result avoids every baseline: there are no predictors, no copies of the agent, no foreign queries, and the agent need halt only on valid inputs.

THEOREM 15.10 (No computable exact optimiser). *No promise-total computable agent is optimal on $\mathcal{P}_{\mathrm{as}}$, even on its binary-choice environments whose realised payoffs lie in $\{0,1\}$, that consult the agent only at the action port on their own description, whose expected values are uniformly computable, and whose expected running time is finite.*

Proof. Let (φ_e) be an effective enumeration of the partial computable functions, and let $K_b = \{e : \varphi_e(e) \text{ halts with output } b\}$ for $b \in \{0,1\}$. The environment E_e receives the action $a \in \{0,1\}$, draws $N \in \mathbb{N}$ with $\Pr(N = n) = 2^{-n-1}$, simulates $\varphi_e(e)$ for N steps, and pays 1 if the simulation halts within them with output a, and 0 otherwise. Every simulation is bounded, E_e terminates almost surely, and its expected running time is finite. If $e \in K_b$ with halting time t, then $V_e(b) = \Pr(N \geq t) = 2^{-t} > 0$ and $V_e(1-b) = 0$; if neither output occurs, both values are 0.

Suppose a promise-total computable exact optimiser T existed. Then $f(e) = T(d(E_e))$ is a total computable function with $f(e) = b$ whenever $e \in K_b$. Let e be an index of the total program $x \mapsto 1 - f(x)$. Then $\varphi_e(e) = 1 - f(e)$, so $e \in K_{1-f(e)}$, and the condition requires $f(e) = 1 - f(e)$. □

The proof is the recursive inseparability of K_0 and K_1 (Odifreddi 1989, Thm. II.2.5). What fails is *exact selection*, not the approximation of expected values: after n simulation steps the unresolved part of each expectation is at most 2^{-n}, so the values are uniformly computable reals with an explicit error bound. What cannot be computed uniformly is which action belongs to the exact argmax when arbitrarily small advantages and genuine ties are both admitted. The obstruction disappears as soon as one of the two is excluded.

PROPOSITION 15.11 (Positive replacements). *On a class of environments with finite policy spaces and uniformly computable values, there are promise-total agents that are*

(i) ε-optimal everywhere, for any fixed rational $\varepsilon > 0$;

(ii) exactly optimal on every environment with a unique maximising policy;

(iii) exactly optimal on every environment whose values are exact rationals.

Proof. (i) Approximate every value within $\varepsilon/3$ and maximise the approximations. (ii) Refine enclosing intervals until one policy's lower bound exceeds every other policy's upper bound, which happens in finite time when the maximiser is unique. (iii) is Theorem 15.5. □

Theorem 15.10 is therefore a true and non-trivial version of the thesis that no computable rule is optimal on all fair problems, on an especially fair class. Its obstruction is classical recursion theory, not a conflict among decision theories. Together with Theorem 15.3 it separates the two things the thesis ran together:

optimal choice, problem by problem, is always possible, and an optimal *program* is not.

16. How non-computable an optimal agent must be

Theorem 15.3 and Theorems 15.6–15.10 leave a precise question: what oracle suffices, and what oracle is necessary, to compute an optimal decision function? The natural guess is the halting problem, since admissibility of a deterministic environment is a halting condition. The guess is wrong. Optimisation does not require recognising admissibility, and the exact strength it requires is that of choosing, among finitely many candidates, one that is never refuted. That strength is a classical degree class, the PA degrees, and it rises to the halting problem and beyond only when the theory is also required to say something definite off the admissible class or to break ties in a prescribed way.

Write $\leq_T$ for Turing reducibility, $\mathbf{0}'$ for the degree of the halting problem and $\mathbf{0}''$ for that of its relativisation. An oracle X *computes* an optimal agent of a given totality if some X-computable agent of that totality is optimal.

16.1. Diagonally non-computable functions

Definition 16.1 (Binary DNC and PA degree). A total function $g : \mathbb{N} \to \{0, 1\}$ is *binary diagonally non-computable* if $g(e) \neq \varphi_e(e)$ whenever $\varphi_e(e)$ halts. A degree is a *PA degree* if it computes such a function.

No binary DNC function is computable, since an index e of it would give $g(e) \neq g(e)$. The name comes from the equivalent condition that the degree computes a complete consistent extension of Peano arithmetic. Equivalently, a degree is PA iff it computes a set separating K_0 from K_1, since $1 - s$ is binary DNC when s separates them, and conversely. These equivalences go back to Scott (1962), Solovay (Odifreddi 1989, Thm. V.5.36), and Jockusch and Soare (1972); Hirschfeldt (2014, §3.1) gives a modern account. The one consequence needed here is elementary.

Lemma 16.2 (Finite selection). *Let a finite nonempty set B of candidates be given, together with a computably enumerable process excluding candidates, uniformly in the data, such that at least one candidate is never excluded. Any binary DNC function selects, with finitely many queries and uniformly in the data, a candidate that is never excluded.*

Proof. If $|B| > 1$, split B into nonempty parts B_0, B_1, and compute an index k of the program that ignores its input, runs the exclusion process, and outputs i as soon as every member of B_i has been excluded. At most one part is ever wholly excluded, because some candidate survives. Continue with $B_{g(k)}$: if it were wholly excluded, $\varphi_k(k) = g(k)$, contrary to the definition of g. So $B_{g(k)}$ contains a survivor, and after fewer than $|B|$ steps a singleton survivor remains. □

The procedure never waits for the exclusion process; it only asks g about programs that would wait. Selecting a survivor is therefore weaker than recognising one; in the language of Brattka et al. (2012) it is finite closed choice.

16.2. The degree theorem

THEOREM 16.3 (Optimality has PA strength). *For an oracle X:*

(i) X computes a globally total optimal agent on $\mathcal{P}_{\mathbb{Q}}$ iff X has PA degree;

(ii) X computes a globally total optimal agent on $\mathcal{P}_{\mathrm{as}}$ iff X has PA degree;

(iii) X computes a promise-total optimal agent on $\mathcal{P}_{\mathrm{as}}$ iff X has PA degree;

while a promise-total optimal agent on $\mathcal{P}_{\mathbb{Q}}$ is computable (Theorem 15.5).

Proof. Upper bound for (i). On an owned query of a description E, take the policies of E as candidates, and let the exclusion process simulate E^{π} for every π; if all simulations halt, it excludes exactly the non-maximising policies, and otherwise it excludes nothing. Some candidate is never excluded on every description, and on admissible E the survivors are exactly the maximisers. Lemma 16.2 selects one with an X-computable binary DNC function; every owned query of E reconstructs the same selection, and the agent answers with its component. It halts on every input, since the lemma makes finitely many queries and never waits.

Upper bound for (ii) and (iii). For a policy π and a time s, let $H_{\pi,s}$ be the probability that E^{π} has halted by step s and $L_{\pi,s}$ the expected reward collected on those executions. Both are rationals computable from the finite execution tree, and with $B_{\pi,s} = L_{\pi,s} + 1 - H_{\pi,s}$ the intervals $[L_{\pi,s}, B_{\pi,s}]$ are nested, because newly halted mass contributes a reward between 0 and that mass. On admissible E,

$$V_E(\pi) \in [L_{\pi,s}, B_{\pi,s}], \qquad B_{\pi,s} - L_{\pi,s} = 1 - H_{\pi,s} \to 0. \tag{4}$$

Exclude π as soon as $L_{\rho,s} > B_{\pi,s}$ for some ρ and s. On admissible environments this excludes exactly the strictly suboptimal policies. On every description some policy survives: let $\ell_{\pi} = \lim_s L_{\pi,s}$, the reward collected on terminating executions; then $L_{\pi,s} \le \ell_{\pi} \le B_{\pi,s}$, so a maximiser π° of ℓ cannot be excluded, since $\ell_{\rho} \ge L_{\rho,s} > B_{\pi^{\circ},s} \ge \ell_{\pi^{\circ}}$ would follow. The quantity ℓ serves only this argument and changes no admissibility condition. Lemma 16.2 now gives a globally total agent, optimal on $\mathcal{P}_{\mathrm{as}}$, and *a fortiori* a promise-total one.

Lower bound for (i). Let D_e run $\varphi_e(e)$ and, if it returns a bit b, pay $\mathbf{1}[a = b]$; if it returns anything else, pay 0 for both actions. D_e is admissible iff $\varphi_e(e)$ halts. For a globally total optimal T, $s(e) = T(d(D_e))$ is total, even where D_e is inadmissible, and equals b whenever $e \in K_b$. So s separates K_0 from K_1 and $1 - s$ is binary DNC.

Lower bound for (ii) and (iii). Every environment E_e of Theorem 15.10 lies in $\mathcal{P}_{\mathrm{as}}$, and an optimal agent, promise-total or global, satisfies $T(d(E_e)) = b$ for $e \in K_b$. Since all E_e are admissible, $s(e) = T(d(E_e))$ is total, and it separates K_0 from K_1. □

Three consequences of the theorem bear on the thesis. First, the deterministic lower bound uses no self-reference and no copy of any agent: D_e is generated independently of the theory evaluated. It contains Theorem 15.6 as the special case

$X = \emptyset$, and shows that the obstruction to a globally total optimiser is not a predictor but the task of completing the promise optimiser of Theorem 15.5 on descriptions that might be admissible. Second, the optimiser does not decide admissibility, which for $\mathcal{P}_{\mathrm{as}}$ is of degree $\mathbf{o}''$ (Proposition 16.5); it only selects a survivor. Third, on $\mathcal{P}_{\mathrm{as}}$ the strength is required even of agents that halt only on the promise, which is why Theorem 15.10 is a negative result about exact selection and not about totality.

Corollary 16.4 (The spectrum of optimal agents). *On $\mathcal{P}_{\mathbb{Q}}$ and on $\mathcal{P}_{\mathrm{as}}$:*

(i) every PA degree is the exact Turing degree of some optimal decision function;

(ii) some optimal decision function T satisfies $\mathbf{o} <_T T <_T \mathbf{o}'$;

(iii) for every non-computable set Y there is an optimal decision function that does not compute Y, so no single non-computable set is computed by every optimal decision function.

Proof. (i) Given X of PA degree, take an X-computable optimiser from Theorem 16.3 and modify it on a computable family of distinct binary environments F_n in which both actions tie, answering $X(n)$ on $d(F_n)$. Ownership confines each modification to F_n, so the result is still optimal and X-computable, and it computes X. (ii) By the low basis theorem (Jockusch and Soare 1972, Thm. 2.1), which applies to the Π^0_1 class of separating sets of K_0 and K_1, there is a PA degree X with $X' \equiv_T \mathbf{o}'$; it does not compute $\mathbf{o}'$, since otherwise $\mathbf{o}'' \leq_T X' \equiv_T \mathbf{o}'$. Apply (i). (iii) By the cone-avoidance basis theorem for Π^0_1 classes (Jockusch and Soare 1972, Thm. 2.5), applied to the class of separating sets of K_0 and K_1, some PA degree does not compute Y. Apply (i). □

So "optimal agents need the halting problem" is false, and there is no least degree of optimality: the PA degrees have no least element.

16.3. Admissibility and prescribed answers

Proposition 16.5 (Complexity of admissibility). *Membership in $\mathcal{P}_{\mathbb{Q}}$ is Σ^0_1-complete, and membership in $\mathcal{P}_{\mathrm{as}}$ is Π^0_2-complete.*

Proof. Since there are finitely many policies, $E \in \mathcal{P}_{\mathbb{Q}}$ iff for some s every E^π halts legally within s steps, a decidable condition on s; an environment that ignores the action, runs a given computation and then pays 0 is admissible exactly when the computation halts. For the stochastic class, $E \in \mathcal{P}_{\mathrm{as}}$ iff $\forall k\, \exists s\, \forall \pi \in \Pi_E : H_{\pi,s} > 1 - 2^{-k}$, and the last quantifier is finite. For hardness, reduce $\mathrm{TOT} = \{e : \varphi_e \text{ is total}\}$: draw a geometric N and run $\varphi_e(N)$ without a bound. Every n has positive probability, so the environment terminates almost surely exactly when $e \in \mathrm{TOT}$. □

The stochastic half is the Π^0_2-completeness of almost-sure termination proved by Kaminski and Katoen (2015); see also Kaminski et al. (2019, §4). A theory that must answer a prescribed default on every inadmissible description, as the existence theorem permits, needs this much more.

Proposition 16.6 (Prescribed answers). *Let the agent be required to return a given computable default on every description that is not an owned query of an admissible environment.*

(i) On $\mathcal{P}_{\mathbb{Q}}$ every such optimal agent computes $\mathbf{0}'$, and one is computable from $\mathbf{0}'$.

(ii) On $\mathcal{P}_{\mathrm{as}}$ every such optimal agent computes $\mathbf{0}''$, and one is computable from $\mathbf{0}''$.

(iii) On $\mathcal{P}_{\mathrm{as}}$, if ties must be broken in a fixed order, every promise-total optimal agent computes $\mathbf{0}'$, and one is computable from $\mathbf{0}'$.

Proof. Take the default to be 0; for another computable default, the environments below compute it on their own description and reward its opposite. (i) Let H_e wait for $\varphi_e(e)$ and, if it halts, pay the action $a \in \{0, 1\}$. If $\varphi_e(e)$ halts, 1 is the unique optimum; otherwise H_e is inadmissible and the default applies. So $T(d(H_e)) = 1$ iff $\varphi_e(e)$ halts. Conversely, $\mathbf{0}'$ decides admissibility, after which the simulation of Theorem 15.5 applies. (ii) Let the environment draw a geometric N, run $\varphi_e(N)$, and if it halts pay a. It is admissible, with unique optimum 1, iff $e \in \mathrm{TOT}$, so T decides TOT. Conversely, $\mathbf{0}''$ decides admissibility, and on admissible environments $\mathbf{0}'$ decides the Σ^0_1 exclusion predicate of the proof of Theorem 16.3, whose survivors are the maximisers. (iii) Let action 0 pay 0, and let action 1 draw a geometric N, simulate a given computation for N steps and pay 1 if it has halted. Action 1 has value 2^{-t} if the computation halts at step t and value 0 otherwise, so with ties broken towards 0 the agent chooses 1 iff the computation halts. Conversely, $\mathbf{0}'$ decides exclusion, and the first survivor in the fixed order is the prescribed maximiser. □

The PA classification therefore concerns optimality as defined, which permits any maximiser and any answer off the admissible class. Every additional specification of what the theory must say, such as a default outside the class or an order among tied alternatives, specifies something other than the choice among alternatives in admissible problems, and it is what raises the degree.

16.4. Relativisation

Theorem 16.7 (Relativised diagonal). *Fix an oracle X, and let environments and their embedded programs be X-computable, with the same ownership and termination conditions. Then no globally total X-computable agent is optimal, an optimal decision function still exists, and Theorem 16.3 and Proposition 16.6 hold with binary* DNC^X *functions in place of binary DNC functions and with X', X'' in place of $\mathbf{0}'$, $\mathbf{0}''$.*

Proof. The relativised recursion theorem (Odifreddi 1989, Thm. II.2.13 and p. 177) constructs, for each globally total X-computable T, an X-computable environment with $U(E_T^X, A) = \mathbf{1}[A(d) \neq T(d)]$; its fixed computation of $T(d)$ halts, so it is admissible and defeats T. Existence uses only finiteness and ownership. The remaining proofs relativise line by line. □

The oracle of the environments is held fixed in the theorem. If environments may consult the optimiser's stronger oracle, the class is a new one and the diagonal

applies again: for every class of environments closed under an oracle, an optimal agent exists, and every optimal agent that halts on every input lies strictly above that oracle in the sense of computing a DNC^X function. This makes precise the claim that an optimal theory must lie above every environment that can run it.

16.5. Sharpness of the hypotheses

Table 1 records what fails when each hypothesis is dropped. The constructions are short, and the non-obvious ones are given after the table.

TABLE 1. Hypotheses of Theorems 15.3 and 16.3, and what fails without them.

hypothesis dropped	consequence
owned queries	no optimal function of any degree (Prop. 15.4)
embedded copies fixed	the diagonal of Thm. 15.6 disappears; the lower bounds of Thm. 16.3 remain, since they use no copy
termination under every table	with divergence paying a fixed value, exact optimisation decides halting even on deterministic environments
explicit rational values	with values given as computable reals, deterministic exact optimisation needs PA degree even on the promise
rewards bounded in $[0, 1]$	an almost surely terminating family with finite values forces $\mathbf{o}'$, even for ε-optimality with $\varepsilon < \frac{1}{2}$
exact optimality	ε-optimal and unique-maximiser agents are computable on the promise (Prop. 15.11)
finite policy spaces	a maximum need not exist: payoffs $1 - 1/(n+1)$ on actions $n \in \mathbb{N}$

For computable reals, let $v_e(b) = 2^{-t}$ and $v_e(1-b) = 0$ if $\varphi_e(e)$ first halts with bit b at step t, and $v_e \equiv 0$ otherwise. These values have uniformly computable fast Cauchy names, and every one of them is in fact rational; what is missing is its explicit exact representation. An exact optimiser separates K_0 from K_1 as in Theorem 16.3. For unbounded rewards, draw $N \geq 1$ with probability 2^{-N}; let action 0 pay $\frac{1}{2}$ and action 1 pay 2^N exactly when a given computation first halts at step N. Then $V(1) = 1$ if the computation halts and $V(1) = 0$ otherwise, so any agent within $\frac{1}{2}$ of optimal decides halting. The values are finite but not uniformly computable, which is why Proposition 15.11(i) does not apply.

16.6. Unfair problems

The remaining question is the one the thesis began with: which theories face problems built against them? "Built against" names a provenance unless a mathematical witness is specified. The anti-prediction environments E_p of Proposition 15.8 supply one without reference to optimality: E_p is *accurate for* a decision function T if $p(d(E_p)) = T(d(E_p))$. For admissible E_p, accuracy is equivalent to defeat, since T then receives 0 while the optimum is 1.

PROPOSITION 16.8 (Unfairness and computability).

(i) *For every oracle X, every globally total X-computable decision function has an accurate, admissible, X-computable anti-prediction environment, constructed uniformly from its index.*

(ii) *No optimal decision function has an accurate admissible anti-prediction environment.*

(iii) *Some non-computable decision function has an accurate admissible anti-prediction environment.*

Proof. (i) is Theorem 16.7 and (ii) is Proposition 15.8. For (iii), let T^* be optimal and non-computable, by Theorem 16.3 and Corollary 16.4, and let E_0 be the anti-prediction environment of the constant program 0, so that $T^*(d(E_0)) = 1$. Changing this single answer to 0 gives a non-computable function for which E_0 is accurate. □

The proposition refutes the claim that unfair problems exist exactly for computable theories. The defensible direction is

$$\text{global simulability by the environments} \implies \text{uniformly constructible defeat},$$

and neither defeat nor local accuracy implies computability: non-computable theories can make predictable mistakes. Nor can computability be read off the restriction to owned queries, since every such restriction is a finite table and hence computable. Within the anti-prediction family the equivalence of accuracy and defeat is exact, but it characterises the family, not the computable theories. Leike and Hutter (2015a, Thm. 18) make a related observation in a different setting: finite hard-coded behaviour defines a computable environment even when the policy it is aimed at is not computable.

16.7. Resource bounds

Finite clocks or circuit descriptions can support globally total optimisers, because every required evaluation then has a finite bound: on a decidable clocked language with a supplied execution bound, deterministic values are evaluated exhaustively and stochastic ones by summing a finite rational execution tree. They do not make optimisation as cheap as one execution.

Proposition 16.9. *If an agent running in time polynomial in the length of the description is exactly optimal on all environments with n binary information sets and a Boolean circuit as payoff, then* $\mathrm{P} = \mathrm{NP}$. *If it is exactly optimal on all single-choice environments that run in polynomial time with fair coins, then* $\mathrm{P} = \mathrm{PP}$.

Proof. Given a Boolean formula with n variables, form the environment whose information sets are the variables and whose payoff is 1 exactly on satisfying assignments. Query the agent on its n owned queries and evaluate the formula on the resulting policy. By optimality the policy is satisfying whenever a satisfying assignment exists, so this decides satisfiability in polynomial time (Cook 1971). For the second claim, let action 1 evaluate the formula on a uniformly random assignment and pay 1 if it is

satisfied, and let action 0 pay 1 with probability $\frac{1}{2} + 2^{-(n+1)}$. The value of action 1 is a multiple of 2^{-n} and that of action 0 is not, so there are no ties, and action 1 is optimal iff more than half of the assignments satisfy the formula. This decides the PP-complete problem MAJSAT (Gill 1977). □

Each play of these environments is short; the difficulty lies in comparing alternatives, all of whose executions must be accounted for. A resource-bounded form of Theorem 15.6 holds against an agent T whenever the environments' budget accommodates self-description, a simulation of T on that description, and the final comparison. If the budget cannot accommodate the simulation, the diagonal gives no lower bound, so the useful resource condition is that the environments can run the agent, not a ranking of agents and environments by a single bound.

Per-input runtime bounds do not decide program equivalence either; the construction of Theorem 17.5 uses components bounded by $|x|$ steps on input x. The bounded variants of Löb's theorem proved by Critch (2019) yield positive cooperation results for specified proof-responsive programs under proof-length assumptions; they do not establish optimality on a class of environments of the present kind.

16.8. What a fixed semantics settles

Theorem 15.5 extends at once to any environment class that supplies a *reference-independent* effective evaluator: a computable $V(E, \pi)$ on a finite policy space, with exactly comparable values, whether the evaluator works by source analysis, by simulation, or by proof. Finite circuits, explicitly clocked programs with stipulated time-out behaviour, and descriptions carrying checkable certificates of the required evaluations all have such evaluators, and admit optimisers that are globally total on the certified language, returning a default on invalid strings. One might conclude that once a rule for propagating the dependence of the world on the agent's output is fixed, optimal choice is a finite computation, and the only remaining question is which rule.

The conclusion is false for anchored kernels. A fixed, computable kernel makes each comparison computable, but it need not admit an agent whose actual output maximises it. In Proposition 15.9 under O_{const} every baseline yields a finite, unambiguous ranking, and every baseline is ranked below its alternative. Maximising a supplied table V_E is a finite computation; being a fixed point of a baseline-dependent kernel is a separate problem, the ratifiability problem of Part I in program form.

Two further remarks fix the scope of "largest solvable class". For reference-independent problems with disjoint owned queries, every decidable class on which some globally total agent is optimal can be enlarged by a single further problem, by altering the agent on that problem's owned queries only; under $\mathcal{F}_{\text{self}}$ this changes nothing on the original class, so there is no largest such class. Conversely, the class of all terminating, rational-valued, finite-policy templates is a maximal promise domain for that representation, so an unqualified claim that there is no largest solvable class would also be false.

Sections 15 and 16 thus answer the question for a designated semantics completely. Optimal choice exists problem by problem; an optimal function is non-computable exactly when it must halt everywhere or compare computable reals exactly, and then its degree is PA; every further demand on it raises the degree without touching the choice among alternatives. Everything in this answer presupposed that the substitution interface was given. The remaining sections remove it.

17. Inferring dependence from source

Theorem 15.5, and the whole degree analysis of Section 16, work because oracle substitution designates the positions at which the agent is consulted. In an embedded environment, a closed program whose source may contain predictors, simulations, or copies of the agent with no designated interface, the agent must identify those positions itself. This section examines the three natural ways of doing so: recognising the agent's code, recognising programs that compute the same function, and reasoning about the agent's output in a formal theory. Each has a positive theorem on a restricted domain and a characteristic failure outside it.

17.1. Designated slots and literal copies

Suppose an environment is a template with designated agent slots, that syntactic replacement changes exactly those slots, that every slot uses the same owned-query namespace, that all other programs are constants, and that every finite-policy replacement terminates with an exact rational payoff. Compiling the oracle calls of Theorem 15.5 to these slots preserves every counterfactual execution, so

$$U_{\mathrm{syn}}(E, T; A) = U_{\mathrm{oracle}}(E, A),$$

and the uniform optimiser applies. This is a substantive positive answer: the rule evaluates arbitrary code implementing the environment without analysing every consequence of its action; it executes the stipulated replacements. It is also a restricted answer, since the representation identifies the slots.

Without designated slots, literal recognition fails. Fix a globally total T and let g be a closed program with $g(x) = T(x)$ for every x, encoded independently in a typed, atomic component so that its representation contains no eligible occurrence of T's code. (Wrapping T is not enough, since the original would remain an eligible nested occurrence.) Let the environment pay $\mathbf{1}[a \neq g(d(E))]$. Literal substitution changes the live agent and leaves g unchanged, so T receives 0 while an agent taking the opposite action receives 1. Every computation terminates. This is the fixed-copy diagonal of Theorem 15.6 again, now evading a syntactic matcher. Literal matching also violates $\mathcal{F}_{\mathrm{ext}}$: two extensionally equal reference agents with different source code produce different replacement masks. A slot-based representation avoids the problem by preserving the agent's role independently of the code that currently occupies it; code equality and role identity should not be conflated.

17.2. Extensional matching

The natural repair is to replace every component that computes the same function as the agent. To make this precise, let E contain a finite list of typed, atomic components g_j, fix comparison domains X_j and a reference function t, and let

$$S_t(E) = \{j : g_j|_{X_j} = t|_{X_j}\}.$$

To evaluate A, replace exactly the components in $S_t(E)$ by A and execute; write $U_{\text{ext}}(E, t; A)$ for the result. Matching occurs once, before replacement; component boundaries and the description supplied to the agent stay fixed; and every permitted counterfactual execution is required to terminate, which termination of the original environment does not imply. The principal choices are *local* equality, on a specified finite set or a frozen baseline trace, and *global* equality, on the entire input domain.

Proposition 17.1 (Matching on the counterfactual trace). *Requiring replacement exactly when a component agrees with t on the inputs used in the resulting counterfactual execution can leave no consistent replacement set, although every candidate execution terminates.*

Proof. Let $t(0) = t(1) = 0$, $g(0) = 0$, $g(1) = 1$, and $A(0) = 1$, and let the environment call $g(0)$ and then call $g(1)$ exactly when the first result is 1. Without replacement only the input 0 is used, where g agrees with t, so replacement is required. With replacement, $A(0) = 1$ triggers the call on 1, where g disagrees with t, so replacement is forbidden. □

"The inputs it is run on" must therefore name the execution that determines those inputs. Equality of total programs on a fixed finite set is decidable by running them, so local matching needs no appeal to Rice's theorem. It fails for a different reason.

Theorem 17.2 (Local equality can leave no optimal agent). *Under frozen matching on the single input $d(E)$, there is a terminating environment satisfying $\mathcal{F}_{\text{self}}$ and $\mathcal{F}_{\text{ext}}$ on which no deterministic agent satisfies anchored optimality* (2).

Proof. Let E contain the components $g_0 \equiv 0$ and $g_1 \equiv 1$, call both at $d(E)$, query the agent there, and pay 1 exactly when the two component outputs agree. For a reference agent with $t(d(E)) = a$, exactly g_a matches. With t itself substituted, the outputs are a and $1 - a$, so $U_{\text{ext}}(E, t; t) = 0$. With an agent A taking $1 - a$, the matched component g_a becomes A and both outputs equal $1 - a$, so $U_{\text{ext}}(E, t; A) = 1$. □

This is not an undecidability theorem. It is a pathology of identifying computations through accidental agreement on one input, and it should not be presented as a theorem about functional decision theory without that identification.

Proposition 17.3 (Rematching does not compose). *Let $R_{t\to a}$ replace by a the components that match t. With components $g_0 \equiv 0$, $g_1 \equiv 1$ and reference $t \equiv 0$, and with matches recomputed after each replacement, $R_{a\to b}(R_{t\to a}(E)) \neq R_{t\to b}(E)$ for $a \equiv 1$ and $b \equiv 0$.*

Proof. $R_{t\to a}$ turns g_0 into 1, giving the pair $(1,1)$. Both components now match a, and $R_{a\to b}$ gives $(0,0)$. Direct replacement of t by $b = t$ leaves $(0,1)$. □

Keeping an original dependency label attached to the first component restores composition, but the label is additional structure; equality of the current functions does not recover it. Global matching avoids accidental agreement, but it identifies the wrong thing.

THEOREM 17.4 (Global equality leaves the diagonal intact). *For globally total agents, global-equivalence replacement of atomic components admits no universal optimiser when fixed algorithms are allowed.*

Proof. Fix T, compile an atomic program $h_T(x) = 1 - T(x)$, and let the environment pay $\mathbf{1}[a = h_T(d(E))]$. Since h_T disagrees with T everywhere, it is not replaced. T receives 0, an agent taking the opposite action receives 1, and all calls terminate. □

The diagnosis is that "depends functionally on T" is not "is extensionally equal to T". If the intended counterfactual turns $1 - T$ into $1 - A$, it transports dependence through a program context, which is a stronger operation than equality-based replacement. Global matching also brings in undecidability, and the expected route needs one correction. For a uniformly total family g_e and a fixed total T, non-equivalence has a finite witness, so $\{e : g_e = T\}$ is co-c.e.; one cannot uniformly make equality hold exactly when a machine halts, but one can make it hold exactly when it does not.

THEOREM 17.5 (Global matching can decide halting). *Assume atomic global matching, frozen descriptions, and globally total binary agents, and let P be a decidable class of descriptions that contains all the environments constructed below and omits arbitrarily long strings. No agent satisfies anchored optimality on all environments in P.*

Proof. Fix a candidate T and let

$$g_e(x) = \begin{cases} 1 - T(x), & x \notin P \text{ and machine } e \text{ halts within } |x| \text{ steps}, \\ T(x), & \text{otherwise.} \end{cases}$$

Each g_e is total, and $g_e = T$ exactly when e does not halt. Let E_e, with $d_e = d(E_e) \in P$, receive the action a, compute $b = g_e(d_e)$, and pay $u(a,b) = 2(1-b) + a$:

	action 0	action 1
$b = 0$	2	3
$b = 1$	0	1

Because $d_e \in P$, $g_e(d_e) = T(d_e)$ whether or not e halts. If e does not halt, global matching replaces g_e by the candidate A, so $b = A(d_e)$ and the values of the two actions are 2 and 1; optimality requires $T(d_e) = 0$. If e halts, g_e stays fixed, and with b held fixed action 1 pays one unit more; optimality requires $T(d_e) = 1$. The computable map $e \mapsto T(d_e)$ would decide the halting problem. □

The payoff table is Newcomb's, and the reduction works by making it undecidable whether its predictor counts as linked to the agent. Every execution terminates and every payoff is rational, but the result uses global equality, atomic components compiled from T, and a class already admitting fixed-constant diagonals; it satisfies executed-query locality and violates kernel locality. It is a valid reduction, not a semantics-independent impossibility arising from the mere presence of embedded programs. Under local matching g_e always matches at d_e, so action 0 always wins and the reduction disappears. Two further qualifications bound any positive subclass. Clocks do not decide global equivalence, since even a program that simulates machine e for only $|x|$ steps can equal zero everywhere exactly when e never halts. And "replace when a theory S proves equality" changes the semantics: it agrees with global matching only where certified matching information is complete, since positive certificates do not show that uncertified components are unequal.

17.3. Proofs and logical counterfactuals

Three questions must be separated: whether proofs can establish the values of independently specified counterfactual programs; whether implications about the actual agent can serve as counterfactual values; and whether logical uncertainty supplies useful finite-time estimates. The answers are yes, no, and not by itself.

Proposition 17.6 (Ground proof-based optimisation). *Let S be a sound, effectively axiomatised arithmetic theory that verifies finite program traces. The agent that, on an environment in $\mathcal{P}_{\mathbb{Q}}$, enumerates S-proofs until it has, for every $\pi \in \Pi_E$, a proof of* $\mathrm{Eval}(E, \pi) = r_\pi$ *for some rational r_π, and then maximises the r_π, is promise-total and optimal under O_{oracle}.*

Proof. Each terminating execution E^π has a finite trace, the trace has a proof in S, and soundness makes every proved value correct. Only finitely many proofs are needed. □

The incompleteness of a fixed theory does not prevent it from verifying every particular terminating computation; what it may fail to prove is a uniform termination statement or a global equivalence. The proposition evaluates the same counterfactual programs as Theorem 15.5 and infers no new semantics from proof search.

Material implication is different. Write H_a for $T(d(E)) = a$ and Y_u for $U(E, T) = u$. If $S \vdash \neg H_a$, then $S \vdash H_a \to Y_u$ for every u, so the relation

$$\mathcal{V}(E, T, a) = \{u \,:\, (H_a \to Y_u) \text{ is true}\}$$

contains only the actual payoff at the actual action and every rational at every other action. It is a relation, not an operator; any numerical counterfactual drawn from it needs a selection principle that the implication does not supply. For an actually terminating computable agent, a sufficiently strong theory verifies the agent's execution and so refutes every non-actual antecedent, and conditioning a coherent probability over its models on such an antecedent is undefined. Proof search can

also select a spurious value, as in the "five-and-ten" problem (Demski and Garrabrant 2019, §2.1), which can be realised by a total program.

THEOREM 17.7 (A computable five-and-ten failure). *Let E_0 pay the action chosen from $\{5, 10\}$, and let S be an effectively axiomatised theory satisfying the derivability conditions for Löb's theorem (Löb 1955). There is a globally total agent T that, on $d(E_0)$, takes* 5 *because it has found a proof that taking* 10 *yields* 0*. It receives* 5*, while the constant agent* 10 *receives* 10*.*

Proof. By the recursion theorem let T return 10 on descriptions other than $d(E_0)$, and on $d(E_0)$ enumerate S-proofs until it finds a proof of

$$\psi = \big(T(d(E_0)) = 5 \to U(E_0, T) = 5\big) \wedge \big(T(d(E_0)) = 10 \to U(E_0, T) = 0\big),$$

and then return 5. Let q be the sentence $T(d(E_0)) = 5$. The definitions of T and E_0 give $S \vdash q \to \psi$, since under q the first conjunct holds and the second is vacuous. Hence $S \vdash \Box_S q \to \Box_S \psi$. Formalising the search gives $S \vdash \Box_S \psi \to q$. So $S \vdash \Box_S q \to q$, and Löb's theorem yields $S \vdash q$, whence $S \vdash \psi$. A proof of ψ therefore exists, the search halts, and T returns 5. □

This refutes the displayed implication-based selection scheme, not every logical-counterfactual procedure. Benson-Tilsen (2014, §2) analyses the mechanism, and his classification of outcomes by proof-search order concerns a specified transparent-universe model with a provability oracle; it should not be quoted as a theorem about every computable proof-based agent. A different response, the "chicken rule", has the agent take any action it can prove it will not take (Slepnev 2011); within Benson-Tilsen's model it is the only universally successful search strategy (Benson-Tilsen 2014, Corollary 4.6), and it blocks spurious proofs at the cost of counterfactuals that can differ from what another agent would actually obtain (Fallenstein 2015a).

A precise logical semantics that avoids spurious values does so by becoming partial.

DEFINITION 17.8 (Guarded modal counterfactuals). Let (A_i) and (U_j) be closed modal formulas, provably mutually exclusive and exhaustive, describing the actions and outcomes of an agent and a problem. Action i *leads to* outcome u_j if

$$\mathrm{GL} \vdash A_i \to U_j \qquad \text{and} \qquad \mathrm{GL} \nvdash \neg A_i,$$

and has no value otherwise (Fallenstein 2015b).

The outcome is unique when defined and is undefined if the action is provably impossible or no outcome is determined; the semantics "fails visibly" rather than assigning a spurious value. Fallenstein uses it to state the optimality of modal updateless decision theory on problems where every action has a value (Fallenstein 2015b). This is optimality relative to the specified counterfactuals, not relative to physical agent replacement, and the two can diverge: on a variant of five-and-ten where every action has a value, these counterfactuals certify as optimal an action

that another agent would beat (Fallenstein 2015a). In provability logic GL (Boolos 1993) the guard is informative. For ordinary programs it is not.

PROPOSITION 17.9 (Guarded counterfactuals for terminating programs). *Replace* GL *by an arithmetic theory S that verifies terminating computations and proves the determinism of its interpreter. For an agent that terminates on d, every non-actual action has no value.*

Proof. If the actual action is b, the finite execution trace gives $S \vdash T(d) = b$, and determinism gives $S \vdash T(d) \neq a$ for every $a \neq b$, so every non-actual action fails the guard. □

This reconstruction is not attributed to the source of Definition 17.8; it explains why positive results for modal agents, or agents with provability oracles, do not transfer directly to ordinary terminating programs.

The modal setting has its own diagonal and its own optimality theorem, and their quantifiers matter. Consider fixed, fully modalised decision schemata $T(\mathbf{u})$ and problems that are *provably extensional*, in the sense that $\mathrm{GL} \vdash (\mathbf{a} \leftrightarrow \mathbf{b}) \rightarrow (P(\mathbf{a}) \leftrightarrow P(\mathbf{b}))$. Every such fixed theory with at least two actions and two outcomes fails on some provably extensional problem (Fallenstein 2014a): the problem rewards any action other than the one the theory takes when facing the constant second-best outcome, modal fixed-point uniqueness makes that action the theory's own, and a different constant action wins. This is the fixed-constant diagonal in modal form. Conversely, for every finite provably extensional problem P there is a true sentence φ such that modal updateless decision theory using $\mathrm{PA} + \varphi$ selects an optimal action on P (Fallenstein 2014b,c). The quantifiers are $\forall P\, \exists \varphi$: there is no single proof system that works for all problems, and the general procedure of Soares and Fallenstein (2015) that queries proofs uses a halting oracle. GL itself is decidable, so none of this shows that restricted modal computation is uncomputable.

Logical induction (Garrabrant et al. 2016) supplies computable, improving assessments of logical sentences. It does not specify which counterfactual program or intervention to evaluate, and it imposes no exact constraint on any finite stage: by its Theorem 4.6.1, changing finitely many market days preserves the logical induction criterion. A decision procedure can use an inductor to estimate the values of policy-substituted programs, but Theorem 15.10 still applies to any resulting computable rule. Decision procedures built on logical induction typically add *exploration*, taking an action whenever its probability falls below a threshold so that the conditionals on it remain defined (Garrabrant 2017); Section 18.8 shows that what such a procedure computes depends on the exploration scheme it is given.

17.4. What inference supplies

The pattern across these results is uniform. Wherever a substitution is designated—by an oracle interface, by slots, or by an evaluator whose results a proof certifies—optimisation is computable on the corresponding termination promise. Wherever the agent must decide for itself which parts of a closed program would change with its

output, each recognition rule fails in its own way: literal matching misses encoded copies, local matching manufactures dependence from accidental agreement, global matching misses transformed dependence and can be undecidable, and implication-based reasoning about the actual agent yields either spurious values or none. None of these failures is a theorem that no coherent rule exists. They show that the rule is the open problem, which the next section takes up axiomatically.

18. Axioms for counterfactual operators

Section 17 leaves one question: is there a coherent way to assign outcomes to the non-actual outputs of a computation? This is the problem of *logical counterfactuals*. This section states candidate desiderata precisely and determines every consistent and inconsistent combination of them. There is a sharp impossibility among three individually satisfiable axioms, and it identifies exactly which notion of program equivalence is responsible. Under a different notion, all the desiderata hold together.

18.1. Preliminary repairs

Three features of a naive formulation produce failures before the substantive question is reached. They are recorded here and excluded below by stipulation.

Transported contexts need not terminate. Let a counterfactual operator be required to *transport* dependence: a component $C[T]$ that computes a context C applied to the agent is evaluated, under the alternative, as C applied to the altered agent. It is natural to promise only that the host environment terminates when its components are replaced by tables. That promise is too weak. Let $T \equiv 0$, let $C[P](x)$ return 0 if $P(x) = 0$ and diverge if $P(x) = 1$, treat $g = C[T]$ as one opaque component, and let the host return $g(d)$. The host terminates under every whole-component table replacement, and g is the total constant 0, yet the transported execution $C[T^{d \leftarrow 1}](d)$ diverges. On this domain totality and execution-faithful transport are already inconsistent. The domain is therefore required to be *intervention-total*: every supplied computational context terminates under every permitted substitution, as can be ensured by finite tables, a terminating language, certificates, or an explicit promise. Atomic boundaries matter, since replacing an opaque outer component by a table can hide divergence inside it.

Source-changing replacement changes inputs.

Proposition 18.1. *If the description supplied to the agent is the literal source and replacing a component changes it, baseline correctness is inconsistent with invariance under replacement of a component by an extensionally equal one.*

Proof. Let E ignore the agent's action and return $g(d(E))$ with $g(x) = |x| \bmod 2$. Replacing g by an extensionally identical implementation padded by one character changes the parity of the source, and hence the actual payoff, while a constant agent takes the same action in both versions. □

The components are equal but are called on different inputs. Below, comparisons of implementations preserve the distinguished inputs, and counterfactual transformations retain the original observation.

Single-input transport changes one input. An intervention at d replaces T by

$$T^{d\leftarrow a}(x) = \begin{cases} a, & x = d, \\ T(x), & x \neq d, \end{cases}$$

and strong transport evaluates $C[T^{d\leftarrow a}]$. A context $C[T] = T \circ h$ does not in general become the constant a: if $h(d) \neq d$, then $C[T^{d\leftarrow a}](d) = T(h(d))$. Replacing the whole agent by a constant is a different, policy-level intervention.

18.2. The finite context domain

The main theorems use a finite domain in which termination and program equivalence raise no difficulty, so that what remains is the semantic question alone.

Definition 18.2 (Context environment). A *context environment* is $E = (u; c_1, \dots, c_k)$, where the action set is $A = \{0, 1\}$, each component $c_i : A \to B_i$ is a table with values in a finite set B_i, and the host $u : A \times \prod_i B_i \to \mathbb{Q}$ is a finite payoff table. The argument of c_i is the *live parameter*: the agent's output at the distinguished input, as seen by component i. E is *independent* if every c_i is constant. Write

$$V_E(a) = u\big(a, c_1(a), \dots, c_k(a)\big), \qquad W_{E,b}(a) = u\big(a, c_1(b), \dots, c_k(b)\big).$$

V_E is coherent substitution through every context; $W_{E,b}$ changes the action at the port and freezes every component at its value under the baseline b. The definition of independence is not circular: it is invariance under every substitution of the declared live parameter, not the absence of an undefined relation of "depending on T". The domain is the causal-matrix domain with its factorisation made explicit.

Proposition 18.3. *Let $M_E[b, a] = W_{E,b}(a)$. Then M_E is the causal matrix of the one-shot problem in which an infallible predictor forecasts the agent's output b and the components respond to the forecast, and $M_E[a, a] = V_E(a)$. Every real 2×2 matrix with rational entries is M_E for some context environment, and E is independent exactly when all rows of M_E coincide.*

Proof. With structural states $b \in A$, $\mu_a = \delta_a$, and $U(b, a) = u(a, c(b))$, the causal matrix is $\sum_s \mu_b(s)U(s, a) = W_{E,b}(a)$. Given M, take one identity component $c(z) = z$ and $u(a, x) = M[x, a]$. If every c_i is constant, $W_{E,b}$ does not depend on b; if some row differs from another, some component's value differs between the two baselines. □

Strictly, the converse in the last clause holds for the matrix, not for the environment: distinct component values may be invisible to the host. Nothing below uses the converse.

Definition 18.4 (Counterfactual kernel). A *kernel* assigns to every context environment E, baseline $b \in A$, and action $a \in A$ a value $\kappa(E, b, a) \in \mathbb{Q} \cup \{\bot\}$, where $\bot$ means undefined. The baseline is the agent's actual output, $b = T(d(E))$. A deterministic agent τ on context environments is *κ-ratifiable* at E if $\kappa(E, \tau(E), \cdot)$ is defined at every action and is maximised at $\tau(E)$.

That the kernel depends on T only through the local baseline is an explicit stipulation; the program extension in Section 18.7 keeps the same interface. Agent extensionality is then automatic, because agents computing the same function have the same baseline. Two notions of extensionality for the environment must be distinguished.

Definition 18.5 (Grounded and parametric equivalence). $E \sim_b E'$ (*grounded equivalence at b*) if E and E' have the same host and interfaces and $c_i(b) = c_i'(b)$ for every i. $E \approx E'$ (*parametric equivalence*) if they have the same host and interfaces and $c_i = c_i'$ as functions on A.

Grounded equivalence has a direct reading for programs. Once the actual agent is inserted, a component is a closed computation returning $c_i(b)$, and two closed computations can be extensionally equal although their open contexts would respond differently to another action. The function $x \mapsto T(x)$ and the constant function with value $T(d)$ agree at d, and after insertion of T both are closed programs with output $T(d)$. Parametric equivalence identifies contexts only when they agree under every substitution, and keeps the information that grounded equivalence discards.

Definition 18.6 (Desiderata). A kernel may satisfy the following conditions, for all context environments E, E' and all $a, b \in A$.

CF1 *Baseline.* $\kappa(E, b, b) = V_E(b)$.

CF2 *Totality.* $\kappa(E, b, a) \neq \bot$.

CF3 *Independent calibration.* If E is independent, $\kappa(E, b, a) = V_E(a)$.

CF4 *Transport.* If $\kappa(E, b, a)$ is defined, it equals $V_E(a)$.

CF5 *Grounded extensionality.* If $E \sim_b E'$, then $\kappa(E, b, a) = \kappa(E', b, a)$, including equality of definedness.

CF6 *Conservativity.* Every defined value has the form $u(a, z_1, \dots, z_k)$ with $z_i = c_i(b)$ for every constant context c_i.

CF7 *Composition.* A transformed environment is the record $(E, b; a)$, whose evaluator returns $\kappa(E, b, a)$, and a further intervention overwrites the target: $(E, b; a)[a \leftarrow z] = (E, b; z)$.

CF8 *Computability.* κ, with $\bot$ returned explicitly, is computable.

CF9 *Realisability.* Every $\kappa(E, b, a)$ is defined and is the payoff that some agent taking a obtains under coherent substitution of the live parameter.

CF10 *Ratifiability.* Some computable τ is κ-ratifiable at every E.

Three remarks fix the reading. CF4 transports the whole supplied context and preserves the host, and it is conditional on definedness; a version requiring the transported values to exist would incorporate CF2. Every agent taking a obtains exactly $V_E(a)$ under slot substitution, and constant agents supply the alternatives, so

$$\mathsf{CF9} \iff \mathsf{CF2} + \mathsf{CF4}.$$

And CF7 is deliberately weak: every kernel has such a lift, since the record exists even when its evaluator returns $\bot$. A history-free requirement, or one demanding fresh rematching after each intervention, would be a different axiom (Section 18.8). CF6 constrains only the constant contexts; how the others respond is left open. On the finite domain CF8 holds for every kernel defined by a finite recipe from the tables, and CF7 for every kernel; both are recorded because they become substantive on program domains.

18.3. Four kernels

Definition 18.7 (Witness kernels).

$$\kappa_{\text{slot}}(E, b, a) = V_E(a), \qquad \kappa_{\text{port}}(E, b, a) = W_{E,b}(a),$$

$$\kappa_{\text{base}}(E, b, a) = \begin{cases} V_E(b), & a = b, \\ \bot, & a \neq b, \end{cases} \qquad \kappa_{\text{flat}}(E, b, a) = V_E(b).$$

By Proposition 18.3, κ_{slot} reads the diagonal of the causal matrix and κ_{port} reads row b: they are ex-ante and causal evaluation, now appearing as counterfactual semantics. κ_{base} values only the actual output, as partial semantics do; κ_{flat} assigns the actual payoff to every alternative, so every agent is ratifiable under it. All four satisfy CF1, CF7, and CF8. Their other properties are as follows.

Lemma 18.8. *κ_{slot} satisfies every desideratum except* CF5*, and the agent choosing the first maximiser of V_E ratifies it. κ_{port} satisfies* CF2*,* CF3*,* CF5*, and* CF6*. κ_{base} satisfies* CF4*,* CF5*, and* CF6*. κ_{flat} satisfies* CF2*,* CF5*, and* CF10*.*

Proof. κ_{slot} is total and equals V_E, which gives CF2–CF4 and CF9; its values $u(a, c(a))$ have $c_i(a) = c_i(b)$ for constant contexts, giving CF6; the first maximiser of V_E is computable from the tables. κ_{port}, κ_{base}, and κ_{flat} depend on E only through u and the baseline outputs $c_i(b)$, which grounded equivalence preserves, so all three satisfy CF5. κ_{port} is total; on an independent environment $W_{E,b} = V_E$; and its values have the form required by CF6. κ_{base} is defined only at $a = b$, where its value $V_E(b) = u(b, c(b))$ satisfies CF4 and CF6. κ_{flat} is total and constant in a, so every agent ratifies it. □

Nothing more holds: κ_{port} violates CF4, CF9, and CF10 on the environments of Theorems 18.10 and 18.11 below; κ_{base} is not total and so violates CF2, CF3, CF9, and CF10; and κ_{flat} violates CF3, CF4, CF6, and CF9 on any independent environment whose two actions have different payoffs.

18.4. The projection lemma and two theorems

For a context environment E and a baseline b, let E^b replace every component by the constant with its baseline value, $c_i^b(a) = c_i(b)$. Then $E \sim_b E^b$, E^b is independent, and $V_{E^b}(a) = W_{E,b}(a)$.

LEMMA 18.9 (Projection). CF3 *and* CF5 *together imply* $\kappa = \kappa_{\text{port}}$. CF5 *and* CF6 *together imply* $\kappa(E, b, a) = W_{E,b}(a)$ *whenever* $\kappa(E, b, a)$ *is defined.*

Proof. CF3 defines the values at E^b as $V_{E^b}(a) = W_{E,b}(a)$, and CF5 transfers them, with their definedness, to E. For the second claim, transfer a defined value to E^b by CF5. Every component of E^b is constant, so CF6 fixes every component value at $c_i(b)$, and the value is $u(a, c(b)) = W_{E,b}(a)$. □

This is the main structural fact: *calibration together with the erasure of dependency information forces causal evaluation.* The two semantic requirements reproduce, without any appeal to decision-theoretic intuition, the frozen-environment counterfactual that C3 uses.

THEOREM 18.10 (Transport and grounded equivalence). {CF2, CF4, CF5} *is minimally inconsistent.*

Proof. Let $u(a, x) = x$, and compare $c(z) = z$ with $c'(z) = 0$ at the baseline 0. The instantiated component outputs agree, so $E \sim_0 E'$, but $V_E(1) = 1$ and $V_{E'}(1) = 0$. CF2 defines both values, CF4 makes them unequal, and CF5 makes them equal. Omitting CF2, CF4, or CF5, the remaining pair is satisfied by κ_{base}, κ_{port}, or κ_{slot} respectively. □

No computability or undecidability enters; the obstruction is the loss of information under an equivalence relation. The second theorem connects that loss with the instability of Part I.

THEOREM 18.11 (Calibration and ratifiability). {CF3, CF5, CF10} *is minimally inconsistent, and each of its proper subsets is satisfied by a total, computable kernel satisfying* CF1, CF2, CF7, *and* CF8.

Proof. Take the copied-action environment $u(a, x) = \mathbf{1}[a \neq x]$, $c(z) = z$. For either baseline, $W_{E,b}(b) = 0$ and $W_{E,b}(1 - b) = 1$. By Lemma 18.9 CF3 and CF5 force this table, so every actual action recommends its opposite, contradicting CF10. Omitting CF3, CF5, or CF10, the remaining pair is satisfied by κ_{flat}, κ_{slot}, or κ_{port}, all total and computable. □

The witness environment is Death in Damascus with an infallible predictor (Proposition 4.3). That no pure action is ratifiable there is familiar; what the theorem adds is a derivation of the causal kernel from two semantic requirements that make no reference to causation, and the exact minimality of the triple. Unlike the within-problem maps of Section 5, it is a minimal inconsistent family of three individually satisfiable conditions.

18.5. The complete map

THEOREM 18.12 (Consistency map). *On the finite context domain, the inclusion-minimal inconsistent subsets of* {CF1, … , CF10} *are exactly*

$$\{\mathsf{CF5}, \mathsf{CF9}\}, \quad \{\mathsf{CF2}, \mathsf{CF4}, \mathsf{CF5}\}, \quad \{\mathsf{CF3}, \mathsf{CF4}, \mathsf{CF5}\},$$
$$\{\mathsf{CF3}, \mathsf{CF5}, \mathsf{CF10}\}, \quad \{\mathsf{CF4}, \mathsf{CF5}, \mathsf{CF10}\}, \quad \{\mathsf{CF5}, \mathsf{CF6}, \mathsf{CF10}\},$$

and the maximal consistent subsets are exactly

witness	maximal consistent subset
κ_{slot}	all except CF5
κ_{port}	{CF1, CF2, CF3, CF5, CF6, CF7, CF8}
κ_{base}	{CF1, CF4, CF5, CF6, CF7, CF8}
κ_{flat}	{CF1, CF2, CF5, CF7, CF8, CF10}

Proof. Inconsistency. {CF5, CF9}: CF9 gives CF2 and CF4, and Theorem 18.10 applies. {CF3, CF4, CF5}: in the environment of Theorem 18.10, CF3 sets the value of action 1 in the constant-context environment E' to 0, CF5 transfers it to E, and CF4 requires 1. {CF4, CF5, CF10}: take $u(a, x) = x$, $c(z) = z$, and let CF10 select b. The alternative $1 - b$ is defined; CF5 transfers definedness to E^b, whose component is the constant b, and CF4 requires the values $1 - b$ at E and b at E^b, which CF5 makes equal. {CF5, CF6, CF10}: by Lemma 18.9, defined values are those of κ_{port}, and CF10 defines both values at the selected baseline in the environment of Theorem 18.11. The other two sets are Theorems 18.10 and 18.11.

Minimality. Deleting the members of {CF3, CF4, CF5} in turn leaves pairs satisfied by κ_{base}, κ_{port}, and κ_{slot}; for {CF4, CF5, CF10} by κ_{flat}, κ_{slot}, and κ_{base}; for {CF5, CF6, CF10} by κ_{slot}, κ_{flat}, and κ_{port}. CF5 and CF9 are satisfied singly by κ_{port} and κ_{slot}. The remaining two cases are in Theorems 18.10 and 18.11.

Exhaustiveness. Let Σ contain none of the six sets. If $\mathsf{CF5} \notin \Sigma$, κ_{slot} witnesses Σ. Otherwise $\mathsf{CF9} \notin \Sigma$. If $\mathsf{CF3} \in \Sigma$, then CF4 and CF10 are excluded and κ_{port} witnesses Σ. If $\mathsf{CF4} \in \Sigma$ and $\mathsf{CF3} \notin \Sigma$, then CF2 and CF10 are excluded and κ_{base} witnesses Σ. If neither CF3 nor CF4 is in Σ but CF10 is, CF6 is excluded and κ_{flat} witnesses Σ; if CF10 is also absent, κ_{port} does. Each displayed set is maximal, since adding any further desideratum completes one of the six cores. □

Every core contains CF5, and deleting CF5 leaves a single kernel satisfying everything else.

COROLLARY 18.13 (A coherent semantics). *If* CF5 *is replaced by* parametric extensionality, $E \approx E' \Rightarrow \kappa(E, b, a) = \kappa(E', b, a)$, *then* κ_{slot} *satisfies all ten desiderata, and the agent choosing the first maximiser of* V_E *ratifies it.*

Proof. V_E depends on E only through u and the functions c_i, which parametric equivalence preserves; the rest is Lemma 18.8. □

The impossibility therefore concerns the erasure of dependency information, not extensionality in every sense. The same positive model arises from a supplied

dependency graph: a finite acyclic graph of typed variables with total structural equations, a designated node for the decision computation, and fixed exogenous data. Intervening on that node and recomputing its descendants, in the manner of structural causal models (Pearl 2009), retains the independent mechanisms and yields the finite table V_E. The graphical formulation of functional decision theory intervenes in exactly this way on a node representing the agent's computation (Yudkowsky and Soares 2017, §§5–6); what the formulation does not provide, and what no axiom here supplies, is a procedure that extracts the graph from an arbitrary closed program.

In matrix terms the map says the following. Grounded extensionality allows the kernel to see only the host and the baseline outputs of the components, that is, the row of the causal matrix belonging to the actual output, together with the payoff table. Calibration then pins the kernel to that row (Lemma 18.9), and Theorem 18.11 is the observation that the diagonal entry need not be maximal in its row. Parametric extensionality allows the kernel to see the whole matrix, and then the diagonal is available. The choice between causal and ex-ante evaluation of Part I, which there was a choice between two laws, reappears here as a choice between two notions of when two computations are the same.

18.6. Other conventions

The map depends on how undefined values and realisability are treated, and these conventions are not cosmetic. Suppose realisability is weakened to CF9′: *whenever* $\kappa(E, b, a)$ is defined, it is attainable under slot substitution, so that CF9′ is equivalent to CF4. Suppose ratifiability is weakened to CF10′: some computable τ has $\kappa(E, \tau(E), \tau(E))$ defined and maximal among the defined values.

Proposition 18.14. *With* CF9′ *and* CF10′ *in place of* CF9 *and* CF10*, the minimal inconsistent subsets are exactly*

$$\{\mathsf{CF2}, \mathsf{CF4}, \mathsf{CF5}\}, \quad \{\mathsf{CF2}, \mathsf{CF5}, \mathsf{CF9'}\}, \quad \{\mathsf{CF3}, \mathsf{CF4}, \mathsf{CF5}\},$$
$$\{\mathsf{CF3}, \mathsf{CF5}, \mathsf{CF9'}\}, \quad \{\mathsf{CF3}, \mathsf{CF5}, \mathsf{CF10'}\}, \quad \{\mathsf{CF2}, \mathsf{CF5}, \mathsf{CF6}, \mathsf{CF10'}\},$$

and the maximal consistent subsets are those of Theorem 18.12 with the primed axioms substituted, except that κ_{base} *now witnesses*

$$\{\mathsf{CF1}, \mathsf{CF4}, \mathsf{CF5}, \mathsf{CF6}, \mathsf{CF7}, \mathsf{CF8}, \mathsf{CF9'}, \mathsf{CF10'}\}.$$

Proof. κ_{base} satisfies CF9′ and, with any computable τ, CF10′, so the cores $\{\mathsf{CF4}, \mathsf{CF5}, \mathsf{CF10}\}$ and $\{\mathsf{CF5}, \mathsf{CF6}, \mathsf{CF10}\}$ do not survive. The cores containing CF4 are duplicated with CF9′, which is equivalent to it. CF3 and CF5 make κ total and equal to κ_{port}, which fails CF10′ in the copied-action environment. CF2, CF5, and CF6 likewise give κ_{port} by Lemma 18.9. The four-element core is minimal: deleting CF2, CF5, CF6, or CF10′ leaves a set satisfied by κ_{base}, κ_{slot}, κ_{flat}, or κ_{port}. For exhaustiveness, let Σ be core-free and contain CF5 (otherwise κ_{slot} witnesses it). If $\mathsf{CF3} \in \Sigma$, κ_{port} witnesses it. If CF4 or CF9′ is in Σ, CF2 is excluded and κ_{base} witnesses it. Otherwise, if CF10′ is in Σ, either CF6 is absent and κ_{flat} witnesses Σ, or CF2 is absent and κ_{base} does; if CF10′ is absent, κ_{port} does. □

The four-element core reflects the splitting of "defined and calibrated" into separate requirements. Theorem 18.11 is less sensitive to such bookkeeping, since its witnesses are all total.

The word "realisable" also needs a benchmark. If realisability freezes every embedded component at its reference implementation, an alternative taking a obtains $W_{E,b}(a)$, and full realisability in that sense forces $\kappa = \kappa_{\text{port}}$; slot realisability forces $\kappa = \kappa_{\text{slot}}$. Neither follows from the word, and the benchmark must be supplied.

18.7. Program domains and global totality

The finite theorem extends to a program domain. Let the components be programs $c_i(d, a) \in B_i$, promised to terminate on every string d and every legal a, with a finite rational host, and with the environment's description supplied as d. A fixed algorithm is allowed as a component $c(d, a) = F(d)$, and it is independent in the declared sense, being invariant in the live parameter, even when F happens to be the agent under evaluation. When CF10 asks only for a promise-total τ, the four kernels remain promise-computable, the slot optimiser halts on every valid description, and Theorem 18.12 holds unchanged; no algorithm deciding membership in the promise class is needed. Requiring τ to be globally total changes the map.

Proposition 18.15. *On the program domain, with* CF10 *requiring a globally total τ, each of* $\{\text{CF3}, \text{CF10}\}$, $\{\text{CF4}, \text{CF10}\}$, $\{\text{CF6}, \text{CF10}\}$, *and* $\{\text{CF9}, \text{CF10}\}$ *is inconsistent. The minimal inconsistent subsets are these four together with* $\{\text{CF5}, \text{CF9}\}$, $\{\text{CF2}, \text{CF4}, \text{CF5}\}$, *and* $\{\text{CF3}, \text{CF4}, \text{CF5}\}$, *and the maximal consistent subsets are* $\{\text{CF1}, \text{CF2}, \text{CF3}, \text{CF4}, \text{CF6}, \text{CF7}, \text{CF8}, \text{CF9}\}$ *(witnessed by κ_{slot}, without a globally total optimiser) and the sets of κ_{port}, κ_{base}, and κ_{flat} in Theorem 18.12.*

Proof. For a globally total candidate τ, the component $c(d, a) = \tau(d)$ is admissible. With the host $u(a, x) = \mathbf{1}[a \neq x]$ and $b = \tau(d(E_\tau))$, the component is constant in the live parameter, so CF3, CF4, CF6, and CF9 each force the values 0 at b and 1 at $1 - b$, contradicting CF10. The finite-domain cores remain, and those containing CF10 are now subsumed. If CF10 is in a core-free set, CF3, CF4, CF6, and CF9 are excluded, and the set lies in that of κ_{flat}, which every globally total agent ratifies; without CF10, the case analysis of Theorem 18.12 applies. □

This is the fixed-copy diagonal of Theorem 15.6, factored through the desiderata, and there is no conflict with promise-total optimisation: embedding the promise-total optimiser as F need not yield a component that terminates on every string, so the diagonal need not belong to the domain. If CF3 uses a syntactically narrower independence class that excludes embedded agent algorithms, $\{\text{CF3}, \text{CF10}\}$ need not be inconsistent; the proposition uses invariance in the live parameter.

18.8. The candidate semantics located

The semantics of Section 17 can now be placed.

Transport through compositions. For postcomposition alone, $h_0 \circ T = h_1 \circ T$ means that h_0 and h_1 agree on the image of T, which determines their values at every legal action exactly when T is surjective. So the relevant property is surjectivity, not injectivity, and in any case no function from infinitely many descriptions to a finite action set is injective. Arbitrary contexts restore the ambiguity for every agent: with $t = T(d)$, the contexts $C_0[P](x) = 0$ and $C_1[P](x) = \mathbf{1}[P(d) \neq t]$ are both the constant 0 at $P = T$ and differ under a change of the output at d. This is the environment of Theorem 18.10 in program form. With tagged contexts, conditional transport, and grounded extensionality the core is {CF2, CF4, CF5}; with transport that asserts existence it is {CF4, CF5}; with parametric extensionality the contexts are not identified and κ_{slot} works.

Local matching records each component's actual queried inputs, selects those components that agree with T there, and freezes the selection. On an appropriate termination promise it satisfies CF1, CF2, agent extensionality, implementation extensionality at fixed inputs, and CF8. In the environment of Theorem 17.2 it violates CF3 and CF6 under fixed-component independence, transport for transformed copies, CF9 under the ordinary fixed-component benchmark, and CF10. That does not make {CF1, CF2, CF5, CF8, CF10} inconsistent, which κ_{flat} satisfies.

Global matching misses $1 - T$, which is never equal to T, while transport requires its value to become $1 - a$. Its kernel can also be uncomputable with every component total: with $T \equiv 0$ and the components g_e of Theorem 17.5, the environment returning $g_e(d(E_e))$ has actual payoff 0, and its value at action 1 is 1 if e never halts and 0 if it halts.

Fresh rematching violates CF7 (Proposition 17.3: the final payoff is 1 where direct intervention gives 0), but no desideratum requires it; retaining the original matching set, or a supplied graph, avoids the failure. Literal syntactic substitution needs a one-pass convention for the same reason, since the replacement code $T^{d \leftarrow a}$ mentions T away from d and repeated rescanning can expand without end.

Material implication yields the relation $\mathcal{V}$ of Section 17.3, not a kernel. In an independent five-and-ten the attainable payoff for ten is ten, so a selector assigning 0 violates CF3 and CF9, but a selector from the relation need not: an additional rule could choose the intervention value. A kernel assigning $(5, 0)$ to an agent that takes five ratifies that agent, so the failure is one of calibration and realisability, not of CF10.

Fixed copies are transported only if the equivalence convention erases their role. If CF5 identifies a designated copy with a fixed program computing the same closed function, the transport owed to the first passes to the second, against conservativity at a non-actual action; if extensionality preserves the open interface, nothing passes. Under the fixed-component realisation benchmark the global fixed-copy diagonal makes that realisability inconsistent with CF10 alone.

Guarded logical semantics for terminating programs is κ_{base}, by Proposition 17.9: it ratifies no agent under CF10 on a binary problem, and every agent under CF10′.

Logical induction with exploration computes, at a stage n, values such as

$$\kappa_n(a) = \frac{1}{n}\sum_{i=0}^{n-1} \mathbb{P}_n\big(\bar{U} > i/n \mid A_n = a\big),$$

using the capped conditional of Garrabrant et al. (2016, Definitions 4.7.1 and 4.8.2), with value 1 when the conditioning probability is zero. Nothing in the logical induction criterion makes these values calibrated at a given stage: setting every price at one stage to zero preserves the criterion by finite-perturbation closure, the capped conditional then assigns 1 to every conditioned proposition, and a constant-zero payoff receives $\kappa_n(0) = \kappa_n(1) = 1$, violating CF1, CF3, and CF9. The scope of exploration is a further input: with $u(a, b) = a + 2b$ at the baseline $a = b = 0$, exploration that changes only the port yields the values $(0, 1)$, while exploration that changes the port and a designated copied bit together yields $(0, 3)$. Conditioning computes the consequences of a supplied exploration mechanism; it does not choose the mechanism.

18.9. Robustness

variation	effect
globally total or promise-total agents	the finite obstruction survives both; arbitrary fixed total algorithms add the pairwise cores of Proposition 18.15
fixed algorithms admitted	fixed-target diagonals return when alternatives leave the target fixed; they are not needed for the finite obstruction
fixed algorithms excluded	requires a concrete grammar: declared agent interfaces, no unrestricted evaluation or source reflection, intervention-total contexts; "nothing derived from T" is not a definition, since every g is $C[T]$ for a context ignoring its argument
informative agents	injectivity is unavailable; surjectivity removes only the $h \circ T$ ambiguity
distribution-valued kernels	exact transport can demand the distinct point masses δ_0 and δ_1 for one grounded representation; uncertainty does not remove the contradiction
mixed agents	some ratifiability failures acquire mixed fixed points, depending on prediction and sampling semantics (Section 6.3); the transport contradiction is untouched
policy-level interventions	every one-input witness embeds; finite policy tables support the slot theorem; infinite policy spaces need separate assumptions
bounded or certified languages	can support a globally total optimiser with a default on invalid descriptions; the unrestricted fixed-copy diagonal need not belong to them

19. The witness problems revisited, and counterpossibles

19.1. Verdicts of the candidate semantics

The witness problems of Section 4 can now be given as programs and evaluated under each candidate semantics. Let 0 stand for one-boxing or cooperating and 1 for two-boxing or defecting, and use the Newcomb and twin payoffs $u(a,b) = 2(1-b)+a$, where b is the predictor's output or the twin's action. With $t = T(d)$ the reference agent's output, write the vectors of values in the action order $(0,1)$ as

$$P_t = \big(2(1-t),\ 2(1-t)+1\big), \quad S = (2,1), \quad D_t = (t,\ 1-t), \quad Z = (0,0), \quad H = (5,10).$$

P_t freezes the predictor; S transports a shared copy; D_t and Z are the frozen and transported vectors of the mismatch diagonal $\mathbf{1}[a \neq b]$ of Theorem 15.6; and H gives the payoffs of five-and-ten. In Newcomb and the twin the dependence is a designated oracle call; in the diagonal the copy of the agent is a fixed component.

semantics	Newcomb	twin	five-and-ten	fixed-copy diagonal
port only	P_t: two-box	P_t: defect	H: ten	D_t: opposite of t
designated slots	S: one-box	S: cooperate	H: ten	D_t: opposite of t
slots and literal copies	S	S	H	Z: tie
matching	S	S	H	Z: tie
supplied graph	S or P_t	S or P_t	H	Z or D_t
guarded, arithmetic	actual only	actual only	actual only	actual only
logical induction	none	none	none	none

The first two rows are κ_{port} and κ_{slot}, that is, O_{const} and O_{oracle}; matching is local or global; logical induction is at an unspecified stage. For the dependency graph the first entry applies when the graph links the predictor or copy to the decision node and the second when it treats it as fixed. "Actual only" means that every non-actual action has no value (Proposition 17.9), so there is no recommendation under CF10. For logical induction, nothing in the criterion fixes a numerical verdict at a given stage; the all-zero-price stage of Section 18.8 assigns the normalised upper bound to both actions, giving $(3,3)$, $(3,3)$, $(10,10)$, and $(1,1)$.

The table makes visible what Section 4 took as given. In Newcomb's problem and the twin prisoner's dilemma the conflict between C1 and C2 is the difference between the second row and the first: whether the predictor's output, or the twin's action, is transported with the agent's output or frozen at its baseline. A semantics that transports it one-boxes and cooperates; one that freezes it two-boxes and defects. The model of Section 2 made this choice by stipulating μ_π; a program does not make it, and the rows below the second show the ways an agent may try to make it from source. On the fixed-copy diagonal, conversely, transporting the copy turns a lost problem into a tie, which is why semantics that recognise copies escape Theorem 15.6 and why Theorem 17.4 needs a copy that is transformed rather than equal.

19.2. Counterpossibles

Every non-actual output of a deterministic computation is, relative to the computation's complete description, an impossibility. Logical counterfactuals are therefore *counterpossibles*, conditionals with impossible antecedents, and the philosophical literature on them bears on the results of Section 18.

On the orthodox semantics of Lewis (1973), a counterfactual with an impossible antecedent is vacuously true. Vacuism is the counterfactual analogue of material implication in Section 17.3: it assigns every consequent to every non-actual output, and so provides a relation, not a kernel. Non-vacuists answer with impossible worlds, in which impossible antecedents hold without everything holding (Brogaard and Salerno 2013; Nolan 1997). Williamson (2015, 2018) defends vacuism, and one of his arguments has the structure of Theorem 18.10. Suppose the intension of $A \mathrel{\square\!\!\rightarrow} B$ is a function of the intensions of A and of $A \wedge B$. If A is impossible, both intensions are empty for every B, so all counterfactuals with antecedent A have the same intension; since $A \mathrel{\square\!\!\rightarrow} A$ is true everywhere, every counterpossible is true. Berto et al. (2018) reply by admitting impossible worlds whose valuations discriminate more finely, and discuss the conflict between reflexivity, closure under classical consequence, and non-trivial treatment of contradictions, together with failures of substitution inside counterfactual contexts.

The analogy is exact at one level: an identification appropriate to actual or possible behaviour need not preserve the distinctions needed under counterpossible suppositions. Williamson's argument identifies antecedents by their possible-worlds intension, which is empty for every impossibility; grounded extensionality identifies components by their instantiated behaviour, which is the same for the identity context and the constant context at the baseline. In both cases the identification collapses exactly the distinctions that non-vacuous counterfactuals need.

Impossible worlds can represent every kernel of Section 18. Given a baseline-correct kernel on finite environments, retain an actual world with the actual action and payoff; for every defined alternative a introduce a world recording action a and payoff $\kappa(E, b, a)$, selected when the antecedent "the output is a" is evaluated; and for an undefined alternative select several worlds with incompatible payoffs, so that no value is determined. The selected worlds are impossible relative to the complete description of the computation. Retaining the origin of each world and overwriting its target realises CF7. The construction represents κ_{slot}, κ_{port}, κ_{base}, and κ_{flat} alike, and therefore derives none of them: impossible worlds can record unattainable values, which CF9 separately excludes, and the selection relation encodes a counterfactual policy chosen in advance. Impossible-world semantics makes room to preserve distinctions; it does not recover dependence information that a closed program's actual behaviour has lost.

Two further precedents are close. Lewis (1976) proved a triviality theorem about representing conditional probabilities as probabilities of fixed conditional propositions; it does not prohibit numerical conditional expectations or any kernel of Section 18, but it is the classical example of a semantic identification that cannot respect the required distinctions. Taylor (2020) discusses changing the agent's source code

together with its hypothetical policy, and the underdetermination of such changes, as an alternative to logical counterfactuals; it is closely related to the slot semantics but contains no axiomatic impossibility. The discussion of embedded agency by Demski and Garrabrant (2019) makes the point that an externally supplied agent slot already contributes disputed structure, without a theorem about numerical kernels.

19.3. What an optimality theorem requires

The results of this part combine into a list of the assumptions that any optimality theorem for an agent reading descriptions must state, because each of them changes some answer above:

(i) the atomic interfaces at which the agent is consulted;
(ii) which contexts are intervention-total;
(iii) whether implementations are compared on preserved inputs;
(iv) whether fixed algorithms are admitted as constants;
(v) whether agents must halt globally or only on a promise class;
(vi) the substitution benchmark by which alternatives are realised;
(vii) how undefined alternatives are treated in ratifiability.

They are mathematical assumptions, not implementation details that may be dropped once the theorem is stated. Supplying them is supplying what Section 2 supplied with μ_π.

Part IV.
Interpretation

20. Philosophical discussion

The preceding sections prove theorems about stated definitions. This section interprets them. Its claims are arguments, not results, and where they go beyond what has been proved they are marked as such. Its aim is the one that motivated the inquiry: to say exactly where the disagreement among decision theories is located, so that the apparent paradoxes can be separated from the substantive question they conceal.

20.1. What a formalisation decides before any theorem is proved

The first lesson concerns method, and the paper is its own evidence. Part I compared decision theories on problems that supply, for every policy π, the law μ_π of the world's response (Definition 2.2). On that class, ex-ante optimality is attained by the most naive construction available, choosing a maximiser problem by problem (Theorem 3.6), and every impossibility for criteria that judge each problem separately reduces to a single problem on which two acceptable sets are disjoint (Lemma 3.8).

The result is cheap for the reason Section 14 makes explicit: causal, evidential, and functional decision theory disagree precisely about where μ_π comes from, and a model that supplies μ_π has already settled their dispute. The well-definedness requirement of Definition 2.3 did the same for problems that refer to a particular theory. The fairness condition admitted them, since fixed algorithms may occur as constants (Definition 2.4), but Remark 3.7 found the corresponding construction ill defined, because it needed a fixed point that no well-defined finite problem supplies. The verdict was reversed once problems became programs (Theorem 15.6).

The general point is this. To state a decision problem formally, one must say how the agent's alternatives are to be evaluated, and that choice is usually the one in dispute. A formal result about decision theories is therefore only as informative as the specification of alternatives it presupposes, and the specification is often made in the format of the problems rather than in the statement of the theorem, where it is easy to miss. Appendix A and Section 19.3 exist to prevent this. Reading the literature with the same question in mind, namely *which part of the answer was built into the format?*, removes much of the appearance that one theory has been refuted by a problem rather than by an assumption.

20.2. The locus of the conflict

The theories compared here agree about more than their dispute suggests. They share the payoffs, the probabilities of the world's own randomness, and the rule of maximising expected value. Section 2 isolates what remains in a single subscript. The causal value $Q_\pi(h, a)$ of an alternative a at h averages with μ_π, the law induced by the policy actually adopted; the ex-ante value $V_D(\pi[h{\leftarrow}a])$ averages with $\mu_{\pi[h\leftarrow a]}$, the law the alternative itself would induce. Every within-problem conflict of Part I is a consequence of that difference. For a single choice, Theorem 4.2 states exactly when it can arise: when, and only when, the law of the states preceding the choice depends on what the agent prescribes. Proposition 6.5 gives the complementary fact: if the world's response does not depend on the agent's policy, every ex-ante optimum is causally optimal and no conflict arises.

The same choice reappears in each part of the paper, in a different form.

- In Part I it is the choice of the law under which an alternative is evaluated.
- In Part II it is the choice between the row and the diagonal of the causal matrix, and the cross-problem results depend on whether the belief perspectives associated with removed options disappear with them (Section 10).
- In Part III oracle substitution recovers the diagonal and port-only intervention a row (Proposition 14.4). At the axiomatic level the choice is one between two notions of when computations are the same (Definition 18.5): grounded equivalence, which identifies components by their behaviour once the actual agent is inserted, and parametric equivalence, which identifies them only if they agree under every substitution. Calibration together with grounded equivalence forces causal evaluation (Lemma 18.9); parametric equivalence admits ex-ante evaluation satisfying every desideratum (Corollary 18.13).

This, then, is the locus of the conflict. The theories do not disagree about how to choose among specified alternatives. They disagree about what the alternatives to the agent's own choice *are*. Causal decision theory evaluates an alternative in a world that keeps the actual past, including whatever in it predicted the agent, and varies only the act: the small miracle of Lewis (1979a, p. 468), located at the moment of choice. Functional decision theory evaluates an alternative in a world in which the agent's decision procedure outputs something else, so that everything computing that procedure varies with it. Neither world is actual, and nothing in the actual course of events selects one of them. The discussion following Proposition 15.9 states this in its simplest form: a fixed program computing 0 and a linked call to an agent that outputs 0 agree on the actual execution and differ under replacement. Theorem 18.10 states it in general: no total semantics can both transport a computation through alternative outputs and identify every presentation with the same actual behaviour, because actual behaviour discards the information that transport needs.

The philosophical literature on counterpossibles meets the same obstruction from the other side (Section 19.2). Williamson's argument for vacuism identifies antecedents by their possible-worlds intension, which is empty for every impossibility; grounded extensionality identifies components by their instantiated behaviour, which is the same for the identity context and the constant context at the baseline. In both cases an identification appropriate to what actually happens collapses the distinctions that non-vacuous counterfactuals need. Wolpert and Benford (2013) reached a related diagnosis of Newcomb's problem: the two classical arguments presuppose incompatible formalisations of how choice and prediction are related. The present results locate that incompatibility and show that it recurs, in the same form, wherever the agent's alternatives must be specified.

20.3. Not arbitrary: the coupling between agent and predictor

If the facts of the actual execution do not decide between the two evaluations, one might conclude that the choice is a convention, and that the dispute between causal and functional decision theory is verbal. The results do not support that conclusion, for two reasons.

First, the choice has consequences. Agents built on the two evaluations, placed before the same reliable predictor, obtain different payoffs (Proposition 4.1). A dispute whose resolution changes what agents receive is not merely verbal.

Second, the candidate ways of making the choice are constrained. Section 17 shows that the natural rules for recognising the agent in a closed program each fail in a characteristic way: literal matching misses encoded copies; local matching manufactures dependence from accidental agreement (Theorem 17.2); global matching misses transformed dependence such as $1-T$ (Theorem 17.4) and can make optimality decide the halting problem (Theorem 17.5); and implication-based reasoning about the actual agent yields spurious values (Theorem 17.7) or none (Proposition 17.9). A semantics is not something one may simply adopt; it must survive these failures.

What, then, could fix the choice? The proposal of this section, which goes beyond anything proved above, is that it is fixed, where it is fixed at all, by the *mechanism*

through which the predictor's output depends on the agent. The actual execution does not reveal that mechanism, but the mechanism is a feature of the world, and in some cases it is known.

- A predictor that runs the agent's own code on the agent's own input, through what Section 14 calls a designated call, depends on the agent's decision function in the parametric sense. Changing the function's output changes the prediction, and oracle substitution, hence ex-ante evaluation, describes the situation literally (Theorem 15.5).
- A predictor whose output is fixed by data that do not compute the agent's function, such as base rates in a population or a record made after the fact, does not depend on that function at all. Port-only intervention describes it correctly, and the two evaluations coincide whenever the induced laws do (Proposition 6.5).
- The difficult cases lie between, and they are the ones thought experiments typically present. A predictor that scans the agent's brain beforehand reads a physical state that will later produce the decision. Whether its output depends on the agent's decision function, or only on a physical state with which the decision is correlated, depends on whether the agent is identified with the computation the brain implements. That identification is not a fact about the prediction mechanism alone; it is a commitment about what an agent is.

On this view, the error of the debate has been to seek a decision rule that is correct independently of the coupling between agent and world. No such rule should be expected. What can be asked of a theory is a correct evaluation *given* the coupling, and causal and ex-ante evaluation each supply one, for different couplings. The irreducibly philosophical residue is then smaller and better defined than the debate suggests. It is not "which decision theory is rational?", but whether, in the cases where the coupling runs through an agent's physical constitution, the agent is that constitution or the function it computes. The extraction problem left open in Section 21, whether a dependency graph can be recovered from an arbitrary closed program, is the formal shadow of this question, and Corollary 18.13 shows that once it is answered nothing further stands in the way of a coherent semantics.

20.4. When evaluation happens: the Bomb and updatelessness

The Bomb is the case most often offered against functional decision theory, and its treatment in Section 4.4 shows the same locus at work. As usually stated, the problem is not well defined (Proposition 4.4): a predictor whose forecast determines the observation, and whose target is the agent's eventual action, cannot be accurate for every policy. Under the policy (L, R), which takes Left after seeing a bomb and Right otherwise, every forecast is wrong. Under the policy (R, L), every forecast is right: a forecast of Right produces a bomb and the agent takes Right, while a forecast of Left produces no bomb and the agent takes Left. The problem therefore leaves undetermined what such an agent receives. The repaired version (Proposition 4.5) fixes the predictor's target as the agent's prescription on seeing a bomb, and under

that stipulation ex-ante evaluation selects (L, L) while causal evaluation selects (R, L). The force of the case thus depends on an assumption that its informal statement does not make.

Even repaired, the disagreement concerns the vantage point of evaluation, not the value of outcomes. Section 4.4 records both readings. Taken as the policy problem, evaluated before the prediction, the Bomb favours Left: a risk of death of probability $\varepsilon = 10^{-24}$ against a near-certain cost of 1. Taken as the already-conditioned problem, "a bomb is visibly present", ex-ante evaluation itself selects Right, since the options are then worth $-M$ and -1. Functional decision theory evaluates from before the prediction and declines to update on what it sees, which is what *updatelessness* means (Dai 2009; Yudkowsky and Soares 2017). The objection to it is that the problem posed is the second one, so that a calculation conducted from the first answers a different question. Counterfactual mugging (Section 4.5) separates the same two vantage points with a prediction about an information set that is unreached on half the paths.

The policy vantage point is not unreasonable. The literature on resolute choice and dynamic consistency (Gauthier 1994; McClennen 1990) and the planning-stage analysis of the absent-minded driver (Aumann et al. 1997; Piccione and Rubinstein 1997) defend executing an ex-ante plan even where it is not optimal ex post. And a commitment made before the prediction is worth exactly the ex-ante value of the policy committed to (Proposition 5.2), so even a causal decision theorist, offered the chance to commit at that stage, would commit to Left. The ordinary acceptance of small risks of death, as in daily travel, makes the policy calculation look sensible. The analogy fails, however, at exactly the point in dispute: in ordinary risk the bad outcome is not observed at the moment of choice, whereas in the Bomb the agent chooses after the improbable event has occurred and is in view. Whether the agent's choice now *is* the output the predictor read earlier, so that choosing Right now amounts to having been an agent for whom bombs are placed, is the coupling question of Section 20.3 in temporal form. If it is, the policy vantage point is correct and the bomb is the price of a good policy. If it is not, the agent simply stands before a bomb, and the policy calculation describes a choice that is not being made. The Bomb is therefore neither a refutation of functional decision theory nor a mere appeal to emotion; it is a test of whether one accepts the identification on which that theory rests.

20.5. Fair and unfair problems

Functional decision theory is claimed to do well on fair problems, those in which the outcome depends only on the agent's behaviour, and problems built against a particular theory are set aside as unfair (Yudkowsky and Soares 2017). The results bear on this restriction in three ways.

First, unfairness is a matter of evaluation semantics, not of the problem's actual behaviour. The fixed-copy diagonal of Theorem 15.6 defeats every globally total computable agent when the embedded copy is held fixed; if the copy is declared linked, the payoff becomes $\mathbf{1}[A(d) \neq A(d)] = 0$ and every agent is optimal. Since the two declarations agree on every actual execution, the boundary between fair and

unfair problems cannot be drawn from behaviour; it is the linked–fixed distinction of Section 20.2 again. The fairness criterion presupposes a classification it does not supply, as Yudkowsky and Soares (2017, §9, n. 15) acknowledge when they note that it remains unclear how to transform a problem unfair to one theory into an analogous problem for another. Defences that remove counterexamples by declaring them unfair, or by declaring them outside a theory's intended domain, do the work that the well-definedness requirement did by stipulation in Part I.

Second, the obstacle to an optimal theory is not choosing well but being computable by the environments. An optimal decision function exists on both admissible classes, fixed copies of arbitrary programs included (Theorem 15.3). What fails is an optimal agent that halts on every input, because any such agent can be run by an environment that rewards the opposite (Theorem 15.6); the oracles that compute a globally total optimal agent are exactly the PA degrees (Theorem 16.3). Optimal choice is always possible problem by problem; no decision procedure that the world can run is optimal everywhere. This vindicates the intuition that a counterexample to any predefined theory is always easy to construct, and it locates that intuition precisely: the counterexample needs the theory to be a program the environment can execute.

Third, the intuition should not be overstated. Problems built against a theory exist for every theory the environments can compute, but not only for those: a non-computable decision function can also be predictably wrong (Proposition 16.8). What protects an optimal function is optimality, not non-computability. The slogan that unfair problems exist exactly for computable theories is false; the defensible statement is that global simulability by the environments guarantees a constructible defeat.

20.6. Thought experiments as instruments

Idealisation is not the difficulty with these thought experiments. Physics proceeds through frictionless planes, and a reliable predictor is a legitimate device for isolating one feature of a decision, its predictability. The difficulty is that the device is usually specified in exactly the respect that matters least and left unspecified in the respect that matters most. A problem states the predictor's accuracy, sometimes to twenty-four decimal places, and says nothing about its mechanism, its target, or how its errors arise. The results above show that these determine the verdicts.

- The *mechanism* fixes whether the prediction is linked to the agent (Sections 20.2 and 20.3).
- The *target* decides whether the Bomb is even well defined (Proposition 4.4), and restrictions on admissible targets decide which conflicts survive (Section 6.4).
- The *error kernel* is a stipulation (item 8 of Appendix A). "Wrong once in 10^{24} cases" is an overall rate. Whether errors are independent of the agent's policy, concentrated in particular situations, or correlated with the agent's disposition changes the ex-ante calculation, and no statement of the problem fixes it.

Two further habits compound the underspecification. Extreme magnitudes, such as an error rate of 10^{-24} against a death cost of 10^{9}, push intuitions into a range in which they have no evidential basis. And informal arguments import structure that the problem does not contain: repeated play, where the problem is one-shot; or, in the other direction, the stipulation that the agent knows it is not inside the predictor's simulation, which the problem may not license. Section 19.3 lists the assumptions that any optimality theorem for an agent reading descriptions must state. The informal debate would benefit from the corresponding discipline: no verdict on a predictor problem should be accepted until the predictor's mechanism, target, and error kernel are fixed, because until they are, the problem has no determinate answer to dispute.

20.7. What the results leave untouched: the utility presupposition

Every result in this paper assumes that outcomes have real-valued payoffs (Definition 2.1). Causal, evidential, and functional decision theory share this assumption, and the critique it invites applies to all of them equally.

In the strict sense required by the representation theorems of von Neumann and Morgenstern (1944) and Savage (1954), an agent has a utility function when a single, menu-independent ranking of outcomes, complete, transitive, and continuous, accounts for its choices in every situation. Agents with incomparable or lexicographic values fail this. Consider a forced choice between two children, the predicament of Styron (1979), and model the chooser as a coalition of two sub-agents, each protecting one child at any price. A small bonus attached to either option does not tip the choice, so the options are incomparable rather than tied, which is the small-improvement argument of Chang (2002); and a lexicographic preference has no real-valued representation at all (Debreu 1954, 159, with n. 1 on p. 164). If the coalition responds to deadlock by refusing, it brings about the outcome both sub-agents rank last. The menu dependence such an agent exhibits violates exactly the contraction condition C of Part II, Sen's property α: an option chosen from a larger menu is not chosen from a smaller one that contains it. Harsanyi's aggregation theorem (Harsanyi 1955, §III) adds that if the coalition and its members were all expected-utility maximisers and the coalition respected their unanimous preferences, its utility would be a weighted sum of theirs, so it would trade one commitment against the other, which is precisely what such an agent refuses to do.

There is a weaker sense in which every consistent agent maximises something: indexing outcomes by the situation in which they occur, or redescribing them finely enough, always yields a function that the agent's choices maximise (Broome 1991, ch. 5). That sense is always available and therefore says nothing. It is the counterpart, for utilities, of the lookup tables of Theorem 3.6. Enriching outcomes with factors such as guilt or responsibility can restore a strict representation, sometimes rightly, since the obvious ranking of outcomes may not be the agent's actual one. But each such enrichment must make predictions across situations, or it becomes the vacuous representation again.

Two consequences follow for the scope of the paper. Its results are conditional on agents with well-defined payoffs, and they do not show that real agents have them. And the epistemic objection to applying decision theory, that an agent's utilities are opaque, shaped by culture and unavailable to introspection, is distinct from the metaphysical one, that for some agents there is no utility function to be known. The first limits application; the second limits the subject matter of the theory.

20.8. Practical scope

Where the world's response does not depend on the agent's policy, all the evaluations considered here coincide (Proposition 6.5). That covers most practical decisions, and in that domain the shared core, expected-value reasoning under uncertainty, has a long record of use. The distinctive content of the theories is confined to cases of policy dependence, and those are the cases in which each fails some criterion: causal evaluation cannot always be ratified (Proposition 4.3), ex-ante evaluation acts against what the agent observes (Proposition 4.5), and every agent that halts on all inputs can be diagonalised by an environment that runs it (Theorem 15.6). The following is an argument, not a result: a theory whose value lies only in the cases where its rivals disagree, and which fails some reasonable criterion in those same cases, has no demonstrated practical advantage over its rivals. Nothing in this paper suggests otherwise.

The one setting in which the coupling view of Section 20.3 promises more is that of agents whose code can be inspected or run by others. There the mechanism of prediction can be known, the dependence can literally be a designated call, and Theorem 15.5 gives a computable optimiser for it. Even there, Theorem 15.6 applies whenever the environment can run the agent with its output held fixed. For human agents facing hypothetical predictors of unspecified mechanism, the dispute between causal and functional decision theory has, on the view defended here, no determinate answer, because the information that would decide it has not been given.

20.9. Summary

The disagreement between causal and functional decision theory is not a disagreement about how to choose. It is a disagreement about what the alternatives to one's own choice are. Formally, it is the choice between the row and the diagonal of the causal matrix (Proposition 14.4); semantically, between grounded and parametric equivalence of computations (Definition 18.5). The actual course of events does not settle that choice (Theorem 18.10), and the choice is not arbitrary (Section 17). This section has proposed that it is settled, where it is settled at all, by the mechanism coupling the agent to whatever depends on it, with an irreducible residue concerning whether an agent is identified with the computation it implements. Where a thought experiment leaves that mechanism unspecified, the dispute has no determinate answer, and the appearance of paradox is what that underspecification leaves behind.

21. Conclusion

The question was whether a decision theory can be optimal on every fair decision problem, and the expectation was that conflicting criteria of optimality would rule it out. The answer has four parts, the first three for problems that supply their counterfactuals and the fourth for problems that do not.

First, *ex-ante optimality is always attainable.* On every class of well-defined finite problems, choosing an ex-ante maximiser in each problem is a decision theory (Theorem 3.6). Because lookup tables count as decision theories, an impossibility for criteria that judge each problem separately is never more than a single problem on which their acceptable sets are disjoint (Lemma 3.8).

Second, *within a problem the conflicts are shallow.* Among ex-ante optimality, structural causal dominance, ratifiability, existence, commitment stability, and a reference-policy evidential criterion, every inconsistency has a core of one or two criteria, and the maximal consistent combinations have explicit total witnesses (Theorems 5.7 and 5.10). The central conflict is Nozick's. It is possible exactly when the law of the states preceding a choice depends on the prescription (Theorem 4.2); it survives imperfect prediction, randomisation, and every restriction on prediction targets short of excluding anticipation; and it obeys an exact trade-off between ex-ante and causal regret (Proposition 6.1). The conflict between dominance and commitment stability, by contrast, is confined on its witness to infallible prediction, whereas its expectational counterpart persists whenever one-boxing has the higher ex-ante value (Proposition 5.11).

Third, *across problems, the depth depends on the semantics of menus.* On the causal matrix, with common affine invariance, the cross-problem axioms reduce to singleton and pairwise cores (Theorem 8.1). Row-translation invariance—evaluation insensitive to the baseline of each belief state—produces a minimal four-axiom impossibility: neutrality, row translations, and contraction force universal choice (Theorem 9.3), exactly on domains satisfying (DR) (Theorem 9.7), a condition that survives imperfect prediction, bounded action sets, and, with affine invariance, bounded payoffs. Principal binary independence, with neutrality, affine invariance, ordering, non-triviality, and monotonicity, characterises ex-ante evaluation (Corollary 11.2). Both results presuppose principal restriction, on which removing an option also removes the belief state it would induce. Under column or fixed-reference restriction, fixed-prior causal evaluation satisfies every axiom considered except diagonal optimality, self-sovereignty, and principal binary independence (Proposition 10.7), and contraction under any semantics is compatible with $2^{\mathfrak{c}}$ rules that do not maximise ex-ante value (Proposition 11.5).

Fourth, *when the dependence must be inferred, the obstruction is the semantics of alternatives.* For agents that read source code, a designated substitution interface makes an optimal decision function exist on both admissible classes considered, fixed copies of arbitrary programs included (Theorem 15.3), and a termination promise makes it computable (Theorem 15.5). Three impossibilities hold without any counterfactual subtlety: queries about other problems (Proposition 15.4), fixed copies of the agent against agents that halt on every input (Theorem 15.6), and exact se-

lection among computable expected values, which is uncomputable even with no predictor (Theorem 15.10) but becomes computable as soon as either ties or arbitrarily small advantages are excluded (Proposition 15.11). The last two have the same exact strength: the oracles that compute an optimal agent halting everywhere, or an exact optimiser on almost surely terminating environments, are the PA degrees (Theorem 16.3), some of them strictly below the halting problem (Corollary 16.4). Optimisation needs only to select a never-refuted policy, not to recognise admissible problems; a prescribed default off the admissible class, or a prescribed tie order, raises the requirement to $\mathbf{o}'$ or $\mathbf{o}''$ (Proposition 16.6). The diagonal relativises to every oracle (Theorem 16.7), and problems built against a theory exist for every theory the environments can compute but not only for those (Proposition 16.8). The ways of recognising the agent in code each fail: literal matching misses encoded copies, local matching creates dependence from accidental agreement (Theorem 17.2), global matching misses transformed copies and can make optimality decide the halting problem (Theorems 17.4 and 17.5), and proof-based counterfactuals about the actual agent are spurious (Theorem 17.7) or undefined (Proposition 17.9). On the axiomatic level, every inconsistency among the desiderata for counterfactual operators contains grounded extensionality (Theorem 18.12). Together with calibration it forces causal evaluation (Lemma 18.9); with calibration and ratifiability it forms a minimal inconsistent triple whose witness is Death in Damascus (Theorem 18.11); and replacing it by parametric extensionality leaves ex-ante evaluation satisfying every desideratum (Corollary 18.13). The fixed-copy diagonal reappears as four pairwise cores when the optimiser must halt on every input (Proposition 18.15).

The thesis that the paper can defend is therefore the following.

> *On any class of well-defined, fair problems containing a sufficiently predictive Newcomb problem, no assignment of policies satisfies both ex-ante optimality and causal dominance, although an ex-ante optimal assignment exists. Within problems, every inconsistency among the natural criteria has a core of size at most two. Across problems, on the unrestricted causal-matrix domain, principal binary independence together with neutrality, ordering, affine invariance, non-triviality, and monotonicity characterises ex-ante ranking, and principal contraction together with neutrality is incompatible with non-trivial, baseline-invariant choice on every domain satisfying (DR). Neither cross-problem result survives when a smaller menu retains the belief model of the larger one. For computable agents that read descriptions, optimality is attainable on the promise domain of any designated substitution semantics; an optimal decision function exists on that domain whatever fixed programs the environments contain, and computing one that halts everywhere, or selects exactly among computable expected values, requires and suffices with a PA degree. Apart from these degrees of non-computability, optimality fails only through the semantics itself: no total counterfactual operator can both transport a computation through alternative outputs and identify every presentation with the same actual behaviour, and calibration with such an identification is inconsistent with*

ratifiability. In every case the obstruction lies in how the agent's alternatives are specified, not in choosing among them.

Section 20 drew the interpretive consequence. The disagreement between causal and functional decision theory is a disagreement about what the alternatives to one's own choice are; the actual course of events does not settle it, and it is settled, where it is settled at all, by the mechanism through which the world's response depends on the agent. That proposal is an argument, not a theorem, and it leaves a residue the paper does not resolve: whether an agent whose predictor reads its physical constitution is that constitution or the function it computes.

Several questions remain open. The first is normative rather than mathematical: which semantics of menu restriction is the right one. Principal restriction is forced by the causal matrix once law-preserving removal is assumed, but an announced restriction may change a predictor's behaviour, and a fixed belief model is the natural reading of fixed-prior causal theories. The cross-problem impossibilities stand or fall with that choice, and the paper has shown only how they depend on it. The second is whether structural dominance and commitment stability are jointly satisfiable on a whole class of problems with imperfect prediction: Proposition 5.11 removes the Newcomb witness, and Proposition 5.12 decides dominance among commitments, but no general consistency theorem has been proved. The third concerns the reduced form. The causal matrix discards structural states and the local structure of multi-stage problems, so that column Pareto is not structural dominance and global ratifiability is not local ratifiability. The context environments of Section 18 are one richer reduced form, which keeps the factorisation of the matrix through the components, and on it the semantic choice that the matrix leaves implicit becomes an axiom; a cross-problem theory of rules on that form has not been developed. The fourth concerns regularity: the cardinality results use the freedom to define rules arbitrarily on orbits, and the classification of rules subject to continuity or measurability is open. The fifth is a complete consistency map for mixed policies, which requires a choice among several inequivalent semantics for output, prediction, and contraction. The sixth is whether parametric extensionality can be implemented for unrestricted programs: Corollary 18.13 assumes the contexts are supplied, and extracting them from a closed program is the problem of extracting a dependency graph, for which no procedure is known and which global matching shows cannot be equality of computed functions. The seventh is whether a counterfactual semantics can be chosen on grounds other than the desiderata considered here, for instance by a learning process that is calibrated in the limit, and what such a choice would do to the map. The eighth concerns the degree results. They are proved for a designated semantics. For the matching semantics of Section 17, where Theorem 17.5 already makes optimality decide whether a machine runs forever, the exact degree of an optimal agent has not been determined. Finally, the status in the literature of the circulant theorem and of the consistency maps has been checked only by a bounded search; their ingredients are classical, and an earlier statement may exist.

TABLE 2. The principal results of Parts I and II, with their status. "Not found" means not found in the literature searched; no priority is claimed.

result	where	status
ex-ante optimality vs dominance (Newcomb)	Prop. 4.1	Nozick (1969)
no ratifiable pure act (Death in Damascus)	Prop. 4.3	Gibbard and Harper (1978); ratifiability literature
universal ex-ante selection; pointwise reduction	Thm. 3.6, Lem. 3.8	elementary
characterisation of the dominance conflict	Thm. 4.2	not found
Bomb: ill-posedness and repair	Props. 4.4, 4.5	case from MacAskill (2019)
within-problem maps	Thms. 5.7, 5.10	not found
noisy commitment and couplings	Props. 5.11, 5.12	uses Strassen (1965)
regret trade-off	Prop. 6.1	not found
mixed existence under continuity	Thm. 6.3	Halpern (2020); Bell et al. (2021)
realisability of every causal matrix	Thm. 7.2	elementary
R as factorisation; R is not causal rationality	Thm. 7.7, Prop. 7.8	not found
Pareto vs self-sovereignty	Prop. 8.5	Briggs (2010)
pairwise reduction under F	Thm. 8.1	not found
circulant theorem; map with R	Thms. 9.3, 9.6	not found; ingredients in Fink and Ruiz (1986), Brandt et al. (2016)
exact domain criterion (DR); physical restrictions	Thm. 9.7, Props. 9.8–9.11	not found
menu semantics; weight preservation; column maps	Sec. 10	not found; cf. Gallow (2026)
binary characterisation of ex-ante ranking	Thm. 11.1, Cor. 11.2	core in Podgorski (2022)
contraction without the diagonal; $2^{\mathfrak{c}}$ families	Thm. 11.4, Props. 11.5–12.5	not found
mixed and evidential extensions	Sec. 13	not found

TABLE 3. The principal results of Part III, continuing Table 2.

result	where	status
existence of optimal decision functions	Thm. 15.3	elementary
uniform optimiser on the promise class	Thm. 15.5, Prop. 17.6	elementary
foreign queries; fixed-copy diagonal	Prop. 15.4, Thm. 15.6	Skalse (2021); modal form in Fallenstein (2014a)
no computable exact optimiser	Thm. 15.10	recursive inseparability; application not found
PA degree of optimality; spectrum	Thm. 16.3, Cor. 16.4	not found; uses Scott (1962), Jockusch and Soare (1972)
admissibility; prescribed answers	Props. 16.5, 16.6	Π^0_2 part in Kaminski and Katoen (2015)
relativised diagonal; unfair problems	Thm. 16.7, Prop. 16.8	not found
matching semantics and their failures	Thms. 17.2–17.5, Prop. 17.3	not found
computable five-and-ten	Thm. 17.7	mechanism in Slepnev (2012), Benson-Tilsen (2014)
guarded counterfactuals for programs	Prop. 17.9	semantics from Fallenstein (2015b)
projection lemma; counterfactual-operator maps	Sec. 18	not found; cf. Williamson (2015)

A. Register of stipulations

The results are theorems about the definitions displayed in the text. The following choices are made there, and each could be made differently; where a different choice is known to change a result, the section that discusses it is given.

1. Policies are deterministic unless a mixed variant is stated (Sections 6, 13).
2. Every admissible policy induces a well-defined probability law (Definition 2.3); self-referential problems that lack one are excluded (Remark 3.7).
3. No information set occurs twice on a path (Definition 2.1).
4. Structural environment plans, including responses on branches not taken, are part of the problem (Definition 2.2); how outcomes on alternative branches are coupled is fixed by them.
5. Causal deviations freeze the baseline distribution over plans, and a local deviation keeps the policy's other prescriptions (definition of Q_π).
6. Dominance quantifies over structural states, including those of probability zero under the evaluated policy (Section 3); with supported states only, C2 becomes unsatisfiable in Death in Damascus.
7. Positive reach is measured under the evaluated policy.
8. Predictor error is given by a kernel; the Newcomb calculations use symmetric error.
9. A predictor targets specified policy coordinates; "the same computation" is represented by a stipulated dependence (Sections 4.2, 4.4); Part III removes this stipulation and the supplied laws μ_π with it.
10. Commitment is irrevocable, precedes the environment's response, offers the full policy set, and uses a specified coupling and structural state set (Definition 5.1, Section 5.4).
11. Commitment extensions are tagged and well founded, and the class is closed under them.
12. C5 is the totality already built into the definition of a theory.
13. The evidential criterion uses the reference-policy law and the null-event convention displayed in Section 5.3; other evidential semantics change the map (Section 13.3).
14. Mixed variants specify sampling, predictor access, information, counterfactual coupling, and continuity (Sections 6.3, 13).
15. Cross-problem rules output a reflexive relation whose maximal elements are required to be nonempty (Definition 7.3); neutrality acts by bijections between label sets, and invariances act on the relation.
16. Ordering concerns the output relation at each input, not rationalisability of each choice set.
17. Menu restriction is principal unless stated otherwise; principal restriction is

forced by law-preserving removal, and column and fixed-reference restriction require additional data (Section 10).

18. Non-triviality is existential unless a universe is specified (Theorem 10.6).
19. On restricted domains, an invariance applies when both a matrix and its transform are admissible (Section 9).
20. Equivalence of problems and parameterisation of rules are made precise before taxonomy claims are stated (Section 12).
21. Agents reading descriptions are either globally total or promise-total on a stated class, and results are stated separately for the two (Definition 14.1); the distinction separates Theorem 15.5 from Theorem 15.6.
22. Owned queries identify their environment and information set, namespaces of distinct environments are disjoint, and repeated queries receive the same answer (Definition 14.2).
23. Payoffs are exact rationals, computable reals, or sampled; the positive results use the first (Section 14).
24. Optimality is a property of decision functions, computable or not, and competitors range over all of them, which by (3) is the same as ranging over computable agents (Definition 15.2).
25. Embedded programs are fixed and run as ordinary computations, with their running time counted in the environment's execution; only designated calls are linked to the agent (Section 15).
26. Stochastic environments branch with exact rational probabilities, their rewards are rationals in $[0, 1]$ enforced by the language, and almost-sure termination is required of the whole execution under every table policy and of nothing else (Definition 15.1).
27. An optimal agent may choose any maximiser and may answer anything off the admissible class; a prescribed default or tie order is a separate requirement and changes the degree (Proposition 16.6).
28. A problem is "built against" a theory when it is an anti-prediction environment accurate for it (Section 16.6).
29. Under syntactic and extensional replacement the description supplied to the agent is held fixed, and implementations are compared on preserved inputs (Definition 14.3, Proposition 18.1).
30. Extensional matching uses typed, atomic components, matches once before replacement, and states its comparison domain (Section 17.2).
31. Counterfactual contexts are intervention-total: every supplied context terminates under every permitted substitution (Section 18).
32. Counterfactual kernels depend on the agent only through its local baseline output (Definition 18.4).
33. Independence of a component means invariance in the declared live parameter, so a fixed copy of the agent counts as independent (Definition 18.2, Section 18.7).

34. Realisability is measured by coherent slot substitution unless a fixed-component benchmark is stated (Section 18.6).
35. Ratifiability requires every alternative to be defined unless the weakened form CF10′ is stated (Proposition 18.14).

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